

7BIT FORM GUIDE

Illustrated step-by-step reference for every exercise in the circuit training programme.

59 movements | 85 exercise slots | 109 variations | 14 sessions
across 4 modes

39 movements carry openly-licensed line art. All 59 carry a full
illustration prompt.

Generated from src/core by tools/form-guide/build.py

HOW TO USE THIS BOOK

WHAT THIS IS

One entry for every distinct movement in the programme. The app's workout card carries a short form guide because it has to fit on a phone mid-set. This is the long version: numbered setup and execution, the position checkpoints that tell you whether a rep counted, the failure modes worth knowing about, and the protocol and evidence behind the prescription.

HOW IT IS ORGANISED

By mode: HOME, CAVE, HANG, MORN. A movement that appears in several sessions is written once and its APPEARS IN line lists every session it belongs to, with that session's own sets, work and rest. The prescriptions are pulled straight out of src/core, so they match the app exactly.

THE ILLUSTRATIONS

Where openly-licensed line art exists for a movement it is printed at the top of the entry, credited on the licence page at the back. It does not exist for the climbing-specific half of this programme. There is no stock artwork, at any licence, for max hangs, repeaters, density hangs, no-hangs, recruitment pulls, campus ladders, critical force blocks or the front lever ladder. Those entries say so, and are drawn from the prompt.

THE ILLUSTRATION PROMPTS

Every entry ends with a complete, self-contained image-generation prompt in an orange box. Paste one into an image model and it produces the illustration for that movement: panel-by-panel body positions, the joint angles that decide whether the position is right, where the arrows go, and a negative prompt. Every prompt shares the same house style block, so a set generated from this book reads as one set rather than a folder of unrelated pictures.

COPYING THE PROMPTS OUT

Copying text out of a PDF mangles line breaks. docs/form-guide-image-prompts.txt has all of them as plain text, one per movement, ready to paste or to loop over in a script.

THE ORANGE RULE

In the prompts, burnt orange is reserved for annotation: arrows, angle marks, callout numbers, ticks and crosses. It never appears on the body. That is what keeps a diagram readable at a glance and separates the instruction from the anatomy.

A NOTE ON THE FINGER WORK

The grip illustrations are the ones worth generating first. Half-crimp, open hand and three-finger drag look similar at a glance and load the pulleys differently, and drifting between them mid-hang is the most common way a finger session quietly becomes something other than what was prescribed. Those prompts specify the joint angles and the thumb position explicitly, and they ask for the wrong versions to be drawn alongside the right one.

CONTENTS

HOME

- 01 TENSION - body tension - trunk (45 min, recovery 24h)
- 02 PULL - maximal pulling (42 min, recovery 24h)
- 03 ARMOUR - antagonist - legs - mobility (45 min, recovery 12h)
- 04 WRISTS - daily - 10 min (10 min, recovery 0h)

Warm-Up Flow, Front Lever Progression, Hanging Leg Raise, Hollow Hold / Arch Hold, Band Pallof Press, Copenhagen Plank, Band External Rotation, Weighted Pull-Ups, Lock-Off Ladder, Inverted Row, Dead Hang, Wrist Extensor + Pronator, Pike Push-Up / Band Overhead Press, Push-Ups, Bulgarian Split Squat, Single-Leg Romanian Deadlift, Cuff Circuit - External Rotation + Y-T-W, Wrist Conditioning, Hip Mobility Flow, Quadruped Wrist Rocks

CAVE

- 01 MAX - fingers - heavy pull (58 min, recovery 48h)
- 02 POWER - limit boulder - contact (58 min, recovery 48h)
- 03 CAPACITY - repeaters - endurance - armour (58 min, recovery 24h)
- 04 ASSESS - test - not train (45 min, recovery 48h)
- 05 ADD-ON - after bouldering - antagonists (26 min, recovery 12h)

Pulse Raiser + Shoulder Prep, Progressive Finger Loading, Max Hang - Half-Crimp 20mm, Limit Bouldering, Campus Ladders, Explosive Pull-Ups, Ring Toe-Hook Hold, TRX Y-T-W, Recruitment Pulls, Repeaters 7:3, Pull-Up Strength Endurance, KB Overhead Press, TEST - Max Hang 20mm Half-Crimp, TEST - Weighted Pull-Up 5RM, TEST - 90 Degree Lock-Off, TEST - Front Lever Hold, TEST - Foot Raise + Straddle, Push-Ups / Ring Dips, Goblet Squat, Russian Twist

HANG

- 01 MAX HANGS - half-crimp - open hand (35 min, recovery 48h)
 - 02 CAPACITY - repeaters - critical force (45 min, recovery 24h)
 - 03 DENSITY - low load - high frequency (12 min, recovery 0h)
- Max Hang - Three-Finger Drag 20mm, Critical Force Block, Density Hangs

MORN

- 01 ABS + OBLIQUES - daily - 12 min (12 min, recovery 0h)
 - 02 SLOW START - daily - 10 min (10 min, recovery 0h)
- 90/90 Breathing Reset, Band Dead Bug, Hollow Body Hold, Band Leg Lowers, Side Plank Reach-Through, Half-Kneeling Band Press-Out, Standing Band Chop, Pull Edge No-Hang, Stomach Vacuum, Supine Spinal Twist, Cat-Cow + Thread the Needle, 90/90 Hip Switch, Band Pull-Apart + Overhead Pass, Bird Dog, Glute Bridge March, Side Plank Hold

HOME

01 TENSION - body tension - trunk

The missing link: hold a line from hands to feet so the fingers stop paying for it.

45 min | recovery 24h | tension, pull

Highest-priority physical session. Strong fingers that still open on a hard move are usually reporting a collapsed body position, not a weak grip. Every hold ends when the lower back arches, not when the timer does.

02 PULL - maximal pulling

The measured limiter: get the weighted pull-up toward 165% bodyweight.

42 min | recovery 24h | pull, shoulder, forearm

Heavy and low rep. If you are getting a pump, the load is too light and the reps too many.

03 ARMOUR - antagonist - legs - mobility

Wrists, hips, shoulders. The session that fixes what climbing broke.

45 min | recovery 12h | press, shoulder, legs, mobility

Currently the highest-priority session in the programme: wrists are symptomatic and hip mobility is the stated weakness on slabs. Twice a week, minimum.

04 WRISTS - daily - 10 min

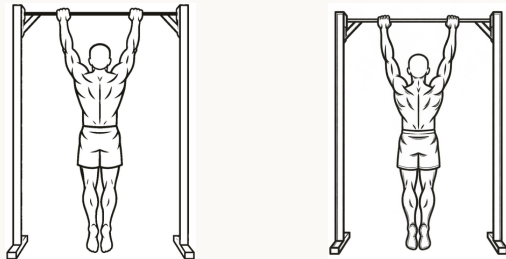
Ten minutes for the joint that is currently complaining.

10 min | recovery 0h | shoulder

Low load by design, so it runs every day including climbing days. Symptomatic tissue responds to frequent light loading, not to occasional heavy loading. This is not a substitute for a diagnosis.

WARM-UP FLOW

BLOCK	WARMUP / EASY
TRAINS	mobility, shoulder
APPEARS IN	HOME 01 TENSION: 1 x 300s, rest 30s HOME 02 PULL: 1 x 300s, rest 30s HOME 03 ARMOUR: 1 x 300s, rest 30s
LOAD	No load
PROGRESSION	Never progressed.
FIXED	Protocol taken as written. Never scaled by level or energy.



Frames show the scapular pull-up, the one component of the flow that is most often done wrong. The rest of the flow is drawn from the prompt.

SETUP

1. Mat down with two metres of clear floor. Band within reach. Pull-up bar accessible.
2. Take your shoes off. You want to feel the floor for the balance work.
3. Set a five-minute timer so you do not cut this short or drift past it.

EXECUTION

1. TWO MINUTES PULSE RAISER. Anything that moves the whole body and raises the breathing rate without loading the fingers: skipping on the spot, star jumps, fast marching with big arm swings, or a brisk stair set. Finish it breathing noticeably harder but able to talk.
2. TEN SHOULDER CIRCLES EACH DIRECTION. Arms straight, draw the largest circle the shoulder allows. Backwards first, then forwards. Slow enough that you can feel where the range runs out.
3. FIFTEEN BAND EXTERNAL ROTATIONS PER SIDE. Elbow pinned to the ribs at 90 degrees, band held across the body, rotate the forearm out and control it back.
4. TEN SCAPULAR PULL-UPS. Dead hang with straight arms. Without bending the elbows, pull the shoulder blades down and together so the body rises two or three centimetres, then release back to a full passive hang. Straight arms throughout.
5. TWENTY SECONDS RELAXED DEAD HANG. Hang, breathe, let the shoulders decompress. This is the readiness check: note anything that feels tender or sharp.

POSITION CHECKS

- Warm and switched on, not tired. You should finish wanting to train.
- The last dead hang tells you what today actually is. Sharp or tender in a finger means the session gets downgraded now, not after the first working set.
- Sweat starting, breathing raised, joints moving freely through full range.

COMMON MISTAKES

- Starting the first working set cold because the session is short. The session being short is exactly why the warm-up is not optional.
- Turning the pulse raiser into a workout. You are not trying to earn anything here.
- Doing the scapular pull-ups with bent elbows, which makes them tiny pull-ups and skips the scapular range entirely.

BREATHING

Nasal and unhurried.

WHY

Five minutes now buys you the session. It also tells you how the body actually feels today.

Progressive Warm-Up. Progressive loading of the flexor tendons and pulleys before maximal work is the most consistently recommended injury-reduction step in climbing, and it doubles as a readiness check: how the fingers feel on the third warm-up hang decides whether the session goes ahead as written.

Protocol: Pulse raiser, then progressive finger loading jug > 30mm > 20mm. Rest 60s between warm-up hangs. 3-4 progressive hangs. 40-60% - enough to prepare, not enough to fatigue. Every session that loads fingers.

Source: RAMP warm-up model; climbing warm-up and injury prevention guidance.

NOTE

Pulse raiser, shoulder circles, band external rotations, scapular pull-ups, dead hang.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal pull-up bar, a plain resistance band drawn as two clean parallel lines, a plain rectangular exercise mat. No other objects.

PANEL 1: A lean climber standing on a mat mid pulse-raiser, marching on the spot with one knee driven up to hip height and opposite arm swinging forward, upright posture, both feet visible.

PANEL 2: The same figure standing, arms straight out from the shoulders, drawing a large backward shoulder circle. Show the arc of the circle as a dashed orange ellipse traced by the fingertips.

PANEL 3: The same figure side-on, upper arm pinned against the ribs with the elbow bent to 90 degrees, holding a resistance band that runs off-frame horizontally, forearm rotated outward away from the belly.

PANEL 4: The same figure hanging from a horizontal bar with COMPLETELY STRAIGHT arms, shoulder blades pulled down and together so the head sits low between the shoulders and the ribcage has lifted slightly toward the bar. Elbows must be straight and locked, this is the critical detail.

KEY ANGLES AND POSITIONS: Panel 3: elbow exactly 90 degrees, upper arm touching the ribcage. Panel 4: elbows straight and locked, scapulae depressed and retracted, the gap between ear and shoulder deliberately large.

ANNOTATION: Orange curved arrow showing the shoulder circle direction in panel 2. Orange arrow showing the outward rotation path of the hand in panel 3. Orange downward arrow at each shoulder blade in panel 4, with a small orange bracket marking the ear-to-shoulder gap.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

FRONT LEVER PROGRESSION

BLOCK	PRIMARY / HARD
TRAINS	tension, pull
APPEARS IN	HOME 01 TENSION: 5 x 10s, rest 120s CAVE 01 MAX: 4 x 10s, rest 90s
LOAD	Hardest progression held with a flat lower back
PROGRESSION	Five clean sets of 12s, then move up a progression and reset to 8s.

NO STOCK ILLUSTRATION. No openly-licensed illustration exists for the front lever ladder. Generate from the prompt.

SETUP

1. Hang from the bar with an overhand grip, hands roughly shoulder width.

2. Depress the shoulders: pull the shoulder blades down and back, away from the ears. The arms stay straight for the whole exercise.
3. Tuck the pelvis under and squeeze the glutes before you pull. Ribs pulled down toward the hips, not flared toward the bar.
4. Pick the progression you can hold with a flat lower back, not the one you can reach. In order: tuck, advanced tuck, one-leg, straddle, full.

EXECUTION

1. Keeping the elbows locked, press down and back into the bar with straight arms. The lats do this, not the biceps. The body rotates up around the shoulders.
2. Bring the body to horizontal in your progression. TUCK: knees tight to the chest, thighs parallel to the floor. ADVANCED TUCK: knees open to roughly 90 degrees at the hip, shins parallel to the floor, back flat. ONE-LEG: one leg straight, the other tucked. STRADDLE: both legs straight and split wide. FULL: legs together.
3. Hold. The line from shoulders to hips must stay flat. The hold ends when the lower back arches, not when the clock does.
4. Lower with control rather than dropping out of it. The descent is free training.
5. Rest the full two minutes between sets. This is a maximal-quality hold, not conditioning.

POSITION CHECKS

- A straight line, or a slight hollow, from shoulder to hip. Never an arch.
- Elbows locked. If they bend, the lats have given up and the arms have taken over.
- Shoulders pulled down. If the shoulders ride up toward the ears the position is already failing.
- You should feel this across the lats, the lower abs and the glutes at once. If it is only abs, you are not pressing into the bar.

COMMON MISTAKES

- Arching the back to make it easier. This is the universal cheat and it removes the exact thing the exercise trains. Regress a progression instead.
- Bending the arms. It turns a lat and trunk exercise into a weak row.
- Skipping ahead a progression before the current one is solid. Five clean sets of twelve seconds is the gate, and it is not negotiable.
- Holding the breath for ten seconds, which raises intra-abdominal pressure but leaves you unable to finish the set.

BREATHING

Short controlled breaths. Do not hold your breath for ten seconds.

WHY

Push the bar away with straight arms rather than pulling with bent ones. This is a lat and trunk exercise, not an arm exercise.

Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

VARIATION LADDER

L1 Tuck Front Lever

L3 Advanced Tuck Front Lever

L5 One-Leg Front Lever

L6 Straddle Front Lever

L7 Full Front Lever

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 5 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4, 5 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A single plain horizontal bar at the top of each panel. Nothing else.

PANEL 1: TUCK FRONT LEVER. A lean climber hanging from a horizontal bar with both arms completely straight, body rotated back so the torso is horizontal and facing the ceiling, knees pulled tight to the chest, thighs parallel to the floor, back flat. Side view, full profile.

PANEL 2: ADVANCED TUCK FRONT LEVER. Same hang, same straight arms, torso horizontal, but the hips have opened to roughly 90 degrees so the thighs point down-ish and the shins are horizontal. The lower back is visibly FLAT, not arched.

PANEL 3: ONE-LEG FRONT LEVER. Same hang, one leg fully straight and horizontal in line with the torso, the other knee tucked to the chest.

PANEL 4: STRADDLE FRONT LEVER. Same hang, both legs straight and split wide apart, body horizontal, toes pointed.

PANEL 5: FULL FRONT LEVER. Same hang, the entire body one perfectly straight horizontal line from hands to pointed toes, parallel to the floor.

KEY ANGLES AND POSITIONS: In EVERY panel: elbows completely straight and locked, shoulder blades depressed, torso horizontal and parallel to the floor, lower back FLAT with a very slight posterior pelvic tilt. Under no circumstances draw an arched lower back.

ANNOTATION: A thin dashed orange horizontal reference line running through the torso in every panel to show the body is parallel to the floor. A short orange arrow at the shoulder in panel 1 pointing down and back, labelled with the number 1, indicating the straight-arm press into the bar.

NOTE: This is a progression ladder read left to right, easiest to hardest. The figure, the bar and the scale must be identical in all five panels so only the leg position changes.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

HANGING LEG RAISE

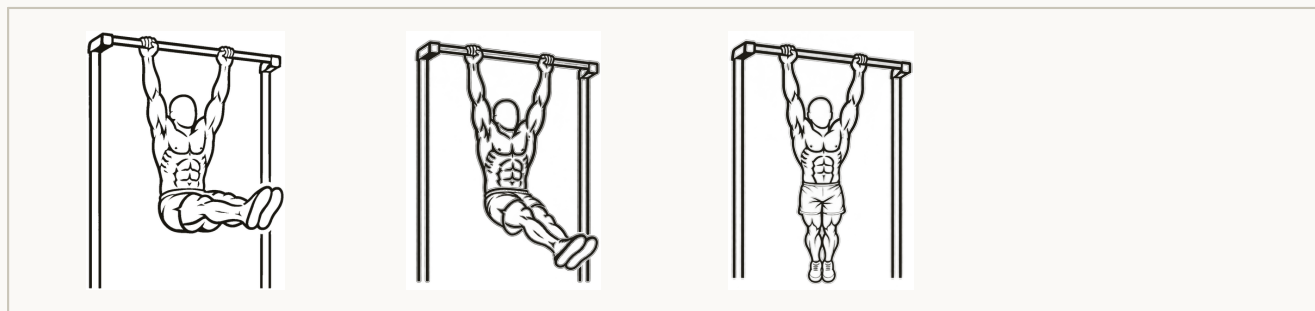
BLOCK SECONDARY / HARD

TRAINS tension

APPEARS IN HOME 01 TENSION: 4 x 8 reps, rest 90s

LOAD Bodyweight, slow and controlled

PROGRESSION Knees to chest, then straight legs to horizontal, then toes to bar, then slow lowering with a 3s eccentric.



SETUP

1. Dead hang from the bar, overhand grip, hands shoulder width.

2. Pack the shoulders before the first rep: pull the blades down and back so you are hanging actively, not sagging in the sockets.
3. Let the body come completely still. Never start the first rep on a swing.

EXECUTION

1. Tilt the pelvis under first, then raise the legs. Leading with the pelvis is what makes this a trunk exercise rather than a hip flexor exercise.
2. Raise to your progression. KNEE RAISE: knees to the chest. STRAIGHT LEG: straight legs to horizontal. TOES TO BAR: continue until the toes touch the bar between the hands.
3. Pause for a beat at the top. The pause is what kills the swing.
4. Lower slowly and under control, resisting the whole way down. On the three-second lower variation, count it out.
5. Come to a complete stop at the bottom. If the body is swinging, kill the swing before the next rep rather than using it.

POSITION CHECKS

- The shoulders stay packed for all eight reps, including the last one.
- No forward-backward swing. If there is a swing, the reps are being thrown.
- The lower abs are doing the work at the top of the rep, not just the front of the hips.

COMMON MISTAKES

- Swinging into each rep. It moves the legs without loading the trunk and it is the reason most people's leg raises never get harder.
- Letting the shoulders go passive at the bottom, which hangs the whole body weight off the shoulder capsule.
- Chasing toes-to-bar before straight-leg raises are clean.
- Rushing the lowering phase, which throws away half the exercise.

BREATHING

Exhale on the way up.

WHY

Hip flexor strength with a locked trunk is what keeps your feet on the wall when it steepens. Speed defeats the purpose.

Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

VARIATION LADDER

L1	Hanging Knee Raise
L3	Hanging Straight Leg Raise
L5	Toes to Bar
L6	Toes to Bar with 3s Lower

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A single plain horizontal bar at the top of each panel.

PANEL 1: HANGING KNEE RAISE. Climber in a dead hang from a bar, shoulders packed down, knees drawn up to the chest, pelvis tilted under. Side view.

PANEL 2: HANGING STRAIGHT LEG RAISE. Same hang, legs completely straight and raised to horizontal, forming a clean L shape with the torso.

PANEL 3: TOES TO BAR. Same hang, body curled so both feet have risen all the way to touch the bar between the hands, legs straight.

KEY ANGLES AND POSITIONS: Shoulders depressed and packed in all three panels, with a visible gap between ear and shoulder. Panel 2: hip at exactly 90 degrees, knees locked straight. Pelvis tucked under in all panels.

ANNOTATION: Orange curved arrow tracing the upward path of the feet in each panel. A small orange arrow at the pelvis in panel 1 showing the posterior tilt.

NOTE: A progression ladder read left to right. Same figure, same bar, same scale.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

HOLLOW HOLD / ARCH HOLD

BLOCK	SECONDARY / MODERATE
TRAINS	tension
APPEARS IN	HOME 01 TENSION: 3 x 30s, rest 60s
LOAD	Bodyweight
PROGRESSION	Extend the arms and legs further, then add time to 45s, then add a light weight overhead.



Stock frame shows the hollow. The arch is drawn from the prompt.

SETUP

1. HOLLOW: lie on your back on the mat, knees to the chest, hands by your ears.
2. Press the lower back hard into the floor and keep it there. That contact is the whole exercise. Squeeze the ribs down toward the hips.
3. ARCH: lie face down, arms extended overhead, forehead resting on the mat.

EXECUTION

1. HOLLOW. With the lower back glued to the floor, lift the shoulder blades a few centimetres off the mat and extend the arms and legs away from each other. Legs low, arms by the ears. Extend only as far as the lower back stays pressed down.
2. Hold the shape for the full thirty seconds. The shape is the exercise. The moment the lower back lifts, shorten the lever: bend the knees or bring the arms forward.
3. ARCH. Face down, squeeze the glutes, then lift the chest and the thighs off the mat at the same time. Arms stay extended overhead, neck long, eyes down at the mat.
4. Hold for thirty seconds. Alternate: one set hollow, next set arch, next set hollow.

POSITION CHECKS

- Hollow: you can slide nothing under the lower back. That is the pass mark.
- Arch: the lift comes from the glutes and the upper back, not from cranking the neck.
- Both positions should be a continuous, even effort across the whole front or back of the body, not a local burn in one spot.

COMMON MISTAKES

- Lower back lifting off the floor in the hollow, which is how people hold a longer lever than they have earned and train nothing.
- Cranking the neck in the arch by looking forward. Keep the eyes on the mat.
- Skipping the arch. Most climbers train the hollow and never the arch, which leaves the front-lever line unbalanced and the posterior chain untrained.

BREATHING

Short breaths, exhale to hold the hollow position.

WHY

Training both sides of the trunk keeps the front-lever line balanced. Most climbers train the hollow and skip the arch.

Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

NOTE

Alternate: one set hollow on the back, one set arch face down.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain rectangular exercise mat in each panel. Nothing else.

PANEL 1: HOLLOW HOLD. Climber lying on his back on a mat, lower back pressed flat into the mat, shoulder blades lifted just off the mat, arms extended straight back beside the ears, legs straight and lifted low, about 15cm off the floor. The whole body forms a shallow banana curve with the belly as the low point. Side view.

PANEL 2: ARCH HOLD. The same climber face down on the mat, arms extended straight overhead and lifted off the mat, chest lifted, both thighs lifted off the mat, glutes engaged, neck long with the gaze down at the mat. The body forms a shallow reverse curve, the mirror image of panel 1. Side view.

KEY ANGLES AND POSITIONS: Panel 1: the lumbar spine MUST be touching the mat with no visible gap, chin slightly tucked, arms straight. Panel 2: only the belly and hips contact the mat, arms and legs both clear of it, neck in line with the spine.

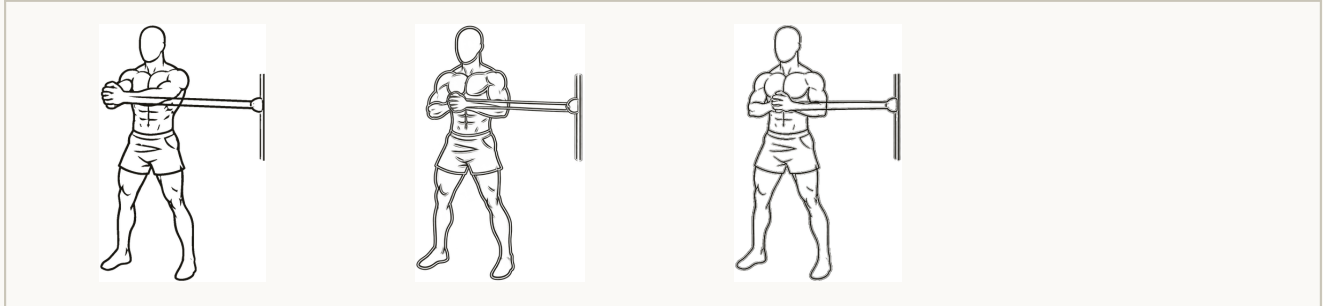
ANNOTATION: In panel 1, a short orange arrow pointing down at the lower back where it meets the mat, with a small orange cross-mark showing there is NO gap. In panel 2, orange arrows at the chest and at the thighs both pointing upward.

NOTE: The two panels are mirror images of each other, front-side and back-side of the same trunk shape. Draw them at identical scale.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BAND PALLOF PRESS

BLOCK	ACCESSORY / MODERATE
TRAINS	tension
APPEARS IN	HOME 01 TENSION: 3 x 10 reps per side, rest 60s
LOAD	Band anchored at chest height
PROGRESSION	Step further from the anchor, then add an overhead press at the end range.



SETUP

1. Anchor the band at roughly chest height: a door handle, a post, or a loop around something solid.
2. Stand side-on to the anchor, feet shoulder width, knees soft.
3. Take the band in both hands and hold it against the sternum. Step away from the anchor until the band is under real tension before you start.
4. Square the hips and shoulders to the front. The band will now be trying to rotate you toward the anchor. Your job is to refuse.

EXECUTION

1. Press both hands straight out from the sternum to full arm extension.
2. The band gets harder the further the hands travel, and the leverage against your trunk peaks at full extension.
3. Hold at full extension for two seconds without letting anything turn.
4. Return the hands to the sternum under control. That is one rep.
5. Complete all ten reps on one side, then turn around and do ten on the other.

POSITION CHECKS

- Hips and shoulders stay square to the front for every rep. Nothing rotates.
- The hands travel in a straight line out from the sternum, level, not drifting toward the anchor.
- The effort is felt in the side of the trunk away from the anchor, and in the deep abdominal wall, not in the arms.

COMMON MISTAKES

- Letting the torso rotate. That is the exercise failing, not the exercise getting harder.
- Using a band so heavy the hips have to counterbalance by pushing out sideways.
- Pressing fast and bouncing back in. The two-second hold at full extension is where the work is.

BREATHING

Exhale on the press.

WHY

Anti-rotation. The trunk resists twisting while the arms move - exactly what happens on a hard reach off a bad foot.

Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A resistance band drawn as two clean parallel lines running off the left edge of the frame at chest height. No anchor point visible, no other objects.

PANEL 1: START. Climber standing side-on to a resistance band that runs horizontally off-frame to the left at chest height. Both hands clasped together holding the band against the sternum, elbows tucked. Feet shoulder width, hips and shoulders square to the viewer. Front three-quarter view.

PANEL 2: PRESSED OUT. The same figure, arms now fully extended straight forward from the sternum, band stretched and under obvious tension. Hips and shoulders still perfectly square, no rotation whatsoever, torso vertical.

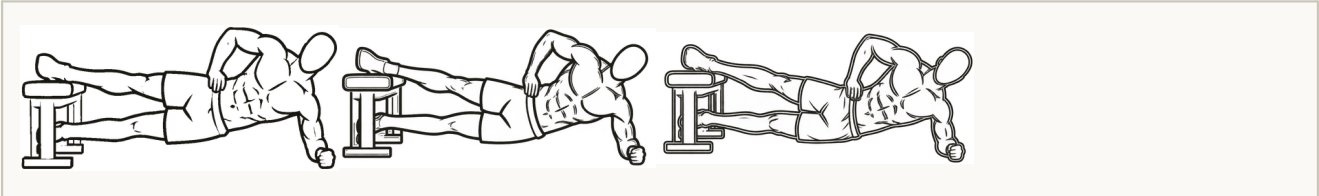
KEY ANGLES AND POSITIONS: Shoulders and hips both perfectly square and level in both panels. In panel 2 the arms are straight and horizontal at sternum height. The spine stays vertical, no side-bend toward the band.

ANNOTATION: An orange straight arrow showing the hands travelling forward from chest to full extension. A separate orange curved arrow at the ribcage showing the rotational pull of the band, crossed through with a small orange X to mark that this rotation must NOT happen.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

COPENHAGEN PLANK

BLOCK	ACCESSORY / MODERATE
TRAINS	tension, legs
APPEARS IN	HOME 01 TENSION: 3 x 20s per side, rest 60s
LOAD	Bodyweight
PROGRESSION	Bent knee on the support, then the full straight-leg version, then add time to 30s.



SETUP

- Set a chair or bench beside the mat, at roughly knee height.
- Lie on your side with the bottom forearm on the mat, elbow directly under the shoulder, forearm pointing forward.
- Rest the TOP leg on the chair. EASY VERSION: the inside of the knee rests on the chair. FULL VERSION: the inside of the ankle rests on the chair, leg straight.
- The bottom leg starts on the floor.

EXECUTION

1. Press down through the top leg into the chair and lift the hips until the body is one straight line from ankle to head.
2. Lift the bottom leg off the floor toward the top one, so both legs are held in the air and the adductors of the top leg carry the load.
3. Hold for twenty seconds, hips high, chest open to the front.
4. Lower under control. Complete the hold on one side, then switch.

POSITION CHECKS

- A straight line from the ankle through the hip to the head, seen from the front.
- The chest stays open and square. The shoulder stack is vertical.
- You feel this hard on the inside of the top thigh. That is the adductor, and that is the point.

COMMON MISTAKES

- Hips sagging. Once the hips drop the adductor unloads and the hold is decorative.
- Rolling the chest toward the floor, which turns it into a different and easier exercise.
- Going straight to the full ankle-supported version. Earn it at the knee first.

BREATHING

Steady.

WHY

The adductors do a lot of work in climbing and almost never get trained directly. This is the fix. Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

NOTE

Adductor strength. Drop knees, bridging and knee health all live here.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain low rectangular bench drawn in simple outline, and a plain exercise mat. Nothing else.

PANEL 1: KNEE VERSION. Climber in a side plank supported on one forearm on a mat, elbow directly under the shoulder. The TOP leg is bent with the inside of the knee resting on a low plain bench. The bottom leg is lifted clear of the floor. Hips lifted so the body is a straight line. Front three-quarter view.

PANEL 2: FULL VERSION. The same side plank, but now the top leg is completely STRAIGHT with the inside of the ankle resting on the bench, and the bottom leg is lifted and held alongside it in the air. Body one rigid straight line from the supporting elbow through the hips to the feet.

KEY ANGLES AND POSITIONS: Supporting elbow at exactly 90 degrees and positioned DIRECTLY under the shoulder in both panels. Hips fully lifted, no sag: draw a straight reference through ankle, hip and shoulder. Chest open and square to the viewer, not rotated toward the floor.

ANNOTATION: A dashed orange straight line running the length of the body from ankle to head in both panels, showing the required straight line. An orange arrow at the inside of the top thigh pointing down into the bench, marking where the load is carried.

NOTE: Panel 1 is the regression, panel 2 the full version. Same figure and scale.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BAND EXTERNAL ROTATION

BLOCK	PREHAB / EASY
TRAINS	shoulder
APPEARS IN	HOME 01 TENSION: 2 x 15 reps per side, rest 45s CAVE 01 MAX: 2 x 15 reps per side, rest 45s CAVE 05 ADD-ON: 2 x 15 reps per side, rest 45s
LOAD	Light band - never to failure
PROGRESSION	Reps to 20, then a marginally heavier band.

NO STOCK ILLUSTRATION. No openly-licensed frame covers band external rotation. Generate from the prompt.

SETUP

1. Anchor a light band at elbow height. Stand side-on so the band runs across the front of your body.
2. Take the band in the hand furthest from the anchor.
3. Pin the elbow to the ribs and bend it to 90 degrees. A rolled towel between elbow and ribs helps enforce this.
4. Start with the forearm across the belly, band already under light tension.

EXECUTION

1. Keeping the elbow pinned, rotate the forearm outward, away from the belly, as if opening a gate. Only the forearm moves.
2. Stop when the forearm is pointing straight ahead or a little beyond. Do not force past comfortable range.
3. Return slowly, resisting the band the whole way back. The return is the half most people throw away.
4. Fifteen reps, then swap sides.

POSITION CHECKS

- The elbow never leaves the ribs. If the towel drops, the set is over.

- The effort is felt at the back of the shoulder, not in the forearm or the lat.
- The band is light enough that rep fifteen looks identical to rep one.

COMMON MISTAKES

- Elbow drifting off the ribs, which lets the shoulder substitute bigger muscles and skips the cuff entirely.
- Going heavy. This is maintenance for a joint, not a lift. It stays light forever.
- Rotating the whole torso away to gain range.
- Dropping it from the programme once the shoulder stops complaining, which is the point at which it is working.

BREATHING

Exhale rotating out.

WHY

Maintenance for the joint that takes the most abuse in climbing. Light and often beats heavy and occasional.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A resistance band drawn as two clean parallel lines leaving the frame horizontally at elbow height. Optionally a small rolled towel wedged between the upper arm and the ribs. Nothing else.

PANEL 1: START. Climber standing, viewed from the front. The right upper arm is pressed flat against the ribcage with the elbow bent to exactly 90 degrees. The right forearm is horizontal and drawn ACROSS the front of the belly. A resistance band runs from the right hand horizontally off the right edge of the frame.

PANEL 2: FINISH. The same figure, upper arm still pressed to the ribs and elbow still at exactly 90 degrees, but the forearm has rotated outward so it now points straight forward, away from the body. The band is visibly stretched.

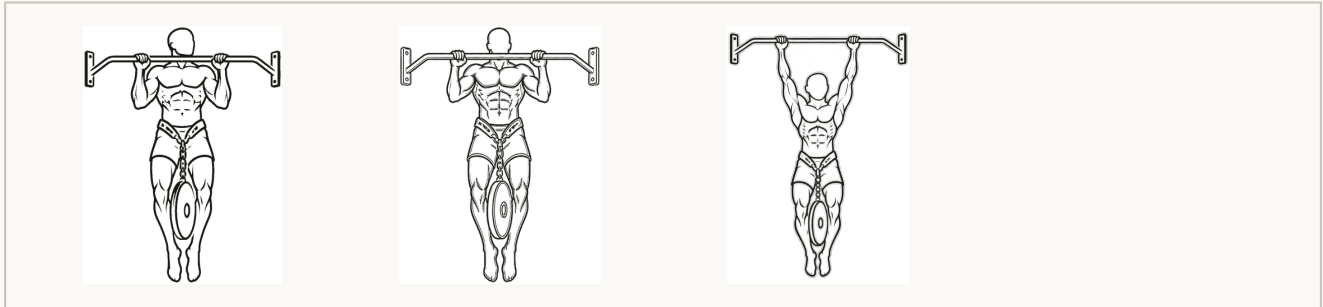
KEY ANGLES AND POSITIONS: The elbow is at EXACTLY 90 degrees in both panels and the upper arm stays in contact with the ribcage in both. Shoulders level and square, torso upright and not rotated.

ANNOTATION: An orange curved arrow showing the outward rotation path of the hand, sweeping like a gate opening. An orange dot marking the elbow as the fixed pivot, with a small orange 'fixed' bracket holding it against the ribs.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

WEIGHTED PULL-UPS

BLOCK	PRIMARY / MAX
TRAINS	pull
APPEARS IN	HOME 02 PULL: 5 x 4 reps, rest 150s CAVE 01 MAX: 4 x 4 reps, rest 180s CAVE 05 ADD-ON: 3 x 4 reps, rest 150s
LOAD	RPE 8-9 - 1-2 reps in reserve. A loaded pack works.
PROGRESSION	Five clean sets of 4, then add 2kg. At 62kg bodyweight, 165% means a working single with +40kg. Anything you add here shows up on the wall.



SETUP

1. Load first: a pack with weight in it, or a dipping belt. Put it on before you touch the bar so the first rep is not a wrestling match.
2. Take an overhand grip just outside shoulder width.
3. Hang completely: arms straight, shoulders relaxed into a full dead hang. This is the start position for every single rep.
4. Before pulling, set the shoulders: pull the blades down and back, ribs down, glutes lightly on so the body is a plank rather than a sack.

EXECUTION

1. Pull by driving the elbows down and back toward your back pockets. Think about pulling the bar toward the chest rather than the chin toward the bar.
2. Continue until the chest is close to the bar and the chin is clearly over it. Half reps do not count and do not build anything.
3. Lower under control all the way to a full dead hang, arms completely straight, shoulders released. Do not bounce out of the bottom.
4. Pause for a beat at the bottom, reset the shoulders, then start the next rep.
5. Four reps at RPE 8 to 9: you should finish each set believing you had one or two more in you, not zero.
6. Rest the full 150 seconds. This is a maximal strength set and the rest is part of the prescription, not a suggestion.

POSITION CHECKS

- Every rep starts from straight arms and a released shoulder.
- No kip, no swing, no leg drive. The body stays a rigid plank throughout.
- The last rep of the last set looks like the first rep of the first set. If it does not, the load is too heavy.

COMMON MISTAKES

- Kipping. It moves the chin over the bar without adding strength and it teaches the shoulder to absorb a snap load.
- Half reps, where the arms never straighten at the bottom. The bottom quarter is where the lat is longest and where the strength you are missing lives.
- Picking a weight that turns a strength set into an endurance set. Four hard reps beats twelve easy ones for this quality.

- Cutting the rest because the set only took ten seconds.

BREATHING

Exhale on the pull.

WHY

Elbows to your back pockets. Every rep starts from a real dead hang.

Maximal Strength. Low reps at high load build force with the least added mass, which is what a strength-to-weight sport needs. Two to three hard sets per pattern, twice weekly, is enough to drive strength in a climber who is also climbing; more volume mostly buys fatigue and mass.

Protocol: 3-5 reps. Rest 180s. 4-5. RPE 8-9, a load you could do 1-2 more reps with. 2x per week per pattern.

Source: Lattice minimum effective dose guidance; systematic review on resistance training, climbing performance and injury prevention (Sports Med Open 2024).

VARIATION LADDER

L1 Pull-Ups - Bodyweight

L2 Pull-Ups - 4s Eccentric

L4 Weighted Pull-Ups

L6 Weighted Pull-Ups - Heavy

L7 One-Arm Pull-Up Progression

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal pull-up bar. A simple weight belt with one round plate hanging on a short strap between the legs. No gym background.

PANEL 1: DEAD HANG START. Climber hanging from a horizontal bar with an overhand grip just outside shoulder width, arms COMPLETELY straight, shoulders released, body vertical and still, legs straight with ankles crossed. A weighted pack is worn on the back, or a simple weight belt with a single plate hanging between the legs on a short strap. Side-on three-quarter view.

PANEL 2: MID PULL. The same figure halfway up, elbows bent to roughly 90 degrees and driving DOWN AND BACK rather than out to the sides, chest beginning to rise toward the bar, body still rigid and vertical with no swing.

PANEL 3: TOP. The same figure at full contraction, chest close to the bar and chin clearly above it, elbows pulled down beside the ribs and back behind the torso line, shoulder blades squeezed down and together.

KEY ANGLES AND POSITIONS: Panel 1: elbows fully locked straight, a visible gap between ears and shoulders. Panel 2: elbows at 90 degrees, upper arms angled back behind the torso, NOT flared wide to the sides. Panel 3: chin above bar height, shoulder blades depressed and retracted. Body vertical in all three panels, no arch, no knees swinging forward.

ANNOTATION: An orange straight arrow at each elbow in panel 2 pointing down and back toward the hips, marking the pull direction. A dashed orange vertical reference line through the whole body in all three panels, showing the body stays on one line and does not swing.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

LOCK-OFF LADDER

BLOCK	SECONDARY / HARD
TRAINS	pull
APPEARS IN	HOME 02 PULL: 3 x 15s per side, rest 120s CAVE 01 MAX: 3 x 15s per side, rest 120s
LOAD	Bodyweight, assisted or weighted
PROGRESSION	Add 2s per position, then go one-armed with the free hand on the wrist, then add weight.

NO STOCK ILLUSTRATION. No openly-licensed frame shows a graded lock-off ladder. Generate from the prompt.

SETUP

1. Overhand grip on the bar, shoulder width. Have a clock or timer you can see.
2. Pull all the way to the top, chin over the bar. This is where the descent starts.
3. One set is one continuous descent through three angles, on one arm's worth of effort. You do not come back up between positions.

EXECUTION

1. Lower until the elbows are at roughly 120 degrees, which is a shallow bend, arms still fairly extended. HOLD FIVE SECONDS.
2. Lower to 90 degrees, elbows square. HOLD FIVE SECONDS. This is the angle most climbers actually fail at on the wall.
3. Lower to roughly 60 degrees, a deep lock with the hands near the collarbones. HOLD FIVE SECONDS.
4. Drop off or lower to a dead hang. That is one set, fifteen seconds of holding.
5. Rest two minutes, then repeat. Three sets per side if you are doing the one-arm version, three sets total if two-armed.
6. To make it harder: add two seconds per position, then go one-armed with the free hand resting lightly on the working wrist, then add weight.

POSITION CHECKS

- Each position is held, not passed through. If you are sinking, the set is already over.
- The elbow angle stays fixed for the full five seconds. Shaking is fine and expected. Drifting is not.
- Shoulders stay packed down. The moment they ride up toward the ears the position has failed.

COMMON MISTAKES

- Sinking through the positions instead of holding them, which turns a set of isometrics into a slow negative.
- On the one-arm version, pulling with the assisting hand. It rests on the wrist, it does not grip and it does not help.
- Only training the deep lock, which is the position that feels strongest and transfers least. Ninety degrees is the one that matters.

BREATHING

Keep breathing.

WHY

Ninety degrees is where most climbers fail. Train the angle you fall from.

Lock-Off Isometrics. Climbing pulling is largely isometric and position-specific. Lock-off capacity at 90 degrees is part of the standard climbing assessment battery and transfers directly to holding a position while the other hand moves.

Protocol: 5-10s hold at 60, 90 and 120 degrees of elbow flexion. Rest 120s. 3 per arm. Hard - the last second should be a fight. 1x per week.

Source: Climbing-specific strength assessment literature; Lattice assessment battery.

NOTE

5s at 120 degrees, 5s at 90, 5s at 60. One descent per set.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar at the top of each panel. Nothing else.

PANEL 1: 120 DEGREE LOCK-OFF. Climber hanging from a horizontal bar, holding a static position with elbows bent only slightly, to about 120 degrees, so the head is still well below the bar. Body still and vertical. Side view.

PANEL 2: 90 DEGREE LOCK-OFF. The same figure holding a static position with elbows bent to exactly 90 degrees, forearms vertical, upper arms horizontal, eyes level with the bar.

PANEL 3: 60 DEGREE LOCK-OFF. The same figure holding a deep static lock with elbows bent to about 60 degrees, hands close to the collarbones, chin above the bar.

KEY ANGLES AND POSITIONS: The elbow angle is the entire subject of this drawing and must be visibly and accurately different in each panel: 120 degrees, then 90 degrees, then 60 degrees. Shoulder blades depressed in all three. Body vertical and motionless, legs straight with ankles crossed, no swing.

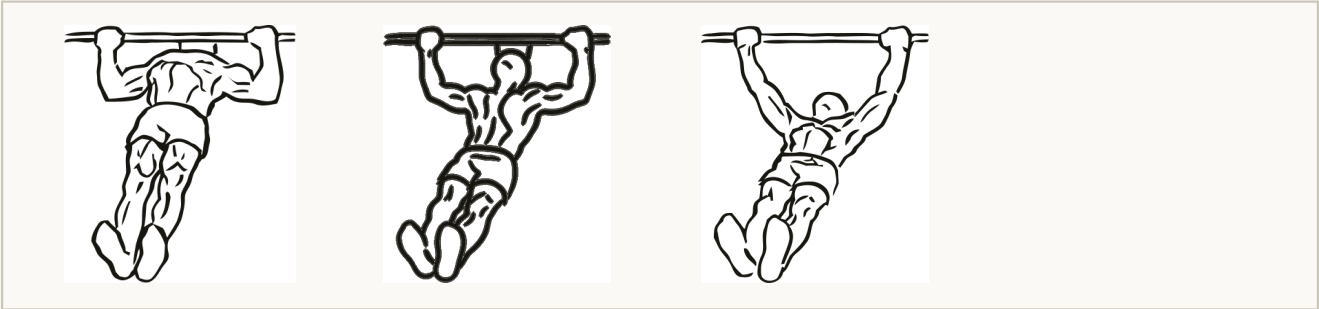
ANNOTATION: A burnt-orange angle arc drawn at the elbow in every panel with the degree figure written beside it: 120, 90, 60. A vertical dashed orange line at the side showing the descending height of the body across the three panels.

NOTE: Read left to right as one continuous descent: the figure gets HIGHER relative to the bar from panel 1 to panel 3 is WRONG. The figure starts high at the top of the bar and descends, so panel 1 (120 degrees) is the LOWEST body position and panel 3 (60 degrees) is the HIGHEST. Draw it that way.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

INVERTED ROW

BLOCK	SECONDARY / MODERATE
TRAINS	pull, shoulder
APPEARS IN	HOME 02 PULL: 3 x 8 reps, rest 90s
LOAD	Body angle sets the load
PROGRESSION	Feet further forward, then feet elevated, then a pack on the chest.



SETUP

1. Set a bar at roughly hip height. No low bar: loop a band around something solid at chest height, or put rings or a TRX at that height.
2. Lie underneath, take an overhand grip slightly wider than the shoulders.
3. Set the body: heels on the floor, legs straight, hips lifted so you are a straight plank from heel to head. Squeeze the glutes to hold it there.
4. Body angle sets the load. The more horizontal you are, the harder it is.

EXECUTION

1. Start with the shoulder blades set: pull them down and back before the elbows bend at all.
2. Pull the chest to the bar by driving the elbows back behind you. Lead with the elbow. The hands are hooks.
3. Touch or come within a fist of the bar, and at the top squeeze the shoulder blades together for a beat.
4. Lower under control to straight arms, letting the shoulder blades spread at the bottom without letting the hips drop.
5. Eight reps. To progress: walk the feet further forward, then elevate the feet on a chair, then wear a loaded pack on the chest.

POSITION CHECKS

- One straight line from heel to head for the entire set. The hips are the first thing to give.
- The chest reaches the bar, not the chin or the belly.
- No shrugging. The shoulders stay down and away from the ears throughout.

COMMON MISTAKES

- Hips sagging, which shortens the lever and quietly makes the exercise easier every rep.
- Stopping short of the bar. The last few centimetres are where the mid-back actually works.
- Shrugging the shoulders up toward the ears, which swaps the mid-back for the traps.

BREATHING

Exhale on the pull.

WHY

Lead with the elbow, not the hand. The hands are hooks.

Maximal Strength. Low reps at high load build force with the least added mass, which is what a strength-to-weight sport needs. Two to three hard sets per pattern, twice weekly, is enough to drive strength in a climber who is also climbing; more volume mostly buys fatigue and mass.

Protocol: 3-5 reps. Rest 180s. 4-5. RPE 8-9, a load you could do 1-2 more reps with. 2x per week per pattern.

Source: Lattice minimum effective dose guidance; systematic review on resistance training, climbing performance and injury prevention (Sports Med Open 2024).

NOTE

Horizontal pulling balances a training week that is otherwise all vertical.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar at about waist height, supported by two simple upright posts. Nothing else.

PANEL 1: BOTTOM. Climber lying face-up beneath a waist-height horizontal bar, gripping it overhand slightly wider than shoulder width, arms COMPLETELY straight, heels on the floor, legs straight, hips lifted so the body is one rigid straight line from heel to head at roughly 30 degrees to the floor. Side view.

PANEL 2: TOP. The same figure pulled up so the CHEST touches the bar, elbows driven back behind the torso, shoulder blades squeezed together, body still one perfectly straight line from heel to head with no hip sag.

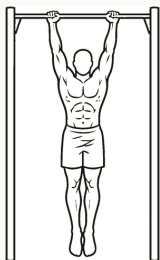
KEY ANGLES AND POSITIONS: The body is a straight rigid line in BOTH panels, heel to hip to shoulder to head, with absolutely no sag at the hips. In panel 2 the upper arms are angled back behind the torso line, not flared out sideways, and the chest reaches bar height.

ANNOTATION: A dashed orange straight line running the full length of the body in both panels, showing the plank. An orange arrow at each elbow in panel 2 pointing backwards. A small orange arrow at the hips in panel 1 pointing UP, crossed with an X in a small inset to mark that sagging is the failure.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

DEAD HANG

BLOCK	ACCESSORY / MODERATE
TRAINS	forearm
APPEARS IN	HOME 02 PULL: 3 x 45s, rest 60s HANG 02 CAPACITY as Dead Hang - Grip Rotation: 3 x 45s, rest 60s
ALSO CALLED	Dead Hang - Grip Rotation
LOAD	Bodyweight on the bar
PROGRESSION	To 60s, then add a small weight rather than more time.



Frame 1 shows the straight-arm hang. Ignore any bent-arm frames in the stock set: those are pull-ups, not hangs.

SETUP

1. Overhand grip on the bar, hands shoulder width.
2. Hang, then immediately engage: pull the shoulder blades down and slightly together so you are supported by muscle, not hanging off the joint capsule.
3. Have a towel ready if you are doing the grip rotation.

EXECUTION

1. Hang for forty-five seconds with the shoulders ACTIVE the whole time.
2. Switch grip every fifteen seconds: overhand on the bar, then a towel draped over the bar, then back. On the HANG-mode version, rotate through half-crimp, open hand and three-finger drag instead.

3. Throughout, try to bend the bar apart: imagine pulling your hands away from each other. This keeps the lats and the cuff switched on.
4. Step off deliberately at the end rather than dropping.
5. Progress to sixty seconds, then add a small weight rather than adding more time.

POSITION CHECKS

- A visible gap between the ears and the shoulders for the full duration.
- The work is felt in the forearms and the lats, not as a stretch in the shoulders.
- You end the hang because the forearms are done, not because the skin hurts.

COMMON MISTAKES

- Dead shoulders, which is where the name misleads people. The shoulders stay active; only the fingers are passive.
- Gritting through pain in the fingers rather than fatigue in the forearms. Finger pain is a stop signal, not a challenge.
- Adding time indefinitely. Past sixty seconds this becomes a skin-tolerance test.

BREATHING

Slow and even.

WHY

Try to bend the bar apart. A passive hang trains skin tolerance and nothing else.

NOTE

Switch grip every 15s: overhand, then a towel over the bar if you have one.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar. Optionally a simple towel draped over the bar in panel 1 with a hand gripping each end. Nothing else.

PANEL 1: ACTIVE HANG, CORRECT. Climber hanging from a horizontal bar with straight arms, but with the shoulder blades pulled DOWN and slightly together so there is a clear visible gap between the ears and the shoulders, chest slightly lifted, body still and vertical. Front view.

PANEL 2: PASSIVE HANG, INCORRECT. The same figure hanging completely relaxed, shoulders shrugged up around the ears so the head appears sunk between them, chest collapsed, body slack.

KEY ANGLES AND POSITIONS: Arms completely straight in BOTH panels. The ONLY difference between the two panels is the shoulder position: depressed and packed in panel 1, shrugged and passive in panel 2. Exaggerate this difference so it reads clearly.

ANNOTATION: An orange downward arrow at each shoulder blade in panel 1 with an orange bracket marking the ear-to-shoulder gap. In panel 2, an orange upward arrow at each shoulder and a single orange X mark in the corner of the panel to flag it as the error.

NOTE: Panel 1 is correct, panel 2 is the common error. Mark panel 2 clearly as wrong with the orange X, and do not add any other text.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

WRIST EXTENSOR + PRONATOR

BLOCK	PREHAB / EASY
TRAINS	shoulder
APPEARS IN	HOME 02 PULL: 2 x 15 reps per side, rest 45s CAVE 03 CAPACITY: 2 x 15 reps per side, rest 45s
LOAD	Band or a light household weight
PROGRESSION	Reps to 20, then slightly heavier. Slow lowering throughout.



SETUP

1. Sit down. Rest the working forearm along the top of your thigh with the wrist and hand hanging just past the knee.
2. Take a light weight or a band in that hand. Light means a 1 to 2kg dumbbell, a full water bottle, or a light band.
3. The forearm stays pressed to the thigh throughout. Only the wrist moves.

EXECUTION

1. WRIST EXTENSION, fifteen reps. Palm facing DOWN. Let the hand drop toward the floor, then curl it up toward the ceiling. Lower over three full seconds every single rep.
2. PRONATION, fifteen reps. Hold the weight by one end so it is off-balance, thumb up, forearm still on the thigh. Rotate the palm slowly DOWN toward the floor, then back to thumb-up under control.
3. Swap sides and repeat both movements.
4. Progress by adding reps to twenty, then a marginally heavier weight. Keep the three-second lowering forever.

POSITION CHECKS

- The forearm never lifts off the thigh. If it does, the load is too heavy.
- The lowering phase is genuinely three seconds. Count it.
- You feel this on the OUTSIDE of the forearm, the side climbing never trains.

COMMON MISTAKES

- Going heavy and fast, which is the one way to make an elbow prehab exercise cause elbow pain.
- Rushing the lowering phase, which is the part that actually remodels the tendon.
- Dropping it from the programme once the elbow stops complaining. This is maintenance, and the quiet period is the evidence it works.

BREATHING

Relaxed.

WHY

Climbing loads the flexors relentlessly and the extensors barely at all. Medial elbow pain lives in that gap.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial

elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain bench, a small short dumbbell. Nothing else.

PANEL 1: WRIST EXTENSION, BOTTOM. Climber seated on a plain bench, forearm resting along the top of the thigh with the wrist hanging just past the knee, palm facing DOWN, holding a small dumbbell. The hand has dropped down toward the floor so the wrist is fully flexed. Side view, close framing on torso, arm and thigh.

PANEL 2: WRIST EXTENSION, TOP. The same position, but the hand has curled UP toward the ceiling so the wrist is fully extended, the forearm still flat on the thigh.

PANEL 3: PRONATION. The same seated position, forearm on the thigh, but now holding the dumbbell by ONE END so it sticks out sideways, with the thumb pointing up. An orange arc shows the hand rotating so the palm turns down toward the floor.

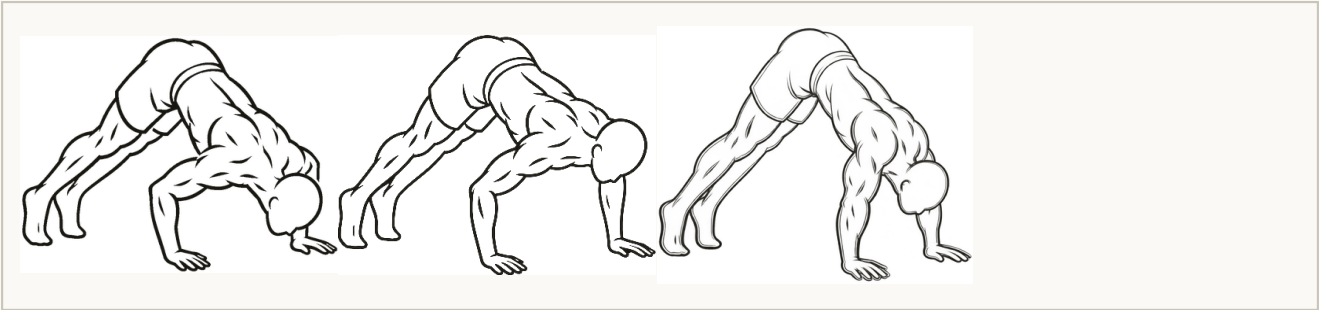
KEY ANGLES AND POSITIONS: The forearm stays in contact with the thigh in ALL THREE panels and the elbow angle never changes. Only the wrist moves in panels 1 and 2; only forearm rotation happens in panel 3.

ANNOTATION: Orange curved arrow showing the upward curl path in panels 1 to 2. Orange rotational arc arrow around the forearm axis in panel 3. A small orange bracket holding the forearm to the thigh in every panel, marking it fixed.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

PIKE PUSH-UP / BAND OVERHEAD PRESS

BLOCK	PRIMARY / HARD
TRAINS	press, shoulder
APPEARS IN	HOME 03 ARMOUR: 4 x 8 reps, rest 90s
LOAD	Bodyweight, or band for the press variant
PROGRESSION	Feet elevated, then a deficit under the hands, then toward a wall handstand push-up.



SETUP

1. Start in a downward dog: hands shoulder width on the floor, feet walked in toward the hands, hips high, legs as straight as your hamstrings allow.
2. Walk the feet in until the torso is close to vertical. The more vertical the torso, the more this becomes a shoulder press rather than a push-up.
3. Shift the weight forward over the hands. Fingers spread, gripping the floor.

4. For the band version instead: stand on the middle of a band, hold an end in each hand at shoulder height, elbows tucked.

EXECUTION

1. Lower by bending the elbows, bringing the CROWN of the head toward the floor between the hands. Not the forehead, not the nose: the top of the head.
2. Keep the hips high throughout. The hips dropping is what turns this into a push-up.
3. Touch lightly or come within a couple of centimetres, then press back up.
4. Finish every rep by pushing the floor away and actively reaching the shoulders up toward the ceiling at the top. That last reach is the part climbers need most.
5. Eight reps. Progress: feet elevated on a chair, then hands on blocks for a deficit, then toward a wall handstand push-up.

POSITION CHECKS

- Hips stay high for all eight reps. Watch the last two, that is when they drop.
- The head travels to a point BETWEEN the hands, not in front of them.
- At the top, the shoulders are actively pushed up, not passively resting.

COMMON MISTAKES

- Hips dropping as fatigue arrives, which silently converts the exercise into a push-up and trains the wrong thing.
- Not reaching depth, so the shoulder never works through the range it is losing.
- Skipping the active reach at the top. Overhead range is the whole reason this movement is in the programme.

BREATHING

Inhale down, exhale on the press.

WHY

Finish every rep with the shoulders actively pushing up. Climbers lose overhead range, and losing it is how shoulders start hurting.

Maximal Strength. Low reps at high load build force with the least added mass, which is what a strength-to-weight sport needs. Two to three hard sets per pattern, twice weekly, is enough to drive strength in a climber who is also climbing; more volume mostly buys fatigue and mass.

Protocol: 3-5 reps. Rest 180s. 4-5. RPE 8-9, a load you could do 1-2 more reps with. 2x per week per pattern.

Source: Lattice minimum effective dose guidance; systematic review on resistance training, climbing performance and injury prevention (Sports Med Open 2024).

NOTE

Overhead pressing is the single most valuable antagonist movement for a climbing shoulder.

VARIATION LADDER

L1	Pike Push-Up
L3	Feet-Elevated Pike Push-Up
L5	Deficit Pike Push-Up
L6	Wall Handstand Push-Up

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. In panel 3 only, a plain low chair. Nothing else.

PANEL 1: TOP. Climber in an inverted-U pike position: hands flat on the floor shoulder width, feet walked in close to the hands, hips high at the apex, legs nearly straight, arms COMPLETELY straight, head between the arms, torso close to vertical. Side view.

PANEL 2: BOTTOM. The same pike shape, but the elbows have bent and the CROWN of the head has lowered to just above the floor between the hands. The hips remain HIGH at the apex, unchanged from panel 1. Elbows track back at roughly 45 degrees, not flared wide.

PANEL 3: FEET ELEVATED. The harder variation: the same pike push-up but with both feet resting on a low plain chair, so the torso is fully vertical and the body forms a sharper, more upright U.

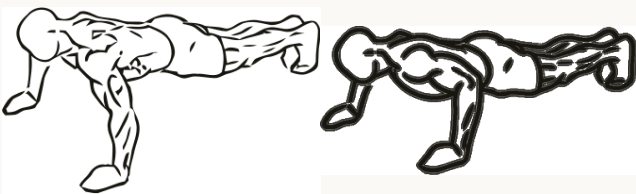
KEY ANGLES AND POSITIONS: The hip is the key checkpoint: in panels 1 and 2 the hips must be at the SAME high apex, proving they did not drop during the rep. The head goes BETWEEN the hands, level with or behind the line of the fingers, never in front of them.

ANNOTATION: An orange dashed horizontal line level with the hips in panels 1 and 2, identical in both, showing the hips do not move. An orange straight arrow showing the head's downward path in panel 2. A small orange upward arrow at the shoulders in panel 1 marking the active reach at lockout.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

PUSH-UPS

BLOCK	ACCESSORY / MODERATE
TRAINS	press
APPEARS IN	HOME 03 ARMOUR: 2 x 12 reps, rest 60s
LOAD	Bodyweight, feet elevated to add load
PROGRESSION	Feet elevated, then a pack on the back, then archer push-ups. Do not chase hypertrophy here.



SETUP

1. Hands flat on the floor directly under the shoulders, fingers spread and pointing forward.
2. Walk the feet back to a plank: one straight line from heels through hips to the crown of the head.
3. Squeeze the glutes and pull the ribs down. Set this before the first rep, not halfway through the set.

EXECUTION

1. Lower the chest toward the floor, keeping the elbows at roughly 45 degrees from the ribs. Not tucked to the sides, not flared to 90.
2. Go down until the chest is about a fist off the floor.
3. Press back to full lockout, arms completely straight.

4. At the top, push the floor away and let the shoulder blades spread apart. That protraction at the top is where the serratus works, and it is the half most people skip.
5. Twelve reps. Progress by elevating the feet, then wearing a loaded pack, then archer push-ups. Do not chase hypertrophy here.

POSITION CHECKS

- The body is one line for the whole set. Hips neither sag nor pike up.
- Elbows at 45 degrees, forming an arrow shape with the body rather than a T.
- Full lockout with the shoulder blades spread at the top of every rep.

COMMON MISTAKES

- Elbows flared to 90 degrees, which loads the front of the shoulder in its least stable position.
- Hips sagging, which is a trunk failure showing up in a pressing exercise.
- Adding endless reps in the belief that more is better. This is enough pressing to stay balanced, not enough to build chest mass you then have to carry up the wall.

BREATHING

Inhale down, exhale up.

WHY

Push the floor away and let the shoulder blades spread at the top. That top position is where the serratus works.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week. Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

NOTE

Enough pressing to stay balanced. Not enough to build chest mass you then have to carry up the wall.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: TOP. Climber in a high plank, hands flat on the floor directly under the shoulders, arms straight and locked, body one perfectly straight line from heels through hips to the crown of the head, shoulder blades spread apart. Side view.

PANEL 2: BOTTOM. The same figure lowered until the chest is about a fist's height above the floor, with the upper arms angled back at roughly 45 degrees from the ribcage. The body remains one straight line, hips neither sagging nor raised.

PANEL 3: OVERHEAD VIEW OF ELBOW PATH. A simple top-down diagram of the same figure at the bottom position, showing the upper arms forming an arrow shape with the torso at about 45 degrees, NOT a T shape at 90 degrees.

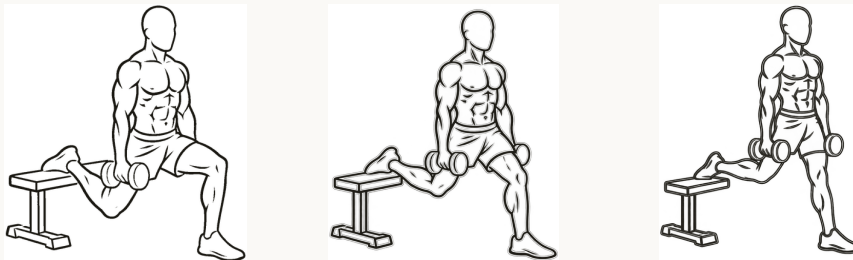
KEY ANGLES AND POSITIONS: Elbows at 45 degrees from the ribs, clearly not 90. The body is a straight rigid line in both side-view panels. Full elbow lockout in panel 1.

ANNOTATION: A dashed orange straight line through the body in panels 1 and 2 showing the plank. In panel 3, an orange angle arc at the armpit marked 45, and a faint greyed-out 90-degree T position drawn behind it with a small orange X.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BULGARIAN SPLIT SQUAT

BLOCK	SECONDARY / MODERATE
TRAINS	legs
APPEARS IN	HOME 03 ARMOUR: 3 x 8 reps per side, rest 90s CAVE 03 CAPACITY: 3 x 8 reps per side, rest 90s
LOAD	Bodyweight, then a loaded pack
PROGRESSION	To 12 reps bodyweight, then add a pack. Strength, not burn.



SETUP

1. Stand facing away from a chair, roughly one long stride in front of it.
2. Place the top of the rear foot on the chair seat, at about knee height.
3. Check the front foot position: when you lower, the front shin should stay close to vertical. If the knee shoots forward, step the front foot further out.
4. Stand tall, chest up, hands on hips or holding a loaded pack to the chest.

EXECUTION

1. Lower straight down by bending the front knee, sending the rear knee toward the floor. The torso stays upright.
2. Descend until the rear knee is just above the floor, or as deep as you can go with the front heel flat.

3. Drive up through the front HEEL, not the toes. Think about pushing the floor away.
4. Return to standing without locking out aggressively, then start the next rep.
5. Eight reps on one side, then swap. Progress to twelve bodyweight reps per side, then add a loaded pack.

POSITION CHECKS

- The front knee tracks over the second toe. It does not cave inward.
- The front heel stays flat on the floor for the whole rep.
- The torso stays upright. Folding forward turns this into a different exercise.
- The rear leg is a kickstand for balance, not a source of power.

COMMON MISTAKES

- Front knee caving inward, which is the single most common fault and the one worth fixing first.
- Torso folding forward to help, which shifts the load off the front leg.
- Turning it into a high-rep burn set. This is strength work. Eight hard reps, not twenty easy ones.

BREATHING

Inhale down, exhale up.

WHY

Drive the floor down through the heel. Single-leg strength carries to high steps, rock-overs and long approaches.

Maximal Strength. Low reps at high load build force with the least added mass, which is what a strength-to-weight sport needs. Two to three hard sets per pattern, twice weekly, is enough to drive strength in a climber who is also climbing; more volume mostly buys fatigue and mass.

Protocol: 3-5 reps. Rest 180s. 4-5. RPE 8-9, a load you could do 1-2 more reps with. 2x per week per pattern.

Source: Lattice minimum effective dose guidance; systematic review on resistance training, climbing performance and injury prevention (Sports Med Open 2024).

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain low chair or bench. Nothing else.

PANEL 1: TOP. Climber standing in a split stance with the top of the REAR foot resting on a plain low chair behind him at about knee height. The front leg is straight, front foot flat on the floor a long stride ahead of the chair. Torso upright and tall, hands on hips. Side view.

PANEL 2: BOTTOM. The same split stance lowered so the REAR knee is just above the floor, the FRONT shin close to vertical, front heel flat on the ground, front thigh roughly parallel to the floor. Torso still upright, not folded forward.

PANEL 3: FRONT VIEW, KNEE TRACKING. The same bottom position seen from the front, showing the front knee tracking directly out over the second toe, hips level.

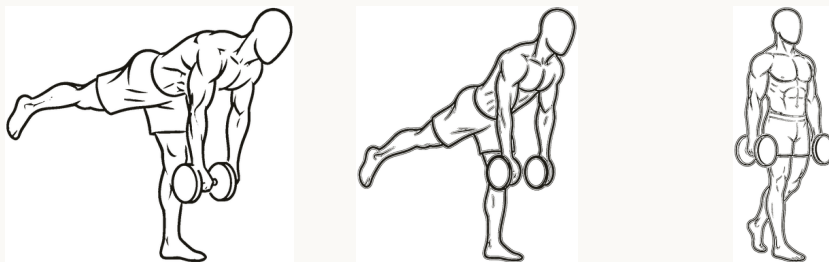
KEY ANGLES AND POSITIONS: In panel 2 the front shin is close to VERTICAL and the front heel is flat on the floor. Torso upright in all panels, not hinged forward. In panel 3 the knee is directly above the second toe, not collapsed inward.

ANNOTATION: An orange dashed vertical line running up the front shin in panel 2, showing it stays vertical. An orange arrow pointing down into the front HEEL marking where the drive comes from. In panel 3, an orange dashed vertical line from knee to second toe, plus a faint greyed inward-collapsed knee drawn behind it with a small orange X.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

SINGLE-LEG ROMANIAN DEADLIFT

BLOCK	ACCESSORY / MODERATE
TRAINS	legs
APPEARS IN	HOME 03 ARMOUR: 2 x 10 reps per side, rest 60s
LOAD	Bodyweight, or a pack held at the chest
PROGRESSION	Add load, then slow the lowering to 3 seconds.



SETUP

1. Stand on one leg with a soft bend in that knee. The knee angle stays the same for the whole rep.
2. Set the back flat and the chest proud. Shoulders pulled back.
3. Square the hips to the floor. Hold something light at the chest, or nothing at all to start.

EXECUTION

1. Hinge at the HIP: push the hips backwards while the torso comes forward and the free leg extends straight back behind you.
2. Torso and free leg move as one rigid line, like a see-saw pivoting at the hip.

3. Descend until you feel a strong stretch in the hamstring of the standing leg, or until the torso is roughly parallel to the floor, whichever comes first.
4. Return by driving the hips FORWARD, not by lifting the chest. Squeeze the glute at the top.
5. Ten reps one side, then swap. Progress by adding load, then slowing the lowering to three seconds.

POSITION CHECKS

- The lower back stays flat throughout. The moment it rounds, you have gone too far.
- The hips stay SQUARE to the floor. The hip of the free leg must not rotate open toward the ceiling.
- The standing knee bend does not change during the rep.
- You feel it in the hamstring and glute of the standing leg.

COMMON MISTAKES

- Rounding the lower back to reach further down. Range you cannot control with a flat back is not range you own.
- Letting the hip of the free leg rotate open, which is the way the body cheats a single-leg hinge.
- Bending the standing knee more and more as you descend, which turns a hinge into a squat.

BREATHING

Inhale on the hinge, exhale on the return.

WHY

Hinge, do not bend. Hips square to the floor throughout.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

NOTE

Posterior chain and single-leg balance. Cheap insurance for the knees and back.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: Nothing. Bodyweight only.

PANEL 1: TOP. Climber standing on one leg, that knee with a soft slight bend, torso upright and tall, the other leg hanging relaxed, hips square. Side view.

PANEL 2: BOTTOM. The same figure hinged forward at the hip so the torso is roughly PARALLEL to the floor, with the free leg extended straight back behind him also parallel to the floor, forming one continuous straight line from the back of the head through the spine to the free heel. The standing knee bend is UNCHANGED from panel 1. The lower back is flat, not rounded.

PANEL 3: REAR VIEW, HIP SQUARENESS. The same bottom position seen from behind, showing both hip bones level and square to the floor, the free leg's hip NOT rotated open toward the ceiling.

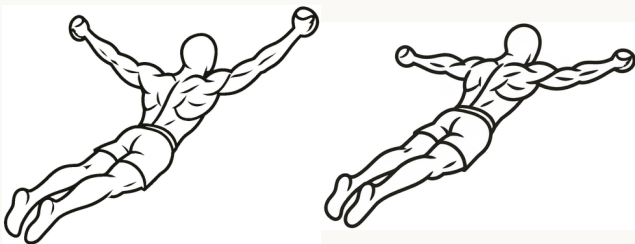
KEY ANGLES AND POSITIONS: The standing knee holds the same soft bend in panels 1 and 2. In panel 2 the head, spine and free leg form ONE straight line parallel to the floor. The lumbar spine is flat, never rounded. In panel 3 the pelvis is level.

ANNOTATION: A dashed orange straight line from the crown of the head through the spine to the free heel in panel 2, showing the see-saw line. An orange arrow at the hips pointing BACKWARD in panel 2, marking the hinge direction. In panel 3, a short orange level-line across both hip bones.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

CUFF CIRCUIT - EXTERNAL ROTATION + Y-T-W

BLOCK	PREHAB / EASY
TRAINS	shoulder
APPEARS IN	HOME 03 ARMOUR: 3 x 12 reps, rest 45s
LOAD	Light band - never to failure
PROGRESSION	Reps to 20, then a marginally heavier band. This stays light forever.



Stock frames show the Y position. The T, W and rotation positions are drawn from the prompt.

SETUP

1. Anchor a light band at elbow height for the rotations. For the Y-T-W you hold the band in both hands with no anchor.
2. Stand tall, ribs down, shoulders relaxed away from the ears. This posture holds for every rep of all four movements.
3. Pick a band light enough that the last rep looks like the first.

EXECUTION

1. EXTERNAL ROTATIONS, 12 per side. Elbow pinned to the ribs at 90 degrees. Rotate the forearm outward, control it back. Only the forearm moves.
2. Y, 12 reps. Hold the band in both hands in front of the thighs. Pull it apart and up overhead into a Y shape, arms straight, thumbs up. Lower under control.

3. T, 12 reps. Band in both hands at chest height, arms straight in front. Pull apart straight out to the sides into a T, squeezing the shoulder blades together.
4. W, 12 reps. Band in both hands, elbows bent and tucked near the ribs. Pull the band apart and drive the elbows back and down, forming a W with the arms.
5. That is one round. Three rounds with 45 seconds between.

POSITION CHECKS

- No shrugging in any of the four movements. The shoulders stay down.
- The movements are precise and slow. You should be able to stop at any point in the range.
- The work is felt around the shoulder blade and the back of the shoulder, not in the lats or the neck.

COMMON MISTAKES

- Shrugging, which is the body recruiting the traps to do the cuff's job.
- Using a heavy band and recruiting the lats. This is scapular and cuff control, not a back workout.
- Rushing. Precision over load, every time. It stays light forever.

BREATHING

Exhale on each pull.

WHY

Precision over load. This is scapular and cuff control, not a back workout.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

NOTE

12 external rotations per side, then 12 each of Y, T and W with the band.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain resistance band drawn as two clean parallel lines. Nothing else.

PANEL 1: EXTERNAL ROTATION. Climber standing, upper arm pinned flat against the ribs, elbow bent to exactly 90 degrees, forearm rotated outward away from the belly, holding a band that runs off-frame horizontally. Front view.

PANEL 2: Y POSITION. The same figure standing, holding a band in both hands, arms straight and raised overhead in a wide V or Y shape, thumbs pointing up, band stretched between the hands, shoulders DOWN and not shrugged.

PANEL 3: T POSITION. The same figure, arms straight and held out horizontally to both sides forming a T, band stretched across the chest, shoulder blades squeezed together, shoulders down.

PANEL 4: W POSITION. The same figure, elbows bent and pulled back and down close to the ribs with the forearms angled up and out, so the arms form a W shape, band stretched between the hands.

KEY ANGLES AND POSITIONS: The shoulders are DOWN and away from the ears in all four panels: this is the single most important detail. Torso upright and still, ribs down, no leaning back. Elbow exactly 90 degrees in panel 1.

ANNOTATION: An orange arrow showing the band's path of travel in each panel. A small orange downward arrow at each shoulder in panels 2, 3 and 4 reinforcing that the shoulders stay depressed and must not shrug.

NOTE: These are four positions of one circuit, read left to right. Same figure, same scale, same band in every panel.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

WRIST CONDITIONING

BLOCK	PREHAB / EASY
TRAINS	shoulder
APPEARS IN	HOME 03 ARMOUR: 3 x 12 reps per side, rest 45s HOME 04 WRISTS: 3 x 12 reps per side, rest 40s
LOAD	Light: a 1-2kg weight, a full bottle, or a band
PROGRESSION	Add reps to 15, then add a very small amount of weight. Slow throughout, 3 seconds on every lowering phase. This block gets heavier more slowly than anything else in the programme.



Stock frames show wrist flexion. Extension, deviation and rotation are drawn from the prompt.

SETUP

1. Sit down. Rest the forearm on a thigh or a table with the wrist just past the edge so the hand can move freely in every direction.
2. Take a light weight in the hand: 1 to 2kg, a full bottle, or a light band.
3. Four movements, twelve reps each, three seconds on every lowering phase. Then repeat the whole block on the other side.

EXECUTION

1. WRIST EXTENSION, palm down, 12 reps. Let the hand drop, curl it up toward the ceiling, lower over three seconds.
2. WRIST FLEXION, palm up, 12 reps. Let the hand drop back, curl it up toward you, lower over three seconds.
3. RADIAL AND ULNAR DEVIATION, 12 reps. Thumb up, forearm on the support. Move the hand up toward the thumb side, then down toward the little-finger side. The forearm does not roll.
4. PRONATION AND SUPINATION, 12 reps. Hold the weight by ONE END so it is off-balance. Rotate the palm slowly down, then slowly up. Control both directions.
5. STOP any movement immediately if it produces sharp or pinching pain rather than muscular effort.

POSITION CHECKS

- The forearm stays supported and still. Only the hand moves.
- Every lowering phase is a genuine three seconds.
- Effort, never pain. Muscular fatigue is the target; sharp or pinching is a stop.

COMMON MISTAKES

- Going heavy. This block gets heavier more slowly than anything else in the programme, on purpose.
- Rushing the lowering phase, which is the part that actually drives the adaptation.
- Working into pain in the belief that it is strengthening something.
- Dropping the block as soon as the symptoms quieten down, which is the point at which it is working.

BREATHING

Relaxed throughout. This should never feel like a lift.

WHY

Climbing loads the wrist in one narrow band of positions under heavy crimp and mantle load, and trains nothing else. Steep crimping in particular is hard on the wrist. This block covers every direction the wrist can go, at a load it can actually adapt to.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

NOTE

Four movements, 12 reps each: wrist extension palm-down, wrist flexion palm-up, radial and ulnar deviation, then slow pronation and supination holding the weight by one end. Never into pain.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain bench or chair, a small short dumbbell. Nothing else.

PANEL 1: WRIST EXTENSION. Climber seated, forearm resting flat along the top of the thigh with the wrist just past the knee, palm facing DOWN, holding a small dumbbell. Hand curled UP toward the ceiling. Side view, close framing.

PANEL 2: WRIST FLEXION. Same seated setup but palm facing UP, hand curled up toward the forearm.

PANEL 3: RADIAL AND ULNAR DEVIATION. Same seated setup, forearm on the thigh, thumb pointing UP, holding the dumbbell by one end. Two ghosted positions of the hand shown: one tilted up toward the thumb side, one tilted down toward the little-finger side.

PANEL 4: PRONATION AND SUPINATION. Same seated setup, holding the dumbbell by ONE END so it projects sideways. An orange rotational arc shows the forearm rolling the palm from facing up to facing down.

KEY ANGLES AND POSITIONS: The forearm remains in contact with the thigh in ALL FOUR panels and the elbow angle never changes. Only the wrist and forearm rotation move. Keep the hand anatomically correct with five clearly separated fingers.

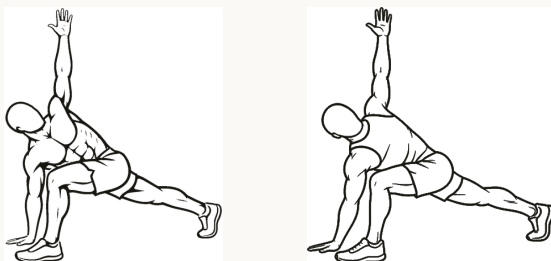
ANNOTATION: Orange curved arrows showing each direction of travel. A small orange bracket clamping the forearm to the thigh in every panel to mark it fixed. In panel 3 use a double-headed orange arrow for the two directions.

NOTE: Four movements of one wrist block, read left to right. Same seated figure, same scale, same close framing throughout.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

HIP MOBILITY FLOW

BLOCK	MOBILITY / EASY
TRAINS	mobility
APPEARS IN	HOME 03 ARMOUR: 2 x 150s, rest 15s
LOAD	End range, loaded where possible
PROGRESSION	Push end range, then add gentle isometric contractions at end range to make the range usable.



Stock frames show a lunge-and-rotate. The four flow positions are drawn from the prompt.

SETUP

1. Mat, clear floor, something solid to hold for balance in the squat.
2. Four positions, roughly forty seconds each, two rounds. Fifteen seconds between rounds.
3. This is loaded end range, not passive stretching. You will contract gently into each position.

EXECUTION

1. 90/90 BOTH SIDES. Sit with both knees bent at 90 degrees, one leg in front, one out to the side. Sit tall. Press the front shin gently into the floor for five seconds, release, sink

a little further. Switch sides.

2. FROG. On hands and knees, knees wide apart, shins in line with the thighs, feet turned out. Rock the hips slowly back toward the heels and forward again. Keep the lower back flat.
3. DEEP SQUAT HOLD. Feet shoulder width, sink into the deepest squat you can with the heels down. Hold onto something if needed. Use the elbows to press the knees gently outward.
4. COUCH STRETCH, per side. Rear foot up against a wall or on a chair, front foot forward, back knee down. Tuck the pelvis under and squeeze the rear glute. That glute squeeze is what makes it work.
5. Then repeat the whole round.

POSITION CHECKS

- The lower back stays flat in the frog and the 90/90. If it rounds, sit up on a folded edge of the mat.
- In the couch stretch the pelvis is tucked under. Without that, you are stretching the lower back rather than the hip flexor.
- You are contracting gently into end range, not hanging passively in it.

COMMON MISTAKES

- Treating it as passive stretching. Range you cannot contract into is range you cannot use on the wall.
- Rounding the lower back to fake more hip range.
- Skipping the second round because the first felt fine.

BREATHING

Long exhales into each position.

WHY

Combined abduction and external rotation is what makes high steps, drop knees and bridging available. Contract gently into end range so the range becomes usable rather than just passive.

Hip and Shoulder Mobility. Combined hip abduction and external rotation is what makes high steps, drop knees and bridging available, and it is measured in climbing assessment batteries via foot-raise height and straddle depth. Cheap to train and it costs no recovery.

Protocol: 45-60s per position, loaded end-range where possible. Rest 15s. 2 rounds. Comfortable end range. 3x per week.

Source: Draper et al., flexibility as a determinant of climbing performance; climbing mobility assessment protocols.

NOTE

90/90 both sides, frog, deep squat hold, couch stretch. Roughly 40s each.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. A plain flat wall in panel 4 only. Nothing else.

PANEL 1: 90/90 SEATED. Climber seated on a mat with BOTH knees bent to 90 degrees: one leg folded in front with the shin across the body, the other folded out to the side, both shins flat on the floor. Sitting TALL with a flat lower back. Three-quarter view from above and in front.

PANEL 2: FROG. Climber on hands and knees with the knees spread very wide, shins in line with the thighs, feet turned outward, hips sinking back toward the heels. Lower back flat. View from the side and slightly above.

PANEL 3: DEEP SQUAT HOLD. Climber in a full deep squat, hips below the knees, heels FLAT on the floor, torso upright, elbows inside the knees pressing them gently outward. Front view.

PANEL 4: COUCH STRETCH. Climber in a half-kneeling lunge with the REAR foot raised behind him against a plain wall, rear knee on the mat, front foot flat and forward, torso upright, pelvis visibly TUCKED UNDER. Side view.

KEY ANGLES AND POSITIONS: Panel 1: both knees at exactly 90 degrees, spine tall and flat. Panel 2: lower back flat, not arched or rounded. Panel 3: heels flat on the floor, hip crease below knee height. Panel 4: pelvis in posterior tilt with the tailbone tucked, torso vertical, rear glute engaged.

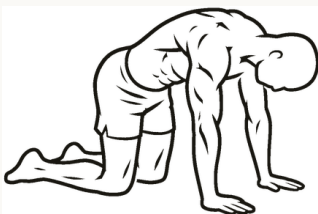
ANNOTATION: An orange arrow at each position showing the direction of gentle pressure: front shin pressing down in panel 1, hips rocking back in panel 2, elbows pressing knees outward in panel 3, pelvis tucking under in panel 4.

NOTE: Four positions of one mobility flow, read left to right. Same figure and scale throughout.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

QUADRUPED WRIST ROCKS

BLOCK	PRIMARY / EASY
TRAINS	shoulder
APPEARS IN	HOME 04 WRISTS: 2 x 10 reps, rest 40s
LOAD	Partial bodyweight through the hands
PROGRESSION	Start on the floor on all fours with almost no weight forward. Progress by shifting weight further over the hands, never by adding reps.
FIXED	Protocol taken as written. Never scaled by level or energy.



Stock frame shows the quadruped base position only. The four hand orientations are drawn from the prompt.

SETUP

1. On hands and knees on the mat. Knees under hips.
2. Place the hands flat with the FINGERS POINTING FORWARD to start. Arms straight, elbows soft but not locked hard.
3. Only put as much weight through the hands as is comfortable. You control the load by how far forward you lean.

EXECUTION

1. FINGERS FORWARD. Rock the body slowly forward over the hands so the wrists bend further into extension, then rock back. Twelve slow rocks.

2. FINGERS OUT TO THE SIDES. Turn both hands so the fingers point outward. Rock gently side to side and forward. Twelve rocks.
3. FINGERS POINTING BACK toward the knees. This is the strongest stretch. Rock back toward the heels very gently. Twelve rocks. Reduce the lean if it is sharp.
4. BACK OF THE HANDS DOWN. Turn the hands over so the backs rest on the mat, fingers pointing toward the knees. Rock gently. This one needs almost no load.
5. Move slowly. Rocking is the exercise; the position is just the setup.

POSITION CHECKS

- You control the intensity entirely with how far you lean. Nothing here should be sharp.
- The arms stay straight but not hyper-locked.
- A broad stretch across the forearm and the palm, never a pinch at the back of the wrist joint.

COMMON MISTAKES

- Loading too fast. This is loaded mobility for a joint that climbing hammers in one narrow position and never trains elsewhere.
- Bouncing rather than rocking slowly.
- Pushing through a pinch at the back of the wrist. Back off the lean instead.

BREATHING

Slow and even.

WHY

This is the position that hurts on mantles and hard crimping. Training it lightly and often is how it stops hurting.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

NOTE

Palms down, then palms reversed with fingers pointing back. Rock gently forward and back.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: FINGERS FORWARD. Climber on hands and knees on a mat, arms straight, hands flat with fingers pointing FORWARD, body rocked forward over the hands so the wrists are in deep extension. Side view with the hand clearly visible.

PANEL 2: FINGERS OUT. The same quadruped position but both hands rotated so the fingers point OUTWARD to the sides, away from the body.

PANEL 3: FINGERS BACK. The same quadruped position with both hands rotated so the fingers point BACKWARD toward the knees, body rocking back toward the heels.

PANEL 4: BACKS OF HANDS DOWN. The same quadruped position with the hands turned over so the BACKS of the hands rest on the mat, fingers pointing back toward the knees, very little weight through the arms.

KEY ANGLES AND POSITIONS: Knees directly under hips and hands roughly under shoulders in all panels. Arms straight but not hyperextended. The spine stays neutral and flat throughout, this is a wrist drill, not a cat-cow.

ANNOTATION: A small orange curved arrow at the torso in each panel showing the rocking direction. An orange inset close-up of the HAND in each panel showing the exact finger orientation, since that is the whole point of the drawing.

NOTE: Four hand orientations of one wrist drill, read left to right in increasing difficulty. Draw the hands large and unambiguous: five clearly separated fingers, correct anatomy, no fusing.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

CAVE

01 MAX - fingers - heavy pull

Raise the ceiling: peak finger force and maximal pulling, both fresh.

58 min | recovery 48h | crimp, pull, tension

Every set here is meant to be hard and well rested. If you are rushing, cut an exercise, not the rest.

02 POWER - limit boulder - contact

Recruit harder: maximal moves, contact strength, explosive pulling.

58 min | recovery 48h | contact, crimp, pull, tension

Quality decides the length of this session. When attempts stop feeling explosive, it is over.

03 CAPACITY - repeaters - endurance - armour

Build the engine: sustain force for longer, and pay the antagonist and leg bills.

58 min | recovery 24h | forearm, pull, press, legs, shoulder

The lowest-intensity session of the three. It should leave you tired but not wrecked.

04 ASSESS - test - not train

Find the limiter. Numbers now, opinions later.

45 min | recovery 48h | crimp, openhand, pull, tension, mobility

Run this every 6-8 weeks on a rest day, fully warm and fresh. Write everything down.

05 ADD-ON - after bouldering - antagonists

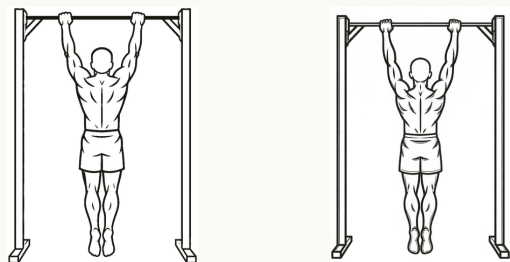
Pay the bills climbing does not: press, legs, rotation. One heavy pull, then the opposite.

26 min | recovery 12h | press, legs, tension, pull

Runs straight after a boulder session, while you are already warm. No finger loading: the bouldering was the finger session. You have just spent two hours pulling, so this deliberately is not more of that - one heavy pull, then the patterns climbing never trains. If you are too cooked, the squat and the cuff work are the two worth doing anyway.

PULSE RAISER + SHOULDER PREP

BLOCK	WARMUP / EASY
TRAINS	shoulder
APPEARS IN	CAVE 01 MAX: 1 x 300s, rest 30s CAVE 02 POWER: 1 x 300s, rest 30s CAVE 03 CAPACITY: 1 x 300s, rest 30s CAVE 04 ASSESS: 1 x 300s, rest 30s HANG 01 MAX HANGS: 1 x 300s, rest 30s HANG 02 CAPACITY: 1 x 300s, rest 30s
LOAD	No load
PROGRESSION	Never progressed. Preparation, not training.
FIXED	Protocol taken as written. Never scaled by level or energy.



Frames show the scapular pull-up. The rest of the sequence is drawn from the prompt.

SETUP

1. Band anchored at elbow height. Floor space. Access to jugs or an easy traverse.
2. Fifteen minutes, in this order. The order is the point: general, then specific.

EXECUTION

1. THREE MINUTES PULSE RAISER. Skipping, star jumps, brisk movement, easy hangboard-free activity. Finish breathing harder but able to talk.
2. FIFTEEN BAND EXTERNAL ROTATIONS PER SIDE. Elbow pinned to the ribs at 90 degrees, rotate the forearm out, control it back.
3. TEN SCAPULAR PULL-UPS. Straight arms throughout. Pull the shoulder blades down and together so the body rises a few centimetres, then release to a full hang.
4. FIVE MINUTES EASY TRAVERSING OR JUGS. Big holds, easy movement, feet on. Let the fingers, elbows and shoulders take progressively more load.
5. Finish warm and switched on, not tired. You should end this wanting to climb.

POSITION CHECKS

- Breathing raised, joints moving freely, hands warm rather than cold and stiff.
- Nothing tender, nothing sharp. If something is, the session downgrades now.
- You have not done anything hard enough to cost you a working set.

COMMON MISTAKES

- Treating it as optional because you feel fine. Feeling fine cold is not the same as being warm.
- Turning it into a session. If the traverse becomes a project, the warm-up failed.
- Skipping straight to the board because the gym is busy.

BREATHING

Nasal, unhurried.

WHY

Warm and switched on, not tired. You should finish wanting to climb.

Progressive Warm-Up. Progressive loading of the flexor tendons and pulleys before maximal work is the most consistently recommended injury-reduction step in climbing, and it doubles as a readiness check: how the fingers feel on the third warm-up hang decides whether the session goes ahead as written.

Protocol: Pulse raiser, then progressive finger loading jug > 30mm > 20mm. Rest 60s between warm-up hangs. 3-4 progressive hangs. 40-60% - enough to prepare, not enough to fatigue. Every session that loads fingers.

Source: RAMP warm-up model; climbing warm-up and injury prevention guidance.

NOTE

Pulse raiser, band external rotations, scapular pull-ups, easy traversing.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain skipping rope, a resistance band, a plain horizontal bar, and a simply outlined climbing wall with plain rounded holds.

PANEL 1: Climber skipping on the spot with a plain skipping rope, both feet just off the ground, upright posture. Full body, front view.

PANEL 2: Climber standing side-on, upper arm pinned to the ribs with the elbow at 90 degrees, forearm rotated outward against a resistance band that runs off-frame.

PANEL 3: Climber hanging from a horizontal bar with COMPLETELY STRAIGHT arms, shoulder blades pulled down and together so the body has risen slightly and there is a large gap between ears and shoulders. Elbows locked straight.

PANEL 4: Climber traversing an easy overhanging climbing wall on large jug holds, both feet on footholds, arms straight, relaxed posture. Draw the wall as a plain outlined plane with simple rounded holds, no texture or background.

KEY ANGLES AND POSITIONS: Panel 2: elbow exactly 90 degrees, upper arm against the ribs. Panel 3: elbows straight and locked, scapulae depressed. Panel 4: arms straight, hips close to the wall, weight through the feet.

ANNOTATION: Orange directional arrows on the rotation path in panel 2 and downward at the shoulder blades in panel 3. An orange sequence number 1 to 4 in the corner of each panel marking the running order.

NOTE: Four stages of one warm-up, read left to right in order.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

PROGRESSIVE FINGER LOADING

BLOCK	WARMUP / EASY
TRAINS	crimp, openhand
APPEARS IN	CAVE 01 MAX: 4 x 10s, rest 60s CAVE 02 POWER: 3 x 10s, rest 60s CAVE 03 CAPACITY: 3 x 10s, rest 60s CAVE 04 ASSESS: 4 x 10s, rest 60s HANG 01 MAX HANGS: 4 x 10s, rest 60s HANG 02 CAPACITY: 3 x 10s, rest 60s
LOAD	Jug > 30mm > 20mm, 40-60%
PROGRESSION	Never progressed.
FIXED	Protocol taken as written. Never scaled by level or energy.

NO STOCK ILLUSTRATION. No openly-licensed frame shows progressive hangboard loading. Generate from the prompt.

SETUP

- Stand at the hangboard with both feet on the floor. Feet on the floor is how you control the load through the whole warm-up.
- Pick three holds in descending size: a jug, a 30mm edge, then the 20mm edge you will work on.
- Set a timer for 10 seconds on, 60 seconds off.

EXECUTION

1. HANG ONE, JUG. Take roughly 40 percent of your weight through the hands, keeping the rest on your feet. Ten seconds. Step off. Rest sixty.
2. HANG TWO, 30MM EDGE. Take a little more weight, maybe 50 percent. Ten seconds. Rest sixty.
3. HANG THREE, 20MM EDGE. Around 60 percent, still with feet helping. Ten seconds. Rest sixty.
4. HANG FOUR, 20MM EDGE. Close to your working position but still submaximal. Ten seconds.
5. THE LAST HANG IS THE READINESS CHECK. Tender, sharp, clicking or aching means the session gets downgraded today, without negotiation. Move to DENSITY instead.

POSITION CHECKS

- Never progressed and never loaded. This is preparation, not training.
- Each hang is in the exact grip position you will use in the working sets.
- Fingers feel warm, blood-filled and responsive by the last hang.

COMMON MISTAKES

- Skipping straight to the working load, which is how most finger injuries start.
- Doing the warm-up hangs in a different grip from the working sets.
- Ignoring what the last hang tells you because the session was already planned.

BREATHING

Normal throughout.

WHY

The last warm-up hang is the readiness check. Tender or sharp means the session gets downgraded, today, without negotiation.

Progressive Warm-Up. Progressive loading of the flexor tendons and pulleys before maximal work is the most consistently recommended injury-reduction step in climbing, and it doubles as a readiness check: how the fingers feel on the third warm-up hang decides whether the session goes ahead as written.

Protocol: Pulse raiser, then progressive finger loading jug > 30mm > 20mm. Rest 60s between warm-up hangs. 3-4 progressive hangs. 40-60% - enough to prepare, not enough to fatigue. Every session that loads fingers.

Source: RAMP warm-up model; climbing warm-up and injury prevention guidance.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard drawn in simple outline with visibly different hold sizes: two large jugs at the outside, a medium edge, and a small edge. A plain floor line. Nothing else.

PANEL 1: JUG, 40 PERCENT. Climber standing beneath a wall-mounted hangboard, BOTH FEET FLAT ON THE FLOOR, hands on the two largest jug holds of the board, arms nearly straight, body upright and taking only part of his weight through the hands. Side view showing clearly that the feet are loaded.

PANEL 2: 30MM EDGE, 50 PERCENT. The same setup on a medium edge, feet still on the floor but with the knees slightly more bent so more weight goes through the hands.

PANEL 3: 20MM EDGE, 60 PERCENT. The same setup on the small 20mm edge, feet still in contact with the floor, more weight through the arms, shoulders engaged and pulled down.

KEY ANGLES AND POSITIONS: THE FEET STAY ON THE FLOOR IN ALL THREE PANELS. This is the defining feature and must be unmistakable. Elbows slightly bent, never locked straight. Shoulders depressed and packed down away from the ears.

ANNOTATION: An orange percentage label beside each panel: 40%, 50%, 60%. An orange downward arrow through the body showing the load line, drawn thicker in each successive panel. A small orange arrow at the feet marking the weight they are still carrying.

NOTE: Three warm-up hangs in ascending load, read left to right. Same figure, same board, same scale.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

MAX HANG - HALF-CRIMP 20MM

BLOCK	PRIMARY / MAX
TRAINS	crimp
APPEARS IN	CAVE 01 MAX: 4 x 10s, rest 180s HANG 01 MAX HANGS: 5 x 10s, rest 180s
LOAD	20mm, 85-90% of tested 10s max
PROGRESSION	All four sets clean at 10s, then add 1-2kg next session.

NO STOCK ILLUSTRATION. No openly-licensed frame shows hangboard grip positions. This one matters more than any other illustration in the book. Generate from the prompt.

SETUP

1. TWENTY MILLIMETRES, ALWAYS. Smaller edges concentrate load over a shorter contact area and are not a testing tool for fingers that are complaining.
2. Set the half-crimp before you load it. SECOND KNUCKLES at roughly 90 degrees. FIRST KNUCKLES (the fingertip joints) EXTENDED, flat or slightly back. THUMB OFF, resting alongside the hand and doing nothing.
3. Elbows SLIGHTLY BENT, never locked. Locked elbows put the load through the joint instead of the muscle.
4. Shoulders down and back, ribs pulled down, glutes lightly engaged. The body is a plank hanging from the fingers, not a sack.
5. Load: 85 to 90 percent of your tested 10 second max, added on a belt or removed with a pulley.

EXECUTION

1. Step up or lift off smoothly. Never snatch onto the edge.
2. Pull the edge TOWARD you rather than just hanging off it. Active fingers, active shoulders.
3. Hold ten seconds, keeping the grip position identical from second one to second ten.
4. Step off DELIBERATELY at ten seconds. Do not drop.

5. Stop the set the moment the position degrades, not when the timer says so. A hang that drifts into a full crimp or an open drag is no longer the exercise you prescribed.
6. Rest the full three minutes. Four to six sets.
7. Progress: all four sets clean at ten seconds, then add 1 to 2kg next session.

POSITION CHECKS

- The knuckle profile is the same in second ten as it was in second one.
- Elbows stay slightly bent. Shoulders stay packed.
- The body hangs still. Kicking and swinging means the load is too high.
- You feel this in the forearm flexors, not as a point of pain in a finger.

COMMON MISTAKES

- Drifting into full crimp (thumb over the index) or open drag under fatigue, which changes the exercise mid-set and loads the pulleys differently.
- Locked elbows, which is the most common source of medial elbow pain from hangboarding.
- Adding load before the current load is clean across every set.
- Training through sharpness, tenderness or joint ache. Any of those: stop the session and do DENSITY instead.

BREATHING

Breathe through it.

WHY

Pull the edge toward you. The body is a plank hanging from the fingers, not a sack.

Max Hangs. Isometric finger strength on a small edge is the single strongest physical predictor of climbing performance in the literature. Sets are short and rest is long because the goal is peak force per set, not accumulated fatigue; three minutes allows near-complete phosphocreatine resynthesis.

Protocol: 10s hang on a 20mm edge. Rest 180s between sets (range 120-300s). 4-6 per grip. 85-90% of tested 10s max load. 2x per week, never on consecutive days.

Source: Lattice Training hangboarding guides; Baláš et al., physical performance testing in climbing (Front Sports Act Living 2023); Winkler/Levernier finger strength vs. grade datasets.

NOTE

Twenty millimetres, always. Smaller edges concentrate load over a shorter contact area and are not a testing tool for fingers that are complaining. Any sharpness, tenderness or joint ache: stop the session and do DENSITY instead.

VARIATION LADDER

L1	Half-Crimp 20mm - Build	Feet assisted. 10s at RPE 7: comfortably hard, never maximal. Live here 6-8 weeks.
L3	Half-Crimp 20mm - Bodyweight	Bodyweight or light assist, 10s at RPE 8
L4	Half-Crimp 20mm - Weighted	20mm + weight, 85-90% of tested 10s max
L6	Half-Crimp 20mm - Heavy	20mm at 140%+ bodyweight (+25kg at 62kg)
L7	Half-Crimp 15mm	15mm. Only past 150% on 20mm, and only after a quiet season.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard in simple outline with a clearly drawn narrow 20mm edge. Nothing else.

PANEL 1: FULL BODY HANG. Climber hanging from a wall-mounted hangboard on a small 20mm edge with both hands, arms nearly straight with a SLIGHT visible bend at the elbows, shoulders pulled DOWN and back, ribs down, body held straight and still like a plank with legs together and slightly forward, toes pointed. Side-on three-quarter view.

PANEL 2: HALF-CRIMP, CORRECT. A large anatomical close-up of ONE HAND on a 20mm edge, seen from the side in profile. The knuckles at the base of the fingers are flexed, the MIDDLE knuckles are bent to approximately 90 degrees, and the FINGERTIP joints are EXTENDED FLAT so the fingertips lie flat on the edge. The THUMB is resting alongside the index finger, NOT wrapped over it. Four fingers, anatomically correct.

PANEL 3: FULL CRIMP, INCORRECT FOR THIS EXERCISE. The same close-up of one hand on the same edge, but now the fingertip joints are HYPEREXTENDED and the THUMB IS WRAPPED OVER THE TOP of the index fingernail, locking the grip.

PANEL 4: OPEN-HAND DRAG, INCORRECT FOR THIS EXERCISE. The same close-up, but the fingers are almost straight and draped over the edge with all joints nearly extended, only the base knuckles flexed, thumb hanging free.

KEY ANGLES AND POSITIONS: Panel 2 is the single most important drawing in this set: middle knuckle at 90 degrees, fingertip joint FLAT, thumb OFF. Panels 3 and 4 must be unmistakably different from panel 2 in exactly those details. In panel 1 the elbows are slightly bent, never locked, and the shoulders are clearly depressed.

ANNOTATION: A burnt-orange 90-degree angle arc on the middle knuckle in panel 2. An orange tick mark in the corner of panel 2 and an orange X in the corners of panels 3 and 4. A small orange arrow pointing at the thumb in each of panels 2, 3 and 4, since the thumb is what distinguishes them.

NOTE: Draw the hands very large and anatomically exact: four fingers plus thumb, correct joint count, no fused or warped digits. The hand close-ups are the reason this illustration exists.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

LIMIT BOULDERING

BLOCK	PRIMARY / MAX
TRAINS	contact, crimp, tension, pull
APPEARS IN	CAVE 02 POWER: 6 x 1 reps, rest 180s
LOAD	Problems you can do in 1-4 tries on a good day
PROGRESSION	Harder problems, not more attempts. Six maximal attempts beats twelve tired ones.

NO STOCK ILLUSTRATION. No openly-licensed frame shows limit bouldering. Generate from the prompt.

SETUP

1. Fully warm. This comes after the warm-up, never at the start and never at the end.
2. Pick two or three problems that are GENUINELY HARD for you: four to six moves, ideally steep ground with small holds.
3. Choose problems that expose you, not ones that suit you. The value is in the positions you are bad at.
4. A good target is a problem you could do in one to four tries on a good day.

EXECUTION

1. ONE FULL ATTEMPT per set. Commit to it completely.
2. Then THREE MINUTES OF REAL REST. Sit down. Breathe. Do not fiddle with your phone under the board and call it rest.
3. Six to eight maximal attempts total across the session.
4. Stop the moment quality drops. Six maximal attempts beats twelve tired ones, and tired attempts train the wrong motor pattern.

5. Progress by choosing HARDER PROBLEMS, not by adding attempts.

POSITION CHECKS

- Every attempt is at genuine maximum intent.
- You are resting long enough that each attempt feels fresh.
- The problems are exposing a weakness rather than confirming a strength.

COMMON MISTAKES

- Cutting rest to three attempts in five minutes, which turns a power session into a mediocre endurance session.
- Continuing once quality drops.
- Choosing problems that suit you. Comfortable projects are a different, easier session wearing this one's name.

BREATHING

Full recovery breathing between attempts. Commit and breathe out on the hard move.

WHY

This is the most transferable power stimulus you have: fingers, tension and movement loaded together in the pattern the sport actually uses.

Limit Bouldering. The most transferable power stimulus available, because it loads fingers, body tension and movement simultaneously in the pattern the sport uses. Long rests keep every attempt maximal; once quality drops the session is over regardless of how many attempts remain.

Protocol: 4-6 move problems at your limit, 2-4 attempts each. Rest 180s between attempts. 6-8 attempts total. Maximal - problems you can do in 1-4 tries on a good day. 1-2x per week, fresh.

Source: Standard practice across Lattice, Power Company and Hoerst methodologies; supported by finger strength / bouldering performance correlations.

NOTE

Four to six move problems at your limit. One attempt per set.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain overhanging climbing wall drawn as one clean angled plane with simple geometric holds. A plain rectangular crash pad. Nothing else.

PANEL 1: Climber mid-move on a steeply overhanging bouldering wall, body tensioned hard: one hand latching a small crimp high above, the other hand pulling on a low hold, hips pulled IN CLOSE to the wall, one foot pressing on a small foothold and the other toe-hooking, back straight, whole body engaged. Side-on three-quarter view. Draw the wall as a plain overhanging plane with simple geometric holds, no texture, no gym background.

PANEL 2: The same climber SEATED on a plain crash pad below the wall, elbows on knees, head up, breathing and resting between attempts, hands loose and open.

KEY ANGLES AND POSITIONS: Panel 1: hips close to the wall, arms not fully extended, body under obvious tension, both feet actively engaged. Panel 2: completely relaxed posture, shoulders down, hands open.

ANNOTATION: An orange '1 attempt' label above panel 1 and an orange '3 min' label above panel 2, with a circular orange arrow between them showing the cycle.

NOTE: The contrast between maximal effort and genuine full rest is the instructional point of this pair.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

CAMPUS LADDERS

BLOCK	PRIMARY / MAX
TRAINS	contact, pull
APPEARS IN	CAVE 02 POWER: 4 x 1 reps, rest 180s
LOAD	1-3-5 ladder, large rungs
PROGRESSION	Add a rung span (1-4-7) before adding sets. Never more than 5 sets.
SAFETY GATE	Campus boarding is the highest injury-risk tool in the gym and loads fingers, elbows and shoulders explosively. It stays locked until your 20mm two-arm hang is at least 130% bodyweight and you have had four clear weeks with no finger or elbow pain. Recruitment pulls give most of the benefit at a fraction of the risk.

NO STOCK ILLUSTRATION. No openly-licensed frame shows campus board work. Generate from the prompt.

SETUP

1. GATED. This exercise stays locked until your two-arm 20mm hang is at least 130 percent of bodyweight AND you have had four clear weeks with no finger or elbow pain. Until then the app substitutes recruitment pulls, which give most of the benefit at a fraction of the risk.
2. LARGE RUNGS ONLY. Small rungs multiply the injury risk and add nothing.
3. Fully warm, AFTER the limit boulders, never at the end of a session.
4. Only two hand positions are safe on rungs: a full open hang, or the chisel half-crimp. Nothing else.

EXECUTION

1. Match both hands on rung 1.
2. Move one hand up to rung 3. Catch it in a controlled position with the shoulder ENGAGED, never hanging loose in the socket.
3. Move the other hand up to rung 5.
4. Come down under control. That is one ladder, and one ladder is one set.
5. Rest three full minutes. Four to five sets, never more.
6. STOP THE EXERCISE the moment a catch feels sloppy. Not the set: the exercise.
7. Progress by adding a rung span (1-4-7) before adding sets.

POSITION CHECKS

- Shoulders stay engaged and pulled down through every catch.
- Every catch is controlled. A catch you barely hold is the one that injures you.
- You stop while it still feels good.

COMMON MISTAKES

- Campusing tired. Every campus injury happens on the set someone should not have done.
- Campusing on small rungs.
- Hanging loose into the shoulder sockets on the catch.
- Doing it because it looks impressive rather than because a test says you need it.

BREATHING

Sharp exhale on each move.

WHY

Stop while it still feels good. Every campus injury happens on the set someone should not have done.

Campus Ladders. Two weekly campus sessions improved bouldering performance in advanced climbers, and four sessions improved rate of force development further. It is also the highest-risk tool in the gym: fingers, elbows and shoulders are the three commonest climbing injury sites and campusing loads all three explosively. Gated on a tested finger-strength standard in this app rather than on

accumulated XP.

Protocol: 1-3-5 or 1-4-7 ladder, one ladder per set. Rest 180s. 4-5. Maximal, stopping well before fatigue. Max 2x per week, in a dedicated block only.

Source: Effects of Two vs. Four Weekly Campus Board Training Sessions, Front Physiol (2021); climbing injury epidemiology reviews.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain overhanging campus board panel with evenly spaced horizontal rungs, drawn in simple outline. Nothing else.

PANEL 1: START. Climber hanging from a steeply overhanging campus board with BOTH hands matched on the same large rung, arms slightly bent, shoulders pulled DOWN and engaged, feet hanging free with legs straight and still. Side view. Draw the campus board as a plain overhanging panel with a ladder of evenly spaced horizontal rungs, numbered 1 to 7 in small burnt-orange figures up the side.

PANEL 2: MID LADDER. The same climber with one hand still on rung 1 and the other hand having moved up and caught rung 3, body extended, shoulder of the high arm clearly ENGAGED and pulled down, not hanging loose.

PANEL 3: TOP. The same climber with one hand on rung 3 and the other having caught rung 5, body extended upward, both shoulders engaged.

KEY ANGLES AND POSITIONS: Elbows NEVER fully locked straight in any panel: always a slight bend. Shoulders depressed and engaged in every panel, with a visible gap between ear and shoulder. Legs straight, together and still, never kicking.

ANNOTATION: Burnt-orange rung numbers 1 to 7 running up the side of the board. Orange arrows tracing each hand's path: 1 to 3 in panel 2, 3 to 5 in panel 3. A small orange arrow at each shoulder pointing DOWN, marking engagement.

NOTE: Three stages of one ladder, read left to right. Same figure, same board, same scale.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

EXPLOSIVE PULL-UPS

BLOCK	SECONDARY / HARD
TRAINS	pull, contact
APPEARS IN	CAVE 02 POWER: 3 x 3 reps, rest 150s
LOAD	Bodyweight, maximal speed
PROGRESSION	Chest to bar, then hands leaving the bar, then a small added load.



Stock frames show the pull-up path. The speed, which is the whole point, is drawn from the prompt.

SETUP

1. Dead hang from the bar, overhand grip, shoulder width.
2. Engage the shoulders BEFORE the pull starts: blades down and back. A pull that starts from a loose shoulder is how shoulders get hurt at speed.

3. Body rigid: ribs down, glutes on, legs still.

EXECUTION

1. Pull AS FAST AS YOU PHYSICALLY CAN, aiming to bring the chest to the bar.
2. Lower under control. The descent is not the exercise but it is not a free drop either.
3. Reset fully at the bottom: full dead hang, shoulders re-engaged, body still.
4. THREE REPS MEANS THREE MAXIMAL-SPEED REPS. The moment speed drops, the set is finished, even at rep two.
5. Rest 150 seconds. Three sets.
6. Progress: chest to bar, then hands leaving the bar at the top, then a small added load.

POSITION CHECKS

- Speed, not height, is the measure. A fast three-quarter rep beats a slow full one here.
- No swing. The speed comes from the pull, not from a kip.
- Shoulders engaged before every rep starts.

COMMON MISTAKES

- Grinding out slow reps once speed has gone, which trains a different quality entirely.
- Using a swing to generate the speed.
- Doing these at the end of a session when the nervous system is flat.

BREATHING

Sharp exhale on each pull.

WHY

Three reps means three maximal-speed reps. The moment speed drops the set is finished.

Maximal Strength. Low reps at high load build force with the least added mass, which is what a strength-to-weight sport needs. Two to three hard sets per pattern, twice weekly, is enough to drive strength in a climber who is also climbing; more volume mostly buys fatigue and mass.

Protocol: 3-5 reps. Rest 180s. 4-5. RPE 8-9, a load you could do 1-2 more reps with. 2x per week per pattern.

Source: Lattice minimum effective dose guidance; systematic review on resistance training, climbing performance and injury prevention (Sports Med Open 2024).

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar. Nothing else.

PANEL 1: BOTTOM. Climber in a dead hang from a horizontal bar, overhand grip, arms completely straight but SHOULDERS ALREADY ENGAGED and pulled down, body rigid and vertical, legs straight and still. Side view.

PANEL 2: EXPLOSIVE PULL. The same figure driving upward at maximum speed, elbows bent past 90 degrees, chest rising toward the bar, body still rigid and vertical with no kip or swing. Convey speed with clean burnt-orange motion lines behind the torso, not with a blurred or distorted body.

PANEL 3: TOP. The same figure at the peak with the CHEST level with the bar, elbows pulled down beside the ribs, body still vertical and controlled.

KEY ANGLES AND POSITIONS: The body stays vertical and rigid in all three panels with no hip kick and no leg swing. Shoulders depressed throughout, including in panel 1.

ANNOTATION: Straight burnt-orange speed lines trailing below the torso in panel 2. A thick orange upward arrow beside the body labelled 'MAX SPEED'. A dashed orange vertical reference line through all three panels showing the body does not swing off the line.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

RING TOE-HOOK HOLD

BLOCK	SECONDARY / HARD
TRAINS	tension
APPEARS IN	CAVE 02 POWER: 3 x 8 reps, rest 90s
LOAD	Bodyweight
PROGRESSION	Longer holds, then a straighter body, then one arm on the rings.

NO STOCK ILLUSTRATION. No openly-licensed frame shows a ring toe-hook. Generate from the prompt.

SETUP

1. Rings set at roughly head height.
2. A low bar, or the ring straps themselves, positioned where you can hook both heels over it.
3. Take the rings, then hook BOTH HEELS over the bar or straps so you are suspended between hands and feet.
4. Shoulders packed down before you load the position.

EXECUTION

1. Pull with the HANDS AND THE FEET AT THE SAME TIME to bring the body into a horizontal line.
2. The heels actively pull toward you while the hands pull the rings toward the hips. This opposing pull is the whole exercise.
3. Hold the horizontal line, hips UP, body straight.
4. For reps, move slowly between a lower and a horizontal position rather than bouncing.
5. Eight reps or holds. Progress: longer holds, then a straighter body, then one arm on the rings.

POSITION CHECKS

- The hips stay up. A dropped hip means the feet have stopped pulling.
- You feel this across the whole posterior chain and the trunk, not just the arms.
- This is the tension climbing actually asks for: pulling with the feet while the hands hold.

COMMON MISTAKES

- Letting the hips drop, which unloads the toe hook and turns it into a hang.
- Relying on the arms and forgetting the feet are meant to be pulling.
- Setting the rings too low, which removes the tension requirement.

BREATHING

Continuous, controlled.

WHY

This is the tension climbing actually asks for: pulling with the feet while the hands hold. Steep rock is won here.

Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

NOTE

Hang from rings, hook both heels over a bar or ring strap, hold body horizontal.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout.

Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: Two gymnastic rings on plain straps, and one plain low horizontal bar. Nothing else.

PANEL 1: Climber suspended horizontally face-up between two gymnastic rings held in his hands above and slightly behind his shoulders, and a low horizontal bar over which BOTH HEELS ARE HOOKED. The body is held in a straight horizontal line from shoulders to heels, hips UP and level with the shoulders, arms bent and pulling the rings in toward the hips. Side view, full profile.

PANEL 2: A close-up of the HEEL HOOK: the back of the heel and the Achilles pressing over the top of the bar, toes pointed back and down, ankle actively flexed to grip. Show clearly that the heel, not the toes, is doing the hooking.

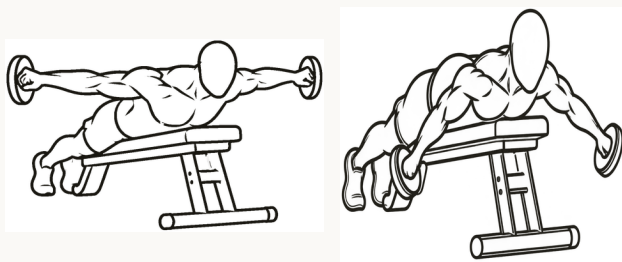
KEY ANGLES AND POSITIONS: The body is HORIZONTAL with the hips level with the shoulders, no sag. Shoulders packed down. Elbows bent, rings drawn toward the hips rather than held out at arm's length.

ANNOTATION: An orange arrow at the heels pointing back toward the body, and an orange arrow at the hands pointing toward the hips, showing the two opposing pulls that create the tension. A dashed orange horizontal line through the body showing the required straight line.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

TRX Y-T-W

BLOCK	PREHAB / EASY
TRAINS	shoulder
APPEARS IN	CAVE 02 POWER: 2 x 10 reps, rest 45s
LOAD	Angle sets the load - stay light
PROGRESSION	Walk the feet forward to steepen the angle. Never to failure.



Stock frames show the T position prone. The suspended Y, T and W are drawn from the prompt.

SETUP

1. TRX or rings at roughly chest height.
2. Take a handle in each hand, arms straight, and lean BACK so your weight is on the straps.
3. Body rigid from heels to head. Walk the feet forward until you feel real resistance, but stay light: the angle sets the load and you want this easy.

EXECUTION

1. Y, 10 reps. Keeping the arms STRAIGHT, pull yourself up by raising both arms into a wide overhead Y, thumbs up. Lower under control.
2. T, 10 reps. Arms straight, pull out to the sides into a T, squeezing the shoulder blades together.
3. W, 10 reps. Bend the elbows and pull them back and down close to the ribs, forearms angled up, so the arms form a W.
4. Two rounds, 45 seconds between.
5. Progress by walking the feet forward to steepen the angle. Never to failure.

POSITION CHECKS

- No shrugging. Shoulders stay down and away from the ears in all three positions.
- The body stays a rigid plank. The hips do not sag or pike.
- Slow and precise beats heavy. This is scapular control, not a back workout.

COMMON MISTAKES

- Shrugging, which swaps the scapular stabilisers for the traps.
- Steepening the angle until it becomes a row.
- Bending the arms on the Y and the T, which does the same thing.

BREATHING

Exhale on the pull.

WHY

Scapular control, not a back workout. Slow and precise beats heavy.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

NOTE

Ten reps in each of the three positions.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A suspension trainer drawn as two plain straps with simple handles, descending from off the top of the frame. Nothing else.

PANEL 1: Y. Climber leaning back with straight body and straight arms, holding two suspension-trainer handles, arms raised overhead in a wide Y shape, thumbs up, shoulders DOWN and not shrugged, body one rigid line from heels to head at roughly 45 degrees to the floor. Front three-quarter view.

PANEL 2: T. The same suspended lean, arms straight and pulled out horizontally to both sides forming a T, shoulder blades squeezed together.

PANEL 3: W. The same suspended lean, but with the elbows bent and pulled back and DOWN close to the ribs, forearms angled upward and outward, arms forming a W shape.

KEY ANGLES AND POSITIONS: The body is one rigid straight line from heels to head in ALL THREE panels, with absolutely no hip sag. Shoulders depressed, never shrugged. Arms straight in panels 1 and 2, bent only in panel 3.

ANNOTATION: A dashed orange straight line through the body in every panel showing the plank. A small orange downward arrow at each shoulder in every panel reinforcing that the shoulders stay down.

NOTE: Three positions of one circuit, read left to right. Same figure, same straps, same body angle in all three.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

RECRUITMENT PULLS

BLOCK	PRIMARY / MAX
TRAINS	contact, crimp
APPEARS IN	CAVE 02 POWER [substitute]: 4 x 5s, rest 150s
LOAD	Maximal intent against an immovable edge
PROGRESSION	Measured peak force on a gauge, or a heavier block. Intent is the variable, not duration.

NO STOCK ILLUSTRATION. No openly-licensed frame exists. Generate from the prompt.

SETUP

1. You need an edge that CANNOT MOVE: a block fixed to a solid anchor, a loading pin bottomed out, or a hangboard edge with your feet planted and a weight belt anchored to the floor.
2. Set the grip you want to train, half-crimp or open, before you start pulling.
3. Elbow slightly bent, shoulder packed down, body braced.
4. This is the substitute the app gives you while the campus gate is closed, and it gives most of the benefit at a fraction of the risk.

EXECUTION

1. Build to MAXIMUM FORCE AS FAST AS YOU CAN. Full effort from the first half-second.
2. Hold that maximal pull for five seconds.
3. Release, rest 120 to 150 seconds. Four to five sets.
4. INTENT IS THE TRAINING VARIABLE, not duration and not load. Contact strength is how FAST you reach force, not how much you eventually reach.
5. Progress by measuring peak force on a gauge, or by using a heavier block.

POSITION CHECKS

- The first half-second is maximal. If you ease into it, you have trained the wrong quality.
- The grip position holds for the full five seconds.

- You are fresh. Maximal intent is simply not available when fatigued.

COMMON MISTAKES

- Easing into the pull, which trains maximum strength rather than rate of force development.
- Doing these fatigued.
- Treating the five seconds as the target. The first half-second is the target.

BREATHING

Brace and exhale hard through the pull.

WHY

Contact strength is how fast you reach force, not how much you eventually reach. Intent is the training variable.

Recruitment Pulls. Contact strength is rate of force development, not peak force. Short maximal-intent efforts train the neural side without the tissue cost of dynamic campus work. A safer first choice than the campus board.

Protocol: 5s maximal pull against an immovable edge or block. Rest 120-180s. 4-5. Maximal voluntary effort. 1x per week, only when fresh.

Source: Levernier & Laffaye, force and RFD differences across climbing ability; campus board frequency RCT (Front Physiol 2021).

NOTE

Pull as hard as possible for 5 seconds. Reaching peak force fast is the point.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A small edge block on a short chain, a heavy plain anchor block or floor plate. Panel 3 is a pure diagram.

PANEL 1: Climber standing with BOTH FEET FLAT AND BRACED on the floor, gripping a small edge block attached by a short chain or strap to a heavy immovable anchor at floor level. He is pulling upward with maximal effort: elbow slightly bent, shoulder packed down and back, back straight, whole body braced and rigid, clear tension through the trunk and legs. The block does NOT move. Side view.

PANEL 2: A large anatomical close-up of the hand on the edge in a half-crimp: middle knuckles bent to about 90 degrees, fingertip joints flat, thumb off and resting alongside. Four fingers, anatomically exact.

PANEL 3: A simple diagram in ink and burnt orange: a force-over-time graph with time on the horizontal axis and force on the vertical. One steep orange curve rising almost vertically to a plateau, labelled 'correct'. A second grey curve rising slowly and gradually to the same plateau, labelled with a small X. No figure.

KEY ANGLES AND POSITIONS: Panel 1: elbow slightly bent, shoulder depressed, spine neutral, whole body braced. The anchor is immovable, so nothing is being lifted.

ANNOTATION: A thick orange upward arrow at the hand in panel 1 with the label 'MAX, from the first half-second'. In panel 3 the steepness of the orange curve versus the grey one is the entire message.

NOTE: Panel 3 is what makes this exercise understandable: it is about how fast force arrives, not how much. Keep the graph flat and schematic.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

REPEATERS 7:3

BLOCK	PRIMARY / HARD
TRAINS	forearm, crimp
APPEARS IN	CAVE 03 CAPACITY: 6 x 60s, rest 150s HANG 02 CAPACITY: 6 x 60s, rest 150s
LOAD	55-65% of max hang load
PROGRESSION	Six sets is the dose. When all six complete without early drops, add 1kg.
FIXED	Protocol taken as written. Never scaled by level or energy.

NO STOCK ILLUSTRATION. No openly-licensed frame shows a repeater protocol. Generate from the prompt.

SETUP

1. Half-crimp on the 20mm edge, same position as your max hangs.
2. Load at 55 to 65 percent of your max hang load. This is deliberately much lighter than a max hang.
3. **TIMER RUNNING BEFORE YOU START.** Seven seconds on, three seconds off, and three seconds is not long enough to set a timer in.
4. This protocol is **FIXED**. It does not scale with level or energy.

EXECUTION

1. Hang seven seconds. Step or drop off for three. Repeat six times. That is **ONE SET**, sixty seconds total.
2. In each three-second gap, **COME OFF THE EDGE AND SHAKE OUT**. Staying on the edge through the rest windows turns a repeater into a long max hang.
3. Rest 150 seconds between sets. Six sets is the dose.
4. The last two repetitions of each set should be a genuine fight.
5. Progress: when all six sets complete without early drops, add 1kg.

POSITION CHECKS

- If set six feels the same as set one, the load is too light.
- The grip position holds through all six pulls of a set. When it degrades, the set is finished.
- You end the block pumped, not injured. This trains strength-endurance, not maximum strength.

COMMON MISTAKES

- Loading it like a max hang, which is the single most common error and makes six sets impossible.
- Staying on the edge through the three-second rests.
- Adding sets. Six is the protocol. If six is easy, add load, not volume.

BREATHING

Rhythmic. Exhale on the releases.

WHY

The last two reps of each set should be a fight. If set six matches set one, the load is too light. Repeaters 7:3. Intermittent dead-hang protocols produce greater grip-endurance gains than maximal holds. The 7:3 rhythm mirrors the contract-release pattern of route climbing and is the same work:rest ratio used in the standard forearm critical-force test.

Protocol: 7s hang, 3s off, x6 = one 60s set. Rest 150s between sets (range 120-180s). 5-6. 55-65% of tested max hang load. 1x per week.

Source: Lopez-Rivera & Gonzalez-Badillo, intermittent vs. maximal dead hangs; Medernach et al., 8-week intermittent hangboard RCT.

NOTE

7s on, 3s off, six times = one set.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard with a visible 20mm edge, and a plain floor line. Nothing else.

PANEL 1: ON, 7 SECONDS. Climber hanging from a wall-mounted hangboard on a 20mm edge in a half-crimp, arms slightly bent, shoulders packed down, body straight and still, feet off the floor. Side view.

PANEL 2: OFF, 3 SECONDS. The same climber standing on the floor directly beneath the board, both arms hanging down and RELAXED at his sides, one hand loosely shaking out, fingers open and loose. He is clearly OFF the board.

KEY ANGLES AND POSITIONS: Panel 1: half-crimp, middle knuckles near 90 degrees, fingertip joints flat, thumb off, elbows slightly bent, shoulders depressed. Panel 2: fully unloaded, hands completely off the board, shoulders relaxed.

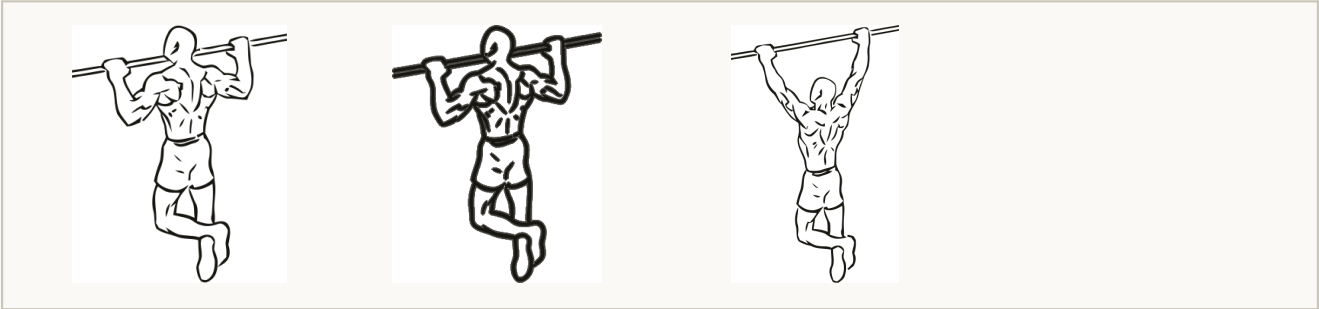
ANNOTATION: A large orange '7s' label above panel 1 and a large orange '3s' label above panel 2. A circular orange arrow looping from panel 2 back to panel 1 with the orange label 'x6' beside it, showing the cycle repeats six times to make one set.

NOTE: Two alternating states of one interval protocol. The contrast between loaded and completely unloaded is the point: panel 2 must show the hands entirely off the board.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

PULL-UP STRENGTH ENDURANCE

BLOCK	SECONDARY / MODERATE
TRAINS	pull
APPEARS IN	CAVE 03 CAPACITY: 4 x 8 reps, rest 120s
LOAD	RPE 7-8 - two or three reps left at the end
PROGRESSION	Add a rep per set to 12, then add load and drop back to 8.



SETUP

1. Bar or rings. Overhand grip, shoulder width.
2. Full dead hang to start. Shoulders engaged before the first pull.
3. Pick a load and a rep count that leaves two or three reps in reserve at the end of each set. RPE 7 to 8.

EXECUTION

1. Eight controlled reps, full range: chin clearly over the bar at the top, arms completely straight at the bottom.
2. No kipping and no bouncing out of the bottom.
3. STOP TWO OR THREE REPS SHORT OF FAILURE on every set. This is deliberately not a max effort.
4. Rest 120 seconds. Four sets.

5. Progress: add a rep per set up to twelve, then add load and drop back to eight.

POSITION CHECKS

- Full range on every rep, including the last one of the last set.
- You finish each set knowing you had more.
- This is the bridge between a heavy single and a thirty-move route.

COMMON MISTAKES

- Taking every set to failure, which wrecks the rest of the session and trains fatigue rather than capacity.
- Shortening the range as the sets go on.
- Kipping once it gets hard.

BREATHING

Exhale on each pull.

WHY

This is the bridge between a heavy single and a 30-move route. It is deliberately not a max effort. Strength Endurance. Bridges maximal strength and route capacity. Kept to one session per week so it does not crowd out the maximal work that actually raises the ceiling.

Protocol: 8-12 reps. Rest 90-120s. 3-4. RPE 7-8. 1x per week.

Source: General resistance training literature applied to climbing.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar. Nothing else.

PANEL 1: BOTTOM. Climber in a full dead hang from a horizontal bar, overhand grip at shoulder width, arms COMPLETELY straight, shoulders engaged and pulled down, body vertical and still, legs straight with ankles crossed. Side view.

PANEL 2: TOP. The same figure pulled up with the CHIN CLEARLY ABOVE the bar, elbows driven down beside the ribs, shoulder blades depressed and retracted, body still vertical with no kip or swing.

KEY ANGLES AND POSITIONS: Panel 1: elbows fully locked straight, this full extension is the checkpoint. Panel 2: chin above bar height. Body vertical in both, no hip kick, no knees swinging forward.

ANNOTATION: A dashed orange vertical reference line through the body in both panels showing there is no swing. An orange bracket at the side spanning the distance between the two positions, labelled 'full range'.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

KB OVERHEAD PRESS

BLOCK	SECONDARY / MODERATE
TRAINS	press, shoulder
APPEARS IN	CAVE 03 CAPACITY: 3 x 8 reps per side, rest 90s
LOAD	KB 12kg, single arm
PROGRESSION	Add reps to 12, then move to a half-kneeling or single-leg stance for a harder position.



SETUP

1. Kettlebell racked at the shoulder: bell resting on the back of the forearm, elbow tucked in close to the ribs, WRIST STRAIGHT and not bent back.
2. Feet shoulder width. Ribs pulled DOWN, glutes lightly squeezed. This rib position stops you pressing by leaning back.
3. Free hand out to the side for balance.

EXECUTION

1. Press the bell straight overhead. The path curves slightly out and then back in over the head, not straight up in front of your face.
2. Finish with the BICEPS BESIDE THE EAR and the shoulder ACTIVELY PUSHING UP at the top. That last push is the range climbers lose.
3. Lower under control back to the rack position.
4. Eight reps, then swap sides. Rest 90 seconds. Three sets per side.
5. Progress: add reps to twelve, then move to a half-kneeling or single-leg stance so the trunk has to work harder.

POSITION CHECKS

- Ribs stay down. If you are leaning back to press, the weight is too heavy.
- The arm finishes fully vertical with the biceps by the ear.
- The wrist stays straight throughout, never bent back under the bell.

COMMON MISTAKES

- Leaning back to press, which converts an overhead press into a standing incline press and skips the range you came for.
- Stopping short of full overhead.
- Flared ribs, which is the same fault as leaning back seen from the front.

BREATHING

Exhale on the press.

WHY

Finish every rep with the shoulder fully overhead. Climbers lose that range, and losing it is how shoulders start hurting.

Maximal Strength. Low reps at high load build force with the least added mass, which is what a strength-to-weight sport needs. Two to three hard sets per pattern, twice weekly, is enough to drive strength in a climber who is also climbing; more volume mostly buys fatigue and mass.

Protocol: 3-5 reps. Rest 180s. 4-5. RPE 8-9, a load you could do 1-2 more reps with. 2x per week per pattern.

Source: Lattice minimum effective dose guidance; systematic review on resistance training, climbing performance and injury prevention (Sports Med Open 2024).

NOTE

Overhead pressing is the antagonist that matters most for a climbing shoulder.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A single plain Kettlebell drawn in simple outline. Nothing else.

PANEL 1: RACK. Climber standing, feet shoulder width, holding a single Kettlebell in the RACK position at the right shoulder: the bell resting against the back of the forearm, elbow tucked down close to the ribs, wrist STRAIGHT, forearm vertical. Free arm out to the side for balance. Ribs pulled down, torso vertical. Front three-quarter view.

PANEL 2: LOCKOUT. The same figure with the Kettlebell pressed fully overhead, arm completely straight and vertical, BICEPS BESIDE THE EAR, shoulder actively pushed up, wrist straight and stacked directly over the elbow and shoulder. The torso is still VERTICAL with no backward lean and the ribs are still down.

PANEL 3: INCORRECT. The same lockout attempt but with the torso LEANING BACK and the lower ribs flared forward, the arm not truly vertical. This is the common fault.

KEY ANGLES AND POSITIONS: Panel 2: the wrist, elbow and shoulder stack in one vertical line, biceps touching the ear, spine VERTICAL. Panel 3 differs from panel 2 only by the backward lean and the rib flare.

ANNOTATION: A dashed orange vertical line running from the bell down through the wrist, elbow, shoulder and hip in panel 2, showing the stack. An orange upward arrow at the shoulder in panel 2 marking the active reach. An orange X in the corner of panel 3 with a small orange arc highlighting the flared ribs.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

TEST - MAX HANG 20MM HALF-CRIMP

BLOCK	TEST / MAX
TRAINS	crimp
APPEARS IN	CAVE 04 ASSESS: 4 x 10s, rest 180s
LOAD	Build up until 10s is the most you can hold
PROGRESSION	Record (bodyweight + added) / bodyweight x 100. At 62kg: 128% = +17kg, 134% = +21kg, 140% = +25kg.

NO STOCK ILLUSTRATION. See MAX HANG for the grip illustration. Generate the test-specific frame from the prompt.

SETUP

1. Fully rested. Not the day after a hard session, and not at the end of one.
2. IDENTICAL SETUP EVERY TEST: same edge, same board, same grip, same shoulder position, same time of day if you can manage it.
3. Full warm-up including progressive finger loading.
4. Have the weights laid out so you are not improvising between attempts.

EXECUTION

1. Start clearly below your expected max. Hang ten seconds in a half-crimp on 20mm.
2. Rest three minutes. Add 3 to 5kg. Hang ten seconds again.
3. Repeat, working up in 3 to 5kg jumps with three minutes between every attempt.
4. STOP AT THE FIRST FAILED 10 SECONDS and record the last successful load.
5. Calculate: (bodyweight + added) / bodyweight x 100. At 62kg bodyweight: 128 percent is +17kg, 134 percent is +21kg, 140 percent is +25kg.
6. Record it. This number opens the campus gate at 130 percent and sets your training loads for the next block.

POSITION CHECKS

- A test only works if it is repeatable. Change nothing between tests except your training.

- Every attempt is a genuine half-crimp. A hang that degrades into a drag is not a result.
- Ten seconds means ten seconds, not eight and a bit.

COMMON MISTAKES

- Testing tired, which produces a low number you then train off for eight weeks.
- Changing edge size between tests and comparing the numbers anyway.
- Taking too many attempts. Every attempt fatigues the next one.

BREATHING

Normal.

WHY

A test only works if it is repeatable. Change nothing between tests except your training.

Assessment Battery. The assess-identify-train-reassess loop is the core of the Lattice method. Without numbers, training targets whatever feels productive rather than whatever is actually limiting. Run it on a rest day, fully warm, and record everything.

Protocol: Standardised tests, full recovery between each. Rest As long as needed - this is a test, not a workout. 1 per test. Maximal on strength tests. Every 6-8 weeks.

Source: Lattice Training assessment methodology.

NOTE

Work up in 3-5kg jumps. Stop at the first failed 10s and record the last success.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard with a 20mm edge, a simple weight belt with round plates. Panel 2 is a pure diagram.

PANEL 1: Climber hanging from a wall-mounted hangboard on a 20mm edge in a half-crimp, wearing a weight belt with two plates hanging between his legs on a short strap, arms slightly bent, shoulders packed down, body straight and still. Side view.

PANEL 2: A simple diagram in ink and burnt orange: a vertical ladder of five rungs, each labelled with an increasing added weight, the lowest three marked with an orange tick and the top one marked with an orange X, and an orange bracket pointing at the highest ticked rung labelled 'record this'. No human figure.

KEY ANGLES AND POSITIONS: Half-crimp: middle knuckles near 90 degrees, fingertip joints flat, thumb off. Elbows slightly bent, shoulders depressed, body still.

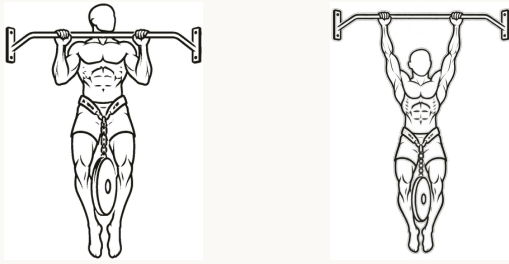
ANNOTATION: An orange '10s' label beside panel 1 and an orange '3 min rest' label between the rungs in panel 2.

NOTE: Panel 2 explains the stepping protocol: work up, stop at the first failure, record the last success. Keep it flat and schematic.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

TEST - WEIGHTED PULL-UP SRM

BLOCK	TEST / MAX
TRAINS	pull
APPEARS IN	CAVE 04 ASSESS: 3 x 5 reps, rest 180s
LOAD	Build to the heaviest clean set of 5
PROGRESSION	Estimated 1RM = SRM load x 1.15. Record as % bodyweight. Lattice male standard is 165%.



SETUP

1. Dead hang start, weight on a belt.
2. Fully rested, fully warm.
3. A 5RM is safer than a true single and almost as informative.

EXECUTION

1. Warm up with progressively heavier sets of five, well short of failure.
2. Perform five STRICT reps: chin clearly over the bar, full extension at the bottom of every rep.
3. Rest three minutes. Add weight. Repeat.
4. Continue until five reps is no longer clean. Record the heaviest clean set of five.
5. Estimate: $1RM = 5RM \text{ load} \times 1.15$. Record as a percentage of bodyweight including your own weight. The Lattice male standard is 165 percent.

POSITION CHECKS

- Every rep starts from a genuine dead hang with straight arms.
- No kipping. A kipped rep does not count and skews the number.
- Under 165 percent total usually means pulling strength is still your limiter.

COMMON MISTAKES

- Counting reps that did not start from a dead hang.
- Testing at the end of a session.
- Taking so many working sets that the last one is fatigue-limited rather than strength-limited.

BREATHING

Exhale on each pull.

WHY

This is the number that tells you whether pulling strength is still limiting you. Under 165% total, it usually is.

Assessment Battery. The assess-identify-train-reassess loop is the core of the Lattice method. Without numbers, training targets whatever feels productive rather than whatever is actually limiting. Run it on a rest day, fully warm, and record everything.

Protocol: Standardised tests, full recovery between each. Rest As long as needed - this is a test, not a workout. 1 per test. Maximal on strength tests. Every 6-8 weeks.

Source: Lattice Training assessment methodology.

NOTE

A 5RM is safer and almost as informative as a true single.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar and a simple weight belt with round plates. Nothing else.

PANEL 1: BOTTOM. Climber in a full dead hang from a horizontal bar, arms COMPLETELY straight, wearing a simple weight belt with two round plates hanging between the legs on a short strap, body vertical and still. Side view.

PANEL 2: TOP. The same figure pulled up with the CHIN CLEARLY ABOVE the bar, elbows down beside the ribs, body still vertical, no kip and no swing.

KEY ANGLES AND POSITIONS: Panel 1: full elbow lockout, this is the checkpoint that makes the rep count. Panel 2: chin above bar height. Vertical body in both.

ANNOTATION: A dashed orange vertical line through the body in both panels showing no swing. An orange bracket at the side labelled 'every rep starts here' pointing at the straight arms in panel 1.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

TEST - 90 DEGREE LOCK-OFF

BLOCK	TEST / MAX
TRAINS	pull
APPEARS IN	CAVE 04 ASSESS: 1 x 30s per side, rest 120s
LOAD	One arm, other hand lightly on the wrist
PROGRESSION	Record seconds per side. Solid is 10s, strong is 15s.

NO STOCK ILLUSTRATION. No openly-licensed frame shows an assisted one-arm lock-off. Generate from the prompt.

SETUP

1. Bar, overhand grip. Timer visible.
2. Pull up to 90 degrees of elbow flexion on ONE arm.
3. Rest the free hand LIGHTLY on the working wrist. It rests. It does not grip and it does not pull.

EXECUTION

1. Hold the 90 degree position. Start the clock.
2. Hold until the elbow angle OPENS PAST 90 degrees. That is the end of the test, not when you drop off.
3. Record the seconds.
4. Rest two minutes, then test the other side.
5. NOTE THE DIFFERENCE between sides. Solid is 10 seconds. Strong is 15 seconds.

POSITION CHECKS

- The assisting hand is genuinely passive.
- The test ends at 90 degrees, not at failure.
- A large left-right gap is worth training out. Climbers accumulate asymmetry without noticing.

COMMON MISTAKES

- Pulling with the assisting hand, which invalidates the result and hides the asymmetry you are testing for.
- Counting the seconds spent sinking below 90 degrees.

- Only testing the strong side.

BREATHING

Keep breathing.

WHY

A large left-right gap is worth training out. Climbers accumulate asymmetry without noticing.

Assessment Battery. The assess-identify-train-reassess loop is the core of the Lattice method.

Without numbers, training targets whatever feels productive rather than whatever is actually limiting. Run it on a rest day, fully warm, and record everything.

Protocol: Standardised tests, full recovery between each. Rest As long as needed - this is a test, not a workout. 1 per test. Maximal on strength tests. Every 6-8 weeks.

Source: Lattice Training assessment methodology.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar. Nothing else.

PANEL 1: Climber hanging from a horizontal bar by ONE ARM, that elbow bent to exactly 90 degrees so the upper arm is horizontal and the forearm vertical. The FREE HAND rests lightly on the working WRIST, fingers open and clearly not gripping. Shoulder of the working arm packed DOWN, body still and vertical, legs straight with ankles crossed. Front view.

PANEL 2: The same figure at the moment of FAILURE: the elbow angle has opened past 90 degrees toward 120, the body has dropped, marking the end of the test.

KEY ANGLES AND POSITIONS: Panel 1: elbow EXACTLY 90 degrees, upper arm horizontal, forearm vertical, working shoulder depressed. Panel 2: elbow clearly more open than 90.

ANNOTATION: A burnt-orange 90-degree angle arc at the elbow in panel 1 with the figure '90' beside it, and a wider arc in panel 2 marked 'test ends'. A small orange arrow pointing at the free hand in panel 1 labelled 'rests only'.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

TEST - FRONT LEVER HOLD

BLOCK	TEST / MAX
TRAINS	tension
APPEARS IN	CAVE 04 ASSESS: 1 x 30s, rest 120s
LOAD	Hardest progression with a flat lower back
PROGRESSION	Record progression and seconds: tuck, advanced tuck, one-leg, straddle, full.

NO STOCK ILLUSTRATION. See FRONT LEVER for the progression ladder. Generate the pass/fail frame from the prompt.

SETUP

1. Straight arms on a bar. Shoulders depressed.
2. Choose your hardest CLEAN progression: tuck, advanced tuck, one-leg, straddle, or full.
3. Have someone watch your lower back, or film it from the side.

EXECUTION

1. Pull into the position with straight arms and hold.
2. Hold to FAILURE OF POSITION, not failure of grip.

3. THE MOMENT THE LOWER BACK ARCHES, THE TEST IS OVER. Arching is how people fake this hold and it is why untested front lever times are meaningless.
4. Record both the progression AND the seconds. A twelve second advanced tuck is a different result from a twelve second straddle.

POSITION CHECKS

- Flat lower back for every recorded second.
- Elbows locked straight throughout.
- The recorded time contains no arched seconds.

COMMON MISTAKES

- Recording a time that included five arched seconds.
- Testing a progression above the one you can actually hold clean.
- Letting the grip end the test when the position failed ten seconds earlier.

BREATHING

Short controlled breaths.

WHY

The moment the lower back arches, the test is over. Arching is how people fake this hold.

Assessment Battery. The assess-identify-train-reassess loop is the core of the Lattice method. Without numbers, training targets whatever feels productive rather than whatever is actually limiting. Run it on a rest day, fully warm, and record everything.

Protocol: Standardised tests, full recovery between each. Rest As long as needed - this is a test, not a workout. 1 per test. Maximal on strength tests. Every 6-8 weeks.

Source: Lattice Training assessment methodology.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain horizontal bar. Nothing else.

PANEL 1: VALID. Climber holding a straddle front lever from a horizontal bar: arms completely straight, body horizontal and parallel to the floor, legs straight and split wide, LOWER BACK FLAT. Side view, full profile.

PANEL 2: INVALID. The same climber in the same position but with the LOWER BACK CLEARLY ARCHED and the hips sagging below the line of the shoulders. This is the moment the test ends.

KEY ANGLES AND POSITIONS: The ONLY difference between the two panels is the spine: flat and horizontal in panel 1, arched with dropped hips in panel 2. Elbows straight in both. Exaggerate the difference so it reads instantly.

ANNOTATION: A dashed orange straight horizontal line through the body in panel 1 showing the flat line, and the same line in panel 2 with the body visibly falling away from it. An orange tick in the corner of panel 1 and an orange X in the corner of panel 2 with the label 'test ends here'.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

TEST - FOOT RAISE + STRADDLE

BLOCK	TEST / MODERATE
TRAINS	mobility
APPEARS IN	CAVE 04 ASSESS: 1 x 120s, rest 30s
LOAD	No load
PROGRESSION	Record heel height in cm per side, and pubic-bone-to-floor distance in a straddle. Average foot raise is about 74cm.



Stock frame shows a seated hip-open position. The two test positions are drawn from the prompt.

SETUP

1. FOOT RAISE: stand next to a wall with a tape measure to hand. Stand tall with the torso upright.
2. STRADDLE: clear floor, tape measure.
3. Warm and loose, but this is a measurement, not a stretching session.

EXECUTION

1. FOOT RAISE. Raise one foot as high as possible with the hip FLEXED, ABDUCTED AND EXTERNALLY ROTATED, knee bent, as if placing a high outside foot on the wall.
2. Keep the torso UPRIGHT. Leaning away inflates the number and tests nothing.
3. Measure floor to heel in centimetres. Both sides. Average is about 74cm.
4. STRADDLE. Sit into a maximal straddle with the legs STRAIGHT and the feet FLAT, not rolled forward.
5. Measure the distance from the pubic bone to the floor. Record it.

POSITION CHECKS

- The torso stays upright on the foot raise.
- Both sides measured, and the difference noted.
- The straddle has straight legs and vertical feet.

COMMON MISTAKES

- Leaning the torso away to fake height.
- Not testing both sides, which hides an asymmetry.
- Bending the knees in the straddle.

BREATHING

Relaxed, exhale into end range.

WHY

Combined abduction and external rotation is what makes high steps, drop knees and bridging available. It is cheap to train and costs no recovery.

Assessment Battery. The assess-identify-train-reassess loop is the core of the Lattice method. Without numbers, training targets whatever feels productive rather than whatever is actually limiting. Run it on a rest day, fully warm, and record everything.

Protocol: Standardised tests, full recovery between each. Rest As long as needed - this is a test, not a workout. 1 per test. Maximal on strength tests. Every 6-8 weeks.

Source: Lattice Training assessment methodology.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain flat wall with a simple vertical measuring tape in panel 1. A plain floor in panel 2. Nothing else.

PANEL 1: FOOT RAISE TEST. Climber standing beside a plain flat wall with his TORSO UPRIGHT and vertical, one foot raised high against the wall with the hip flexed, turned out and opened to the side, knee bent outward, as if placing a high outside foot. A simple vertical measuring tape runs up the wall with an orange marker at the height of the raised HEEL. Front three-quarter view.

PANEL 2: STRADDLE TEST. The same climber seated on the floor in a maximal straddle: legs straight and split as wide as possible, knees locked, feet upright and flat rather than rolled forward, torso upright. Viewed from directly above.

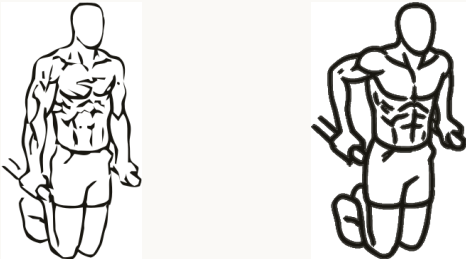
KEY ANGLES AND POSITIONS: Panel 1: the torso is VERTICAL, not leaning away from the wall. The raised hip is simultaneously flexed, abducted and externally rotated. Panel 2: both knees locked straight, feet vertical.

ANNOTATION: An orange horizontal marker line at the raised heel in panel 1 with an orange bracket down to the floor labelled 'measure here'. A faint greyed figure leaning away from the wall behind panel 1, marked with a small orange X. In panel 2, an orange bracket between the pubic bone and the floor line.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

PUSH-UPS / RING DIPS

BLOCK	SECONDARY / MODERATE
TRAINS	press, shoulder
APPEARS IN	CAVE 05 ADD-ON: 3 x 10 reps, rest 90s
LOAD	Bodyweight. Add a pack or a belt once three sets of twelve are easy.
PROGRESSION	Reps to 12 across all three sets, then move up the variation ladder before you add any weight. Range before load on a pressing pattern you have barely trained.



Stock frames show the bar dip. The ring turnout is drawn from the prompt.

SETUP

1. PUSH-UPS: hands under the shoulders, body one straight line from heels to crown.
2. RING PUSH-UPS: rings set just above floor height, shoulders packed down, body a rigid plank.
3. RING DIPS: rings at hip height. Support yourself with arms locked, shoulders pressed DOWN away from the ears, ribs down, legs straight and together.
4. Move up the variation ladder before you add any weight. Range before load on a pattern you have barely trained.

EXECUTION

1. Lower with the elbows tracking BACK at about 45 degrees from the ribs, never flared out to 90.
2. PUSH-UPS: chest to the floor. RINGS: chest to ring height.
3. DIPS: lower only to the point where the SHOULDER STAYS PACKED. A deep dip loads the front of the shoulder hard and the extra range is not worth having.
4. Press to full lockout. On rings, finish each rep by TURNING THE RINGS OUT until the palms face forward. That turnout is the stability work.
5. Ten reps, three sets, 90 seconds rest.

POSITION CHECKS

- One straight line from heels to crown on the push-up variants.
- Shoulders stay down and packed at the bottom of every dip.
- Rings turned out at the top of every ring rep.

COMMON MISTAKES

- Hips sagging on the push-up variants.
- Elbows flaring to 90 degrees.
- On dips, dropping below the point where the shoulder stays packed.
- Chasing the harder ring variation before the easier one is clean.

BREATHING

Inhale down, exhale on the press.

WHY

Climbing is two hours of pulling and zero pressing, and the shoulder pays for that imbalance long before the fingers do. This is the single most useful thing in the session even though it feels like the least climbing-specific.

Strength Endurance. Bridges maximal strength and route capacity. Kept to one session per week so it does not crowd out the maximal work that actually raises the ceiling.

Protocol: 8-12 reps. Rest 90-120s. 3-4. RPE 7-8. 1x per week.

Source: General resistance training literature applied to climbing.

VARIATION LADDER

L1	Push-Ups	
L2	Ring Push-Ups	
L4	Ring Dips	
L6	Weighted Ring Dips	Belt or pack, RPE 8

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: Two gymnastic rings on plain straps descending from off the top of the frame. Nothing else.

PANEL 1: RING SUPPORT, TOP. Climber supporting himself on two gymnastic rings at hip height, arms COMPLETELY straight and locked, shoulders pressed DOWN away from the ears, ribs down, body vertical, legs straight and together below him. The RINGS ARE TURNED OUT so the palms face forward. Side view.

PANEL 2: BOTTOM. The same figure lowered until the upper arms are roughly parallel to the floor and the chest is at ring height, elbows tracking BACK at about 45 degrees close to the ribs, shoulders still packed DOWN and not shrugged up toward the ears. He has not dropped deeper than this.

PANEL 3: CLOSE-UP OF THE TURNOUT: one hand gripping a ring at lockout with the palm rotated to face FORWARD, and beside it a greyed-out version with the palm facing inward marked with a small orange X.

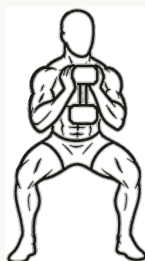
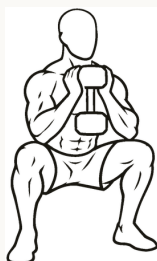
KEY ANGLES AND POSITIONS: Shoulders DEPRESSED in both panels 1 and 2, never shrugged. In panel 2 the upper arm is roughly horizontal and no lower: this is a deliberately limited depth. Elbows at 45 degrees from the ribs, not flared to 90.

ANNOTATION: An orange downward arrow at each shoulder in both panels marking depression. An orange rotational arc at the wrist in panel 1 showing the turnout. A dashed orange horizontal line at the bottom of panel 2 labelled 'stop here', marking the depth limit.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

GOBLET SQUAT

BLOCK	SECONDARY / MODERATE
TRAINS	legs
APPEARS IN	CAVE 05 ADD-ON: 2 x 10 reps, rest 90s
LOAD	KB 12kg at the chest. Bodyweight until the depth is honest.
PROGRESSION	Full depth with a flat back for two sets of twelve first. Then a 3s pause in the hole, then the heaviest bell in the gym. Legs recover fast - this is the one place in the programme you can be impatient.



SETUP

1. Feet shoulder width, toes turned slightly out.
2. Hold the kettlebell at the chest BY THE HORNS, elbows pointing down and tucked inside the knees. The bell at the chest is the counterweight that lets you stay upright.
3. Bodyweight only until the depth is honest.
4. Heels stiff: if the ankles are tight, park the heels on a 2cm plate.

EXECUTION

1. Big breath in at the top and brace the trunk.
2. Sit DOWN between the hips rather than back, keeping the chest tall.

3. Descend until the HIP CREASE PASSES BELOW THE KNEE. That is the depth, and it is not negotiable for the exercise to be worth doing.
4. Keep the heels planted flat the whole way down.
5. Drive up through the WHOLE FOOT, exhaling on the way up.
6. Ten reps, two sets, 90 seconds rest.
7. Progress: two clean sets of twelve at full depth, then a 3 second pause in the hole, then the heaviest bell in the gym. Legs recover fast, so this is the one place in the programme you can be impatient.

POSITION CHECKS

- Hip crease below the knee on every rep.
- Heels flat on the floor throughout.
- Knees tracking out over the toes, not collapsing inward.
- Lower back stays long, not rounded at the bottom.

COMMON MISTAKES

- Cutting depth to move weight, which is the whole reason the movement stops working.
- Heels lifting. Put them on a small plate rather than shortening the range.
- Knees collapsing inward.
- Rounding the lower back at the bottom.

BREATHING

Big breath in at the top, brace, exhale on the way up.

WHY

Climbers get strong at pulling and at one-legged rockovers and stay weak at loaded knee flexion. Two sets is a maintenance dose that costs you nothing on the wall tomorrow.

Strength Endurance. Bridges maximal strength and route capacity. Kept to one session per week so it does not crowd out the maximal work that actually raises the ceiling.

Protocol: 8-12 reps. Rest 90-120s. 3-4. RPE 7-8. 1x per week.

Source: General resistance training literature applied to climbing.

VARIATION LADDER

L1	Bodyweight Squat
L2	Goblet Squat
L4	Goblet Squat - 3s Pause
L6	Goblet Squat - Heavy

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A single plain Kettlebell drawn in simple outline. Nothing else.

PANEL 1: TOP. Climber standing upright, feet shoulder width with toes turned slightly out, holding a Kettlebell at chest height by its horns with both hands, elbows pointing down. Torso tall and vertical. Front three-quarter view.

PANEL 2: BOTTOM. The same figure in a deep squat with the HIP CREASE CLEARLY BELOW THE KNEE, heels FLAT on the floor, chest tall and upright, elbows tucked inside the knees, lower back long and not rounded, knees tracking out over the toes.

PANEL 3: FRONT VIEW. The same bottom position seen from directly in front, showing both knees tracking OUT over the second toes, hips level, feet flat.

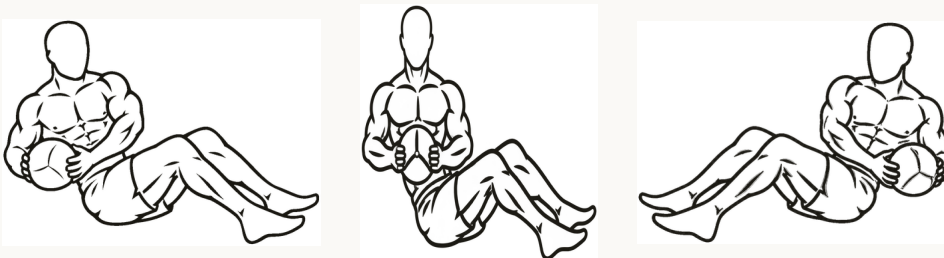
KEY ANGLES AND POSITIONS: Panel 2: the hip crease must be drawn visibly LOWER than the top of the knee, heels flat, torso close to vertical. Panel 3: Knees over toes, not caved inward.

ANNOTATION: A dashed orange horizontal line at the hip crease and a second at the top of the knee in panel 2, with a small orange arrow showing the hip line is below the knee line. An orange arrow pressing down through the whole foot. In panel 3, a faint greyed inward-collapsed knee drawn behind, marked with a small orange X.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

RUSSIAN TWIST

BLOCK	ACCESSORY / EASY
TRAINS	tension
APPEARS IN	CAVE 05 ADD-ON: 2 x 16 reps, rest 60s
LOAD	Bodyweight, then the 12kg bell once the rotation is clean
PROGRESSION	Feet down until 16 controlled reps are easy, then feet up, then hold the bell.



SETUP

1. Sit on the mat, knees bent, torso leaning back to roughly 45 degrees.
2. FEET ON THE FLOOR to start. Lift them only once the rotation is controlled.
3. Chest tall, LOWER BACK LONG rather than rounded. This posture is what makes the exercise work.
4. Hands together at the chest, or holding the bell once the movement is clean.

EXECUTION

1. Rotate the RIBCAGE to one side until the hands pass outside the hip.
2. Rotate to the other side. Left plus right is TWO reps.
3. Move slowly enough that the turn comes from the trunk and the arms are just along for the ride.
4. Sixteen reps, two sets, 60 seconds rest.

5. Progress: feet down until sixteen controlled reps are easy, then feet up, then hold the bell.

POSITION CHECKS

- The ribcage genuinely turns. If only the arms swing, nothing is being trained.
- The lower back stays long, not rounded and collapsed.
- Even count left and right.

COMMON MISTAKES

- Swinging the arms while the ribcage stays still, which trains nothing at all.
- Rounding the lower back and grinding the reps out fast. This should look controlled.
- Going heavy before the movement is clean.

BREATHING

Exhale on each turn, short and sharp.

WHY

The rest of the programme trains the trunk to resist rotation - front lever, Pallof, bird dog. This is the one place it produces rotation, which is what a drop-knee and a big cross-through actually ask for.

Body Tension Isometrics. Ten weeks of twice-weekly core training improved climbing-specific body-lift and lock-off tests by roughly 10-30% in advanced climbers. Isometric and dynamic work performed similarly, so this programme uses both: isometric for the front-lever line, dynamic for hip flexion.

Protocol: 8-15s holds. Rest 90s. 4-5. The hardest progression you can hold with a flat lower back. 2x per week.

Source: Saeterbakken et al., ten weeks dynamic or isometric core training in highly trained climbers, PLOS ONE (2018).

NOTE

Left plus right is 2 reps

VARIATION LADDER

L1	Russian Twist - Feet Down
L4	Russian Twist - Feet Up
L6	KB Russian Twist

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Optionally a small kettlebell held in both hands in panel 3. Nothing else.

PANEL 1: CENTRE. Climber seated on a mat, knees bent, feet flat on the floor, torso leaning back to about 45 degrees with the LOWER BACK LONG and not rounded, chest tall, both hands clasped together in front of the sternum. Front three-quarter view.

PANEL 2: ROTATED LEFT. The same seated position with the RIBCAGE turned to the left so the clasped hands have travelled past the outside of the left hip, shoulders turned with the ribcage, hips staying relatively square.

PANEL 3: ROTATED RIGHT, FEET UP. The harder variation: the same seated position with BOTH FEET LIFTED off the floor, shins roughly parallel to the ground, ribcage turned to the right and hands past the right hip.

KEY ANGLES AND POSITIONS: The torso leans back at about 45 degrees in all panels with the lumbar spine LONG, never collapsed into a C-shape. In panels 2 and 3 the shoulders and ribcage have clearly rotated together, not just the arms.

ANNOTATION: An orange curved arrow at the RIBCAGE, not at the hands, showing that the rotation originates in the trunk. A faint greyed version of arms swinging alone with a static ribcage, marked with a small orange X, drawn in the corner of panel 2.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

HANG

01 MAX HANGS - half-crimp - open hand

Maintain peak finger force. Eight working hangs, nothing else.

35 min | recovery 48h | crimp, openhand

Lowest-priority session in the programme for this athlete: peak finger force is already ahead of grade, so this maintains rather than builds. Do not run it at all while fingers or wrists are symptomatic. Eight hard hangs is the whole session either way.

02 CAPACITY - repeaters - critical force

Priority finger session: sustain force and recover between moves.

45 min | recovery 24h | forearm, crimp

Strong peak force with slow pump recovery means capacity, not max strength, is the finger quality worth training. This is the session that fixes that. Load is 55-65% of your max hang. At L5 the critical force block is added - if time is short, drop to four sets of repeaters rather than skipping it.

03 DENSITY - low load - high frequency

Ten minutes at 40%. Twice a day if you like. The cheapest strength on offer.

12 min | recovery 0h | crimp, openhand

Feet stay on the floor. Light strain only. This never counts as a hard finger day.

MAX HANG - THREE-FINGER DRAG 20MM

BLOCK	PRIMARY / MAX
TRAINS	openhands
APPEARS IN	HANG 01 MAX HANGS: 3 x 10s, rest 180s
LOAD	20mm three-finger drag, 85-90% of max
PROGRESSION	Same rule as half-crimp: five clean seconds on the last set before adding load.

NO STOCK ILLUSTRATION. No openly-licensed frame shows the three-finger drag. Generate from the prompt.

SETUP

1. Same 20mm edge, same board, same body position as the half-crimp hang.
2. INDEX, MIDDLE AND RING ONLY. The little finger stays off the edge. Using it changes the exercise.
3. Set the drag: fingers DRAPED over the edge, knuckles almost flat, all the finger joints close to straight. Only the base knuckles carry any flexion.
4. Thumb resting alongside, doing nothing.
5. Expect roughly 70 to 80 percent of your half-crimp load. It is a weaker position and that is normal.

EXECUTION

1. Load smoothly. Elbows slightly bent, shoulders packed down, body a rigid plank.
2. Hold ten seconds at 85 to 90 percent of your tested drag max.
3. RESIST THE URGE TO CURL. As fatigue arrives the hand wants to pull into a crimp. The moment it does, the set is over.
4. Step off deliberately. Rest three minutes. Three sets.
5. Progress: five clean seconds on the last set before adding any load.

POSITION CHECKS

- Three fingers on, little finger clearly off.
- The knuckle profile stays flat. Any curl and it has become a crimp hang.
- This grip is the one that keeps working when the crimp is sore, so train it honestly rather than as an afterthought.

COMMON MISTAKES

- Sliding into a crimp mid-hang, which is the whole failure mode of this exercise.
- Using the pinky, which changes the exercise and invalidates the comparison to your recorded drag numbers.
- Loading it like a half-crimp hang, which it will not support.

BREATHING

Steady through the hang.

WHY

This grip is the one that keeps working when the crimp is sore. Train it like it matters, because it does on real rock.

Max Hangs. Isometric finger strength on a small edge is the single strongest physical predictor of climbing performance in the literature. Sets are short and rest is long because the goal is peak force per set, not accumulated fatigue; three minutes allows near-complete phosphocreatine resynthesis.

Protocol: 10s hang on a 20mm edge. Rest 180s between sets (range 120-300s). 4-6 per grip. 85-90% of tested 10s max load. 2x per week, never on consecutive days.

Source: Lattice Training hangboarding guides; Baláš et al., physical performance testing in climbing (Front Sports Act Living 2023); Winkler/Levernier finger strength vs. grade datasets.

NOTE

Index, middle and ring. Open position, no thumb. Expect roughly 70-80% of your half-crimp load.

VARIATION LADDER

L1	Four-Finger Open Hand 20mm
L4	Three-Finger Drag 20mm
L6	Three-Finger Drag 20mm - Weighted

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard in simple outline with a clearly drawn narrow 20mm edge. Nothing else.

PANEL 1: FULL BODY HANG. Climber hanging from a wall-mounted hangboard on a 20mm edge, arms nearly straight with a slight elbow bend, shoulders packed down, body straight and still. Side-on three-quarter view.

PANEL 2: THREE-FINGER DRAG, CORRECT. A large anatomical close-up of ONE HAND in profile on a 20mm edge. The INDEX, MIDDLE and RING fingers are DRAPED over the edge with all their joints close to STRAIGHT, only the base knuckles flexed, so the hand makes a smooth shallow curve rather than a step. The LITTLE FINGER is clearly lifted OFF the edge. The thumb hangs free alongside.

PANEL 3: CURLED INTO A CRIMP, INCORRECT. The same close-up, but the fingers have curled so the middle knuckles are bent toward 90 degrees, turning the drag into a crimp. This is the failure mode.

KEY ANGLES AND POSITIONS: Panel 2: all finger joints close to extended, the knuckle line nearly flat, little finger visibly raised clear of the edge. Panel 3 differs from panel 2 ONLY in that the fingers have curled.

ANNOTATION: An orange arrow pointing at the raised little finger in panel 2, marking it as deliberately off. An orange tick in the corner of panel 2 and an orange X in the corner of panel 3.

NOTE: Draw the hands very large and anatomically exact, four fingers plus thumb, correct joint count. The distinction between panels 2 and 3 is the entire point of the drawing.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

CRITICAL FORCE BLOCK

BLOCK	SECONDARY / HARD
TRAINS	forearm
APPEARS IN	HANG 02 CAPACITY: 1 x 240s, rest 300s
LOAD	At or just above critical force, roughly 60-70% of max
PROGRESSION	Retest critical force every 6-8 weeks and reset the training load from it.
FIXED	Protocol taken as written. Never scaled by level or energy.

NO STOCK ILLUSTRATION. No openly-licensed frame exists. Generate from the prompt.

SETUP

1. Same edge and same grip as your repeaters.
2. Load at or just above your measured critical force, roughly 60 to 70 percent of max. If you have not tested it, start conservatively.
3. Five clear minutes with nothing to interrupt you. Timer visible.
4. This protocol is FIXED. Brutal and boring, and both of those are by design.

EXECUTION

1. Seven seconds on, three seconds off, CONTINUOUSLY for four minutes. Twenty-four pulls with no set breaks.

2. Settle into a rhythm in the first thirty seconds. The pace you set early decides whether you finish.
3. Keep the effort EVEN. Do not reduce effort in the middle to survive to the end: that produces a number that means nothing and a stimulus that trains nothing.
4. The last minute should be at the edge of failure without falling off.
5. Rest five full minutes between blocks. Two to three blocks.
6. Retest critical force every 6 to 8 weeks and reset the training load from it.

POSITION CHECKS

- A steady, sustainable rhythm rather than a fast start and a collapse.
- Breathing settled early. Panic breathing burns the set.
- This is the closest thing on a board to being pumped mid-route and having to keep going.

COMMON MISTAKES

- Starting too heavy and blowing up at two minutes.
- Quietly reducing effort in the middle to survive to the end.
- Treating it as a max session. The load is moderate on purpose; the duration is what hurts.

BREATHING

Settle into a rhythm early. Panic breathing burns the set.

WHY

This is the closest thing on a board to being pumped mid-route and having to keep going. Train the thing you keep failing on.

Critical Force Intervals. Critical force is the highest intermittent force a forearm can sustain without accumulating fatigue, and is the metric Lattice uses for forearm endurance. For a route climber whose limit is 7a+, sustained capacity at moderate force is usually a larger limiter than peak force. Training at critical force is the most time-efficient way to raise it.

Protocol: 7s on / 3s off continuously for 4 minutes (24 pulls). Rest 300s between blocks. 2-3. At or just above measured critical force, roughly 60-70% of max. 1x per week.

Source: Lattice 4-minute all-out finger-flexor critical force test; Giles et al., validation of the 4-min all-out test in sport climbers (2024).

NOTE

7s on / 3s off continuously for 4 minutes. 24 pulls. Brutal and boring.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard with a visible 20mm edge in panel 1. Panel 2 is a pure diagram with no equipment.

PANEL 1: Climber hanging from a wall-mounted hangboard on a 20mm edge in a half-crimp, arms slightly bent, shoulders packed, body straight, showing visible effort and controlled breathing but good position. Side view.

PANEL 2: A simple diagrammatic TIMELINE drawn in ink and burnt orange: a horizontal bar four minutes long, divided into 24 alternating segments, the longer segments labelled 7s and the shorter ones 3s, with a burnt-orange horizontal dashed line running the full length of the bar labelled 'critical force'. No human figure in this panel.

KEY ANGLES AND POSITIONS: Panel 1: half-crimp with middle knuckles near 90 degrees and fingertips flat, elbows slightly bent, shoulders depressed, body still.

ANNOTATION: In panel 2, an orange bracket spanning the whole bar labelled '4 minutes, 24 pulls'. In panel 1, a small orange arrow at the ribcage indicating steady rhythmic breathing.

NOTE: One figure panel and one diagram panel. Keep the diagram flat, simple and schematic, in the same ink and orange as everything else.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

DENSITY HANGS

BLOCK	PRIMARY / EASY
TRAINS	crimp, openhand
APPEARS IN	HANG 03 DENSITY: 10 x 10s, rest 50s
LOAD	About 40% of max - light strain, feet on the ground
PROGRESSION	Do not chase load. Progress by doing it more often: one block daily, then two blocks at least 6h apart. Load only creeps up as bodyweight support reduces naturally.
FIXED	Protocol taken as written. Never scaled by level or energy.

NO STOCK ILLUSTRATION. No openly-licensed frame shows a feet-on density hang. Generate from the prompt.

SETUP

1. Any edge, anywhere: home doorway board, the gym, a hangboard by the kitchen.
2. Stand UNDER the edge with BOTH FEET ON THE FLOOR. The feet stay there.
3. Take only PART of your weight through the fingers, about 40 percent of max. Light strain only.
4. This protocol is FIXED, and the whole design depends on it staying light.

EXECUTION

1. Ten seconds on, fifty seconds off. Ten times. Ten minutes total, one hundred seconds of actual hanging.
2. Alternate grips across the ten hangs: half-crimp, open hand, three-finger drag, and two-finger pockets if they are healthy.
3. It should feel ALMOST TOO EASY. That is the protocol working as designed, not you slacking.
4. You should be able to hold a conversation throughout.
5. Progress by FREQUENCY, not load: one block daily, then two blocks at least six hours apart. Load only creeps up as bodyweight support naturally reduces.

POSITION CHECKS

- Feet on the floor for all ten hangs.

- Breathing completely normal. If you are bracing, it is too heavy.
- No pump, no fatigue, nothing sore the next morning.

COMMON MISTAKES

- Turning it into a max session because it feels too light. This is the one thing that breaks the protocol.
- Doing it twice a day with less than six hours between blocks.
- Chasing load. Frequency is the progression variable here, not weight.

BREATHING

Completely normal. Have a conversation through it.

WHY

It should feel almost too easy. That is the protocol working as designed, not you slacking.

Density Hangs (Abrahams). Retrospective analysis of logged training found low-intensity, frequent finger loading produced strength gains comparable to max hangs, and gains were additive when both were run together. Consistent with tendon loading work showing collagen synthesis is maximised by short bouts spaced about 6h apart rather than by long single sessions. Low enough load that it does not interfere with climbing or maximal training.

Protocol: 10 x 10s hang, 50s rest - 10 minutes total. Rest 50s between hangs. 1 block, repeatable twice daily at least 6h apart. About 40% of max - feet on the ground, light strain only. Daily, up to 2x per day.

Source: Effects of Different Loading Programs on Finger Strength in Rock Climbers, Sports Medicine - Open (2024) - retrospective Crimpd app cohort, not an RCT; Baar, tendon loading and collagen synthesis.

NOTE

Alternate grips across the ten hangs: half-crimp, open hand, three-finger drag, two-finger pockets if healthy.

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain wall-mounted hangboard in simple outline and a plain floor line. Panel 2 is a pure diagram.

PANEL 1: Climber standing beneath a wall-mounted hangboard with BOTH FEET FLAT ON THE FLOOR, knees slightly bent, hands on a small edge, arms nearly straight, body upright and completely relaxed. He is taking only part of his weight through his hands. His expression and posture are calm and unstrained. Side view, the loaded feet clearly visible.

PANEL 2: A simple diagrammatic TIMELINE in ink and burnt orange: a horizontal bar ten minutes long divided into 10 short segments labelled 10s separated by long segments labelled 50s, with the orange label 'x10' beneath. No human figure.

KEY ANGLES AND POSITIONS: THE FEET ARE FLAT ON THE FLOOR AND OBVIOUSLY CARRYING WEIGHT. This is the defining feature of the exercise and the thing that separates it from every other hang in the book. Posture relaxed and upright, shoulders not strained.

ANNOTATION: An orange '40%' label with a downward arrow through the arms, and a second orange arrow at the feet labelled '60%', showing how the load is split. No strain lines anywhere on the figure.

NOTE: The mood of panel 1 must read as EASY and casual, in deliberate contrast to the max hang illustration. That contrast is instructional.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

MORN

01 ABS + OBLIQUES - daily - 12 min

Trunk under load at the one hour of the day when nothing else is competing for it.

12 min | recovery 0h | tension, mobility

Floor to standing: the first exercise asks you to lie down, the last leaves you upright. Low load by design, so it runs on climbing days too. Alternate with 02 rather than doing this one seven days a week.

02 SLOW START - daily - 10 min

For the mornings you have nothing. Undoes the night and the wall, and still counts.

10 min | recovery 0h | mobility, tension

The low day. Run this after a hard climbing session, on a bad night of sleep, or any morning 01 feels like too much. Mobility first, trunk second, fingers lightest of all.

90/90 BREATHING RESET

BLOCK	WARMUP / TECHNIQUE
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 40s, rest 15s
LOAD	No load
PROGRESSION	Never progressed. Add the heel press once the ribs drop without effort.
FIXED	Protocol taken as written. Never scaled by level or energy.

NO STOCK ILLUSTRATION. No openly-licensed frame shows the 90/90 breathing position. Generate from the prompt.

SETUP

1. On your back on the mat. Hips and knees BOTH at 90 degrees: feet flat on a wall, on a chair seat, or on the floor with the knees up.
2. Arms at your sides, palms up.
3. Press the lower back FLAT into the mat and keep it there for the whole exercise.
4. The only rule for this session: get on the mat.

EXECUTION

1. INHALE through the nose for FOUR counts. Send the air into the SIDES AND BACK of the ribcage, not into the belly. Feel the ribs widen sideways.
2. EXHALE through pursed lips for SIX TO EIGHT counts, as if blowing out a candle slowly. Keep going until the ribs pull DOWN and the abdominal wall switches on by itself.
3. Pause for a beat at the bottom of the exhale, empty, before the next inhale.
4. Roughly four breath cycles, about forty seconds.
5. HEEL PRESS VARIATION: push the heels into the wall at about 20 percent effort throughout.
6. BAND VARIATION: hold a band overhead under light tension and pull it apart a little on each exhale.

POSITION CHECKS

- The lower back stays flat on the mat for every breath, including the inhale.
- If the lower ribs flare toward the ceiling, the exhale was too short.
- By breath three the abdominal wall should tighten without you consciously bracing it. That is the switch this exercise exists to flip.

COMMON MISTAKES

- Belly breathing instead of rib breathing.
- Lower back arching off the mat.
- Rushing the exhale, which is the half that does the work.
- Skipping it because it feels too easy, which loses the switch that makes the next eight exercises work.

BREATHING

This exercise is the breathing. Four seconds in through the nose, six to eight out through the mouth.

WHY

This is not a stretch and not a rest. The long exhale drops the ribcage and turns on the deep abdominal wall, which is the position every other exercise here is built on. If the lower ribs flare toward the ceiling, the exhale was too short.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the

twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

NOTE

The only rule: get on the mat

VARIATION LADDER

L1	90/90 Breathing Reset
L4	90/90 Breathing + Heel Press
L6	90/90 Breathing + Band Pull-Apart

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat and a plain low chair. Nothing else.

PANEL 1: INHALE. Climber lying on his back on a mat, hips and knees BOTH bent to exactly 90 degrees with the lower legs resting horizontally on a plain low chair seat. Arms relaxed at the sides, palms up. The lower back is pressed FLAT against the mat with no gap. The ribcage is shown WIDENING SIDEWAYS. Side view.

PANEL 2: EXHALE. The same position, but the ribcage has visibly drawn DOWN and IN toward the hips, the lower ribs flattened, the abdominal wall drawn tight. The lower back is still flat on the mat.

PANEL 3: INCORRECT. The same position but with the lower ribs FLARED UP toward the ceiling and a clear GAP between the lower back and the mat.

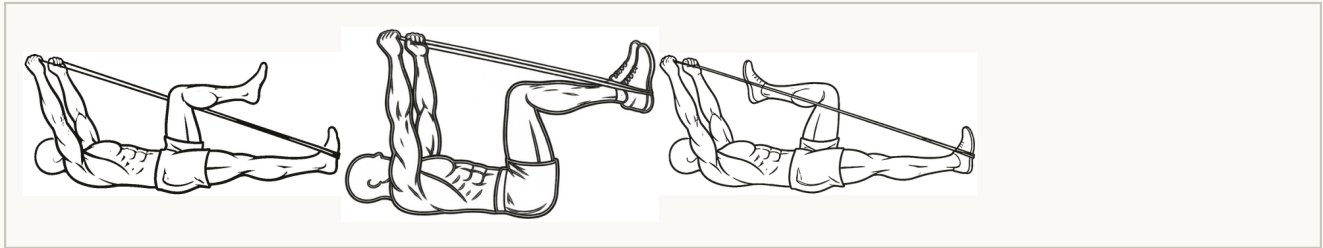
KEY ANGLES AND POSITIONS: Hips and knees at EXACTLY 90 degrees in all three panels. In panels 1 and 2 the lumbar spine touches the mat along its whole length with no gap. Panel 3 shows the gap and the rib flare, and differs in nothing else.

ANNOTATION: In panel 1, orange arrows pointing OUTWARD at both sides of the ribcage with the label '4 counts in, nose'. In panel 2, orange arrows pointing DOWN and IN at the ribcage with the label '6 to 8 counts out, pursed lips'. In panel 3, an orange X and a small orange arc highlighting the gap under the lower back.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BAND DEAD BUG

BLOCK	PREHAB / EASY
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 16 reps, rest 20s MORN 02 SLOW START as Slow Dead Bug: 1 x 14 reps, rest 15s
ALSO CALLED	Slow Dead Bug
LOAD	Light band arch to hand, or none until the back stays flat
PROGRESSION	Hold a flat back for all 16 first. Then add band tension, not reps.



SETUP

1. On your back, arms straight up over the chest, hips and knees at 90 degrees.
2. Press the lower back FLAT into the mat. This contact is the exercise.
3. BAND VERSION: loop the band around both arches and hold one end in each hand, arms vertical, band under light tension. The band loads both directions so the return is working too.
4. SLOW VERSION: no band at all. The tempo is the load, roughly two seconds out and two seconds back.

EXECUTION

1. Extend ONE LEG long and low while the OPPOSITE ARM reaches back overhead.
2. Go only as far as the lower back stays glued down. That limit is different for everyone and it changes day to day.
3. Return under control. Do not let a band snap the limbs back: resist it.
4. Alternate sides. Left plus right is two reps.
5. Sixteen reps in circuit 01 with the band, fourteen slow reps in circuit 02 without.

POSITION CHECKS

- Imagine a strip of paper under your lower back that someone is trying to pull out. Your job for the whole set is to trap it.
- Opposite arm and opposite leg, every rep.
- If you have to hold your breath, shorten the range.

COMMON MISTAKES

- Lower back arching as the leg lowers. The moment the paper slips free, you went too far.
- Arm and leg on the same side.
- Racing the reps, which is especially wrong on the slow version where the tempo IS the load.
- Letting the band snap the limbs back instead of resisting it.

BREATHING

Exhale slowly as the limbs extend, inhale as they return. If you have to hold your breath, shorten the range.

WHY

Imagine a strip of paper under your lower back that someone is trying to pull out. Your job for the whole set is to trap it. The moment it slips free, you went too far.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

NOTE

Alternating - left plus right is 2 reps

VARIATION LADDER

L1	Dead Bug
L4	Band Dead Bug
L6	Band Dead Bug, 3s Hold

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout.

Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Optionally a light band looped around both arches with the ends held in the hands. Nothing else.

PANEL 1: START. Climber on his back on a mat, both arms reaching straight UP toward the ceiling over the chest, hips and knees BOTH at 90 degrees so the shins are horizontal. Lower back pressed FLAT into the mat. Side view.

PANEL 2: EXTENDED. The same figure with the RIGHT leg extended long and low, hovering just above the mat, and the LEFT arm reaching back overhead toward the floor behind the head. The other arm and leg remain in the start position. THE LOWER BACK IS STILL FLAT against the mat.

PANEL 3: INCORRECT. The same extended position but with the lower back ARCHED and a clear gap beneath it, the ribcage flared. The limbs have reached further than the trunk can control.

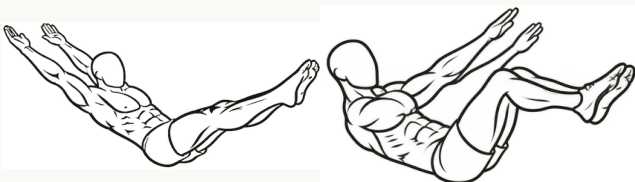
KEY ANGLES AND POSITIONS: Opposite arm and opposite leg in panels 2 and 3, never the same side. In panel 2 the lumbar spine touches the mat along its whole length. Panel 3 differs only by the arch and the gap.

ANNOTATION: A small orange horizontal strip drawn under the lower back in panels 1 and 2 labelled 'trap this', shown pinned flat. In panel 3 the same strip is shown slipping out through the gap, with an orange X in the panel corner.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

HOLLOW BODY HOLD

BLOCK	PREHAB / MODERATE
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 2 x 25s, rest 25s
LOAD	Longest lever you can hold with the lower back glued down
PROGRESSION	Two clean 25s holds, then lengthen the lever: knees out, then arms overhead, then rocks.



SETUP

1. On your back on the mat.
2. Press the lower back into the mat and SQUEEZE THE RIBS DOWN toward the hips. Both, before you lift anything.
3. TUCKED: knees drawn to the chest, hands reaching past the knees, shoulder blades just off the mat.
4. FULL: arms extended overhead by the ears, legs straight and held low.

EXECUTION

1. Hold the shape. The shape is the entire exercise.
2. The longer the arms and legs, the harder the lever. LENGTHEN ONLY AS FAR AS THE LOWER BACK STAYS GLUED DOWN.
3. Twenty-five seconds, two sets, twenty-five seconds rest.
4. ROCKS VARIATION: keep the identical rigid shape and rock from the shoulders to the hips using the CURVE OF THE BACK. Never bend in the middle. The shape does not change; it just travels.
5. Progress: two clean 25 second holds, then lengthen the lever. Knees out, then arms overhead, then rocks.

POSITION CHECKS

- No gap under the lower back at any point.
- Chin slightly tucked but not jammed to the chest, neck long.
- If the lower back lifts on the inhale, bend the knees more.

COMMON MISTAKES

- Lower back arching, which loses the whole exercise.
- Chin jammed to the chest.
- Arms drifting down toward the hips to compensate, which shortens the lever without admitting it.
- Holding to failure on day one and dreading it tomorrow. This is a daily routine.

BREATHING

Short controlled breaths through the mouth. If the lower back lifts on the inhale, bend the knees more.

WHY

The shape is the exercise. This loads the whole abdominal sheet from ribcage to pubic bone at once, which is the part a sit-up misses entirely.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

VARIATION LADDER

L1	Hollow Body Hold - Tucked
L4	Hollow Body Hold
L6	Hollow Body Rocks

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: TUCKED. Climber on his back on a mat, knees drawn up toward the chest, hands reaching forward past the outside of the knees, shoulder blades lifted just off the mat, chin slightly tucked, lower back pressed FLAT into the mat. Side view.

PANEL 2: FULL. The same climber with the arms extended straight back overhead beside the ears and the legs straight and lowered to hover about 15cm above the mat. The body forms one long shallow banana curve with the lower back still pressed flat against the mat.

PANEL 3: ROCKS. The same FULL shape shown mid-rock, tipped slightly so more weight is on the upper back, with the body shape completely UNCHANGED, rocking on the curve of the back like a rocking chair.

KEY ANGLES AND POSITIONS: In ALL THREE panels the lumbar spine touches the mat and the rib cage is drawn down. The body shape in panel 3 is IDENTICAL to panel 2, only its angle on the mat differs: it must not bend in the middle.

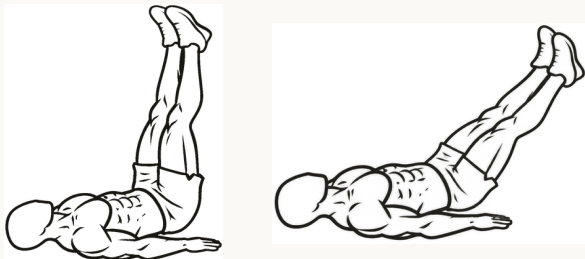
ANNOTATION: A small orange arrow pointing at the lower back where it meets the mat in panels 1 and 2, with a cross-mark showing no gap. A curved orange double-headed rocking arrow beneath the body in panel 3 showing the direction of travel.

NOTE: A progression ladder read left to right: tucked, full, rocks.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BAND LEG LOWERS

BLOCK	PREHAB / MODERATE
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 12 reps, rest 25s
LOAD	Band over both arches, ends held at the chest
PROGRESSION	Straighten the legs before you slow the tempo. Then take five seconds on the lowering.



SETUP

1. On your back, legs up toward the ceiling.
2. Hands flat UNDER THE GLUTES at first. Move them to your sides once the position holds without help.
3. Band over both arches with the ends held at the chest, so it resists the legs coming down.
4. Lower back flat before the first rep.

EXECUTION

1. Lower BOTH LEGS slowly toward the floor.
2. STOP THE INSTANT the lower back starts to arch. That point is your range today.
3. Pause a beat at the bottom of your range.
4. Return to vertical, exhaling hard on the way up. The exhale keeps the ribs down.

5. Twelve reps. Bend the knees to shorten the lever if the straight-leg version breaks the position.
6. Progress: STRAIGHTEN THE LEGS BEFORE YOU SLOW THE TEMPO. Then take five seconds on the lowering.

POSITION CHECKS

- The lower back never leaves the mat.
- The legs move under control both directions. No bouncing at the bottom.
- The head stays down and the neck relaxed.

COMMON MISTAKES

- Arching the lower back to gain range you cannot control.
- Yanking the legs back up with the hip flexors instead of controlling them with the trunk.
- Bouncing at the bottom.
- Straining the neck instead of leaving the head down on the mat.

BREATHING

Inhale as the legs lower, exhale hard as they return. The exhale keeps the ribs down.

WHY

This is the lower abdominal exercise. The bottom segments respond to holding a posterior pelvic tilt under load, which is exactly what you are defending here.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

VARIATION LADDER

L1	Bent-Knee Leg Lowers
L4	Band Leg Lowers
L6	Band Leg Lowers, 5s Lowering

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat and a resistance band drawn as two clean parallel lines from the arches to the chest. Nothing else.

PANEL 1: TOP. Climber on his back on a mat with both legs straight and raised VERTICAL toward the ceiling, a band looped over both arches with its ends held in both hands at the chest, hands flat under the glutes or at the sides, lower back pressed flat. Side view.

PANEL 2: BOTTOM. The same figure with both straight legs LOWERED to about 30 degrees above the floor, band stretched and under tension. The LOWER BACK IS STILL FLAT against the mat, ribs down.

PANEL 3: INCORRECT. The legs lowered further, almost to the floor, but with the lower back ARCHED clear of the mat and the ribs flared. This is what going past your range looks like.

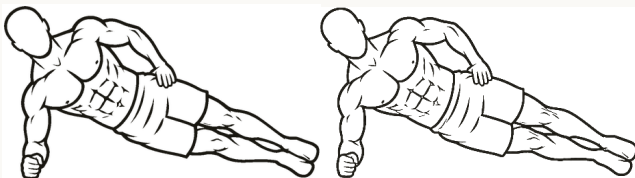
KEY ANGLES AND POSITIONS: Legs straight and knees locked in all panels. In panels 1 and 2 the lumbar spine touches the mat. Panel 3 differs only by the arch and the gap.

ANNOTATION: An orange downward arrow tracing the path of the feet from panel 1 to panel 2. A small orange bracket marking the lower-back contact in panel 2 and the same bracket showing a GAP in panel 3, with an orange X in the corner.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

SIDE PLANK REACH-THROUGH

BLOCK	PREHAB / MODERATE
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 12 reps per side, rest 25s
LOAD	Hips high for every rep
PROGRESSION	Knees down until 12 clean reps, then straight legs, then hold the band in the top hand.



Stock frames show the side plank base. The thread-through rotation is drawn from the prompt.

SETUP

1. On your side, ELBOW DIRECTLY UNDER THE SHOULDER, forearm flat and pointing forward.
2. EASY: knees bent and stacked. FULL: legs straight, feet stacked or staggered.
3. Lift the hips into ONE STRAIGHT LINE. Top arm reaching to the ceiling.
4. Set the hips high before the first rep. They stay there.

EXECUTION

1. Rotate the TORSO and thread the top arm DOWN AND THROUGH the gap under your body, reaching as far behind you as you can.
2. The rotation comes from turning the RIBCAGE against a fixed pelvis. That is where the obliques live.
3. Rotate back and reach the arm to the ceiling. That is one rep.
4. Twelve reps on one side, then switch. Hips stay high throughout.

5. Progress: knees down until twelve clean reps, then straight legs, then hold the band in the top hand.

POSITION CHECKS

- The hips do not dip between reps. If they do, drop to knees down.
- The elbow stays under the shoulder, not creeping forward.
- The ribcage turns, not just the arm.

COMMON MISTAKES

- Hips sagging between reps, which turns this into a shoulder exercise.
- Elbow ahead of the shoulder instead of under it.
- Threading with the arm alone while the ribcage stays still.
- Uneven reps left versus right.

BREATHING

Exhale as you thread through, inhale as you reach up.

WHY

The rotation is the point. Obliques run diagonally and only load fully when the ribcage turns against a fixed pelvis. Keep the hips high throughout, or this becomes a shoulder exercise.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

VARIATION LADDER

L1	Side Plank Reach-Through - Knees Down
L4	Side Plank Reach-Through
L6	Band Side Plank Reach-Through

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: OPEN. Climber in a side plank on one forearm, elbow DIRECTLY under the shoulder, legs straight and feet stacked, hips lifted so the body is one straight line from feet to head. The top arm reaches straight UP toward the ceiling, chest open and facing forward. Front three-quarter view.

PANEL 2: THREADED. The same side plank with the hips STILL HIGH, but the torso has rotated so the top arm has threaded DOWN and THROUGH the gap beneath the body, reaching far behind. The ribcage has visibly turned while the hips stay square and lifted.

PANEL 3: INCORRECT. The same threading position but with the HIPS SAGGING toward the floor, the body no longer a straight line.

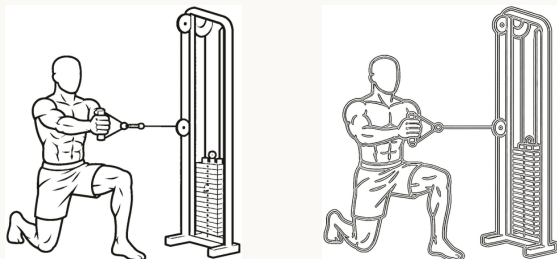
KEY ANGLES AND POSITIONS: The supporting elbow is DIRECTLY under the shoulder in all panels. In panels 1 and 2 the body is one straight line with the hips fully lifted. In panel 2 the rotation is clearly in the RIBCAGE and shoulders, not just the arm. Panel 3 differs only by the sagging hips.

ANNOTATION: A dashed orange straight line through the body in panels 1 and 2 showing the plank. An orange curved arrow at the RIBCAGE in panel 2 showing where the rotation originates. An orange X in the corner of panel 3 with a small downward arrow at the dropped hip.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

HALF-KNEELING BAND PRESS-OUT

BLOCK	PREHAB / EASY
TRAINS	tension, shoulder
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 10 reps per side, rest 25s
LOAD	Band trapped under the down knee - enough tension to feel the pull
PROGRESSION	Add a 3s hold at full extension, then move to tall kneeling, then press overhead.



SETUP

1. Half-kneeling: one knee down on the mat, the other foot forward and flat. Set the front foot far enough forward to give you a real base.
2. TRAP ONE END OF THE BAND under the down knee, or under the front foot on that same side. Other end in both hands at the sternum.
3. The band now pulls DIAGONALLY DOWN AND ACROSS and will try to twist and bend you toward it.
4. Squeeze the DOWN-SIDE GLUTE and stand tall through the spine.

EXECUTION

1. Press both hands STRAIGHT OUT from the chest to full arm extension.
2. The band gets harder as the hands travel, so the hardest point is full extension.
3. Hold a beat at full extension.

4. Return the hands to the sternum under control.
5. Ten reps, then swap knees AND band side.
6. Progress: add a 3 second hold at full extension, then move to tall kneeling, then press overhead.

POSITION CHECKS

- The ribcage does not turn and the torso does not tip toward the anchor.
- The hips stay SQUARE. They must not shift sideways to compensate.
- You are not trying to twist. You are refusing to.

COMMON MISTAKES

- Rotating or side-bending toward the band, which loses the entire point.
- A band too light to feel, which makes the whole set decorative.
- Front foot too close to the knee, killing the base.
- Hips shifting instead of staying square.

BREATHING

Exhale as you press out, inhale as you return. The tension lives in the trunk, not in a held breath.

WHY

Anti-rotation. You are not trying to twist, you are refusing to. This is what an oblique actually does on the wall when you are fighting a barn-door, and it builds the deep waist tension flexion work never will.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

VARIATION LADDER

L1	Half-Kneeling Band Press-Out
L4	Half-Kneeling Band Press-Out, 3s Hold
L6	Tall-Kneeling Band Press-Out + Overhead

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat and a resistance band drawn as two clean parallel lines running from beneath the down knee to the hands. Nothing else.

PANEL 1: START. Climber in a half-kneeling position on a mat: right knee down, left foot forward and flat, torso tall and vertical. A band runs from under the RIGHT KNEE diagonally up to both hands held together at the sternum. Front three-quarter view.

PANEL 2: PRESSED OUT. The same half-kneeling position with both arms extended STRAIGHT forward from the sternum, band stretched taut and under obvious tension. The torso is still perfectly UPRIGHT and SQUARE, with no twist toward the band and no side-bend.

PANEL 3: INCORRECT. The same pressed-out position but with the torso visibly TWISTED and TIPPED toward the band anchor, one shoulder dropped, hips shifted sideways.

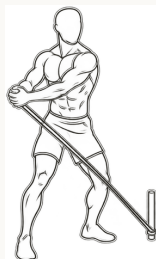
KEY ANGLES AND POSITIONS: Front knee and down knee both at roughly 90 degrees. Torso VERTICAL and square in panels 1 and 2, hips level. Panel 3 differs only by the twist, the tip and the shifted hips.

ANNOTATION: An orange straight arrow showing the hands travelling forward in panel 2. A separate orange curved arrow showing the band's rotational pull, crossed with a small orange X to mark that this rotation must NOT happen. An orange X in the corner of panel 3.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

STANDING BAND CHOP

BLOCK	PREHAB / EASY
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 12 reps per side, rest 20s
LOAD	Standing on one end, tension on from the first rep
PROGRESSION	Add a 2s hold at the top, then move to a split stance so the trunk does more of the work.



SETUP

1. Stand on ONE END of the band with the left foot, feet shoulder width.
2. Other end in both hands, arms fairly straight, hands down by the LEFT HIP, band already under tension before the first rep.
3. Soft knees, tall chest.
4. This is the first standing exercise in the session, by design. You are getting up.

EXECUTION

1. Pull the band DIAGONALLY UP AND ACROSS the body.
2. Finish with the hands HIGH OUTSIDE THE RIGHT SHOULDER and the ribcage rotating to follow.
3. Drive the rotation from the RIBS and the front-side obliques. Your arms are a rope; your trunk is the winch.

4. Control it back down to the left hip. Do not let the band snap you down.
5. Twelve reps on one diagonal, then stand on the other end and reverse.
6. Progress: add a 2 second hold at the top, then move to a split stance so the trunk does more of the work.

POSITION CHECKS

- The torso genuinely rotates. If the arms travel and the chest stays still, nothing is training.
- No leaning back to get the hands higher.
- Equal counts left and right.

COMMON MISTAKES

- Chopping with the arms while the torso stays still.
- Leaning back to gain height, which loads the lower back instead of the obliques.
- Letting the band snap you down on the return.
- Uneven counts left versus right, which is how obliques get built asymmetrically.

BREATHING

Exhale sharply on the way up and across, inhale on the controlled return.

WHY

Drive the rotation from the ribs and the front-side obliques, not from the arms and never from the lower back. Your arms are a rope, your trunk is the winch.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

VARIATION LADDER

L1	Standing Band Chop
L4	Standing Band Chop, 2s Hold
L6	Split-Stance Band Chop

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain resistance band drawn as two clean parallel lines from beneath the left foot to the hands. Nothing else.

PANEL 1: START LOW. Climber standing with feet shoulder width, the LEFT FOOT standing on one end of a resistance band. Both hands hold the other end together, arms fairly straight, hands down beside the LEFT HIP, band already under tension. Knees soft, chest tall. Front three-quarter view.

PANEL 2: MID CHOP. The same figure with the hands travelling diagonally up across the body to roughly chest height, the RIBCAGE beginning to rotate to the right, band stretched.

PANEL 3: FINISH HIGH. The same figure with the hands finished HIGH outside the RIGHT SHOULDER, arms extended, the ribcage clearly rotated to the right, torso still upright with NO backward lean, both feet flat on the floor.

KEY ANGLES AND POSITIONS: The spine stays UPRIGHT in all three panels: no backward lean at the finish. The rotation is visible in the RIBCAGE and shoulders, not just the arms. Both feet stay flat on the floor.

ANNOTATION: A long orange diagonal arrow tracing the path of the hands from the left hip to outside the right shoulder, spanning all three panels. A separate small orange curved arrow at the RIBCAGE in panel 3 marking where the rotation comes from.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

PULL EDGE NO-HANG

BLOCK	PREHAB / EASY
TRAINS	openhands, crimp
APPEARS IN	MORN 01 ABS + OBLIQUES: 1 x 10s per side, rest 25s MORN 02 SLOW START as Pull Edge No-Hang - Open Hand: 1 x 10s per side, rest 20s
ALSO CALLED	Pull Edge No-Hang - Open Hand
LOAD	About 40-50% of max - you set the load through the leg
PROGRESSION	Frequency before load. Daily for a month, then a second block later in the day at least 6h apart, before you ever pull harder than half.

NO STOCK ILLUSTRATION. No openly-licensed frame shows a no-hang device. This is the highest-risk movement in MORN and the illustration matters. Generate from the prompt.

SETUP

1. Loop the sling around ONE FOOT and sit or stand with that leg out in front. Edge in one hand.
2. BECAUSE THE LEG SETS THE LOAD, you decide exactly how hard this is. That is what makes it safe first thing in the morning.
3. OPEN HAND: joints mostly extended, only the base knuckle flexed. HALF-CRIMP: middle joints near 90 degrees, fingertip joints still slightly curved, THUMB OFF the index finger.
4. Elbow slightly bent, shoulder packed down.

EXECUTION

1. BUILD TENSION GRADUALLY over the first two seconds by pressing down through the foot. NEVER SNATCH AT IT.
2. Hold ten seconds at roughly 40 to 50 percent of what you could pull. In circuit 02 the open-hand version is lighter still, 30 to 40 percent: circulation, not strength.
3. Release over a second. Swap hands.

4. One primer set per hand. This is NOT a density block: the full 10 x 10s protocol is HANG 03.
5. ROLL VARIATION: slowly curl the fingers from open hand toward half-crimp and back once during the hold, moving through the range under gentle load.
6. Progress by FREQUENCY, not load. Daily for a month, then a second block at least six hours later, before you ever pull harder than half.

POSITION CHECKS

- Breathing completely relaxed for the full ten seconds. If you are bracing, back the load off.
- Thumb clearly OFF the index finger. A wrapped thumb makes it a full crimp.
- Nothing clicks, pinches or feels sharp.

COMMON MISTAKES

- Going near maximal on cold tendons. Finger tendons are stiffest and least hydrated on waking, and a hard morning pull is how climbers collect pulley injuries.
- Snatching the load on rather than building it over two seconds.
- Thumb wrapped over the index finger.
- Continuing through any click, pinch or sharp point. Stop and skip it for the day: skipping costs nothing in this app.
- Doing it first, before anything else is warm. It goes last, after the trunk work.

BREATHING

Relaxed and normal for the full ten seconds. If you are holding your breath, back the load off.

WHY

This is the one exercise here where more is not better. Finger tendons are cold, stiff and least hydrated on waking. A light daily pull keeps the tissue stimulated and the grip position sharp. A hard one is how climbers collect pulley injuries. Max hangs belong in HANG mode, later in the day. Density Hangs (Abrahangs). Retrospective analysis of logged training found low-intensity, frequent finger loading produced strength gains comparable to max hangs, and gains were additive when both were run together. Consistent with tendon loading work showing collagen synthesis is maximised by short bouts spaced about 6h apart rather than by long single sessions. Low enough load that it does not interfere with climbing or maximal training.

Protocol: 10 x 10s hang, 50s rest - 10 minutes total. Rest 50s between hangs. 1 block, repeatable twice daily at least 6h apart. About 40% of max - feet on the ground, light strain only. Daily, up to 2x per day.

Source: Effects of Different Loading Programs on Finger Strength in Rock Climbers, Sports Medicine - Open (2024) - retrospective Crimpd app cohort, not an RCT; Baar, tendon loading and collagen synthesis.

NOTE

Per hand - one primer set, not a density block. The full 10x10s protocol is HANG 03.

VARIATION LADDER

L1	Pull Edge No-Hang - Open Hand
L4	Pull Edge No-Hang - Half-Crimp
L6	Pull Edge No-Hang - Half-Crimp, Heavier

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain webbing sling, a small flat rectangular pull edge, a plain exercise mat. Nothing else.

PANEL 1: SETUP. Climber seated upright on a mat with one leg extended straight forward. A sling loops around the arch of that extended FOOT and runs up to a small flat pull edge held in one hand at about waist height. The elbow is slightly bent and the shoulder is packed down. He is relaxed and upright, clearly not straining. Side view.

PANEL 2: OPEN HAND, CORRECT. A large anatomical close-up of the hand on the small edge: the fingers DRAPED with all their joints close to STRAIGHT, only the base knuckles flexed, the thumb hanging free and clearly NOT touching the index finger.

PANEL 3: HALF-CRIMP, CORRECT. The same close-up but with the middle knuckles bent to roughly 90 degrees and the fingertip joints slightly curved. The THUMB IS STILL OFF, resting alongside the hand and not over the index finger.

PANEL 4: FULL CRIMP, INCORRECT. The same close-up with the THUMB WRAPPED OVER THE TOP of the index fingernail, locking the grip. This position must not be used here.

KEY ANGLES AND POSITIONS: Panel 1 must make it obvious that the LEG, not bodyweight, sets the load: the foot presses down against the sling and the climber's posture is relaxed. Panels 2, 3 and 4 differ only in finger and thumb position.

ANNOTATION: An orange arrow at the foot in panel 1 pressing DOWN, labelled 'leg sets the load', and an orange '40-50°' label beside the arm. An orange tick in the corners of panels 2 and 3, an orange X in the corner of panel 4, with a small orange arrow pointing at the thumb in all three close-ups.

NOTE: Draw the hands very large and anatomically exact: four fingers plus thumb, correct joint count, no fused or warped digits. The thumb position is the whole instructional point of panels 2 to 4.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

STOMACH VACUUM

BLOCK	PREHAB / TECHNIQUE
TRAINS	tension
APPEARS IN	MORN 01 ABS + OBLIQUES: 2 x 15s, rest 15s
LOAD	No load - empty lungs are the whole difficulty
PROGRESSION	Supine until 15s is comfortable, then on all fours, then standing. Gravity helps less each time.

NO STOCK ILLUSTRATION. No openly-licensed frame shows a stomach vacuum. Generate from the prompt.

SETUP

1. SUPINE: on your back, knees bent, feet flat. QUADRUPED: hands and knees, flat back.
STANDING: feet shoulder width, hands on the thighs.
2. Each position removes a little help from gravity, which is why they unlock in that order.
3. Best done fasted, before breakfast. That is genuinely the easiest time to get a full contraction.

EXECUTION

1. EXHALE EVERY LAST BIT OF AIR. Empty the lungs completely. This has to happen first: you cannot draw the abdominal wall in around a full ribcage.
2. With the lungs EMPTY and WITHOUT BREATHING IN, pull the navel UP AND BACK toward the spine as hard as you can.
3. Hold for fifteen seconds.
4. Release and breathe normally before the second set.
5. Two sets, fifteen seconds each, fifteen seconds rest.

6. Progress: supine until fifteen seconds is comfortable, then quadruped, then standing.

POSITION CHECKS

- The hold happens on EMPTY lungs. No air held in.
- Nothing shortens. Everything draws INWARD. This is not a crunch.
- Come out of it the moment you need to breathe.

COMMON MISTAKES

- Sucking in without exhaling first, which cannot work around a full ribcage.
- Holding until lightheaded.
- Doing it on a full stomach.
- Confusing it with a crunch. The trunk does not flex at all here.

BREATHING

The hold happens on empty lungs. Full exhale, then hold with no air. Come out of it the moment you need to breathe.

WHY

This trains the transverse abdominis, the deep corset layer under the visible abs. It does nothing for the segments themselves, but it is what draws the whole midsection in and makes the obliques read as a defined line rather than a soft edge. Fasted, on waking, is genuinely the easiest time to get a full contraction.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

NOTE

Best done before breakfast

VARIATION LADDER

L1	Stomach Vacuum - Supine
L4	Stomach Vacuum - Quadruped
L6	Stomach Vacuum - Standing

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: FULL EXHALE. Climber on hands and knees on a mat with a flat neutral back, having just blown all the air out: the ribcage is drawn down, the abdomen relaxed and hanging slightly. Side view, close framing on the torso.

PANEL 2: VACUUM. The same quadruped position with the lungs still EMPTY and the abdomen drawn sharply UP AND IN toward the spine, creating a deep visible hollow under the ribcage. The back stays FLAT and neutral: the spine does not round or arch.

PANEL 3: PROGRESSION POSITIONS. Three small figures side by side at reduced scale: one lying supine with knees bent, one on hands and knees, one standing with hands on the thighs, each showing the same drawn-in abdomen.

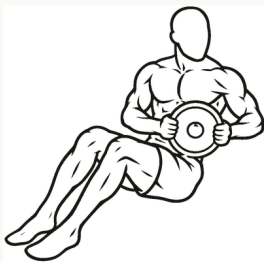
KEY ANGLES AND POSITIONS: The spine stays NEUTRAL and flat in panels 1 and 2: the difference between them is ONLY the abdominal wall drawing inward, never spinal flexion. Do not draw a rounded cat-stretch back.

ANNOTATION: Orange arrows at the abdomen in panel 2 pointing UP and IN toward the spine. An orange 'lungs empty' label with a small crossed-out breath symbol in both panels 1 and 2. Small orange numbers 1, 2, 3 above the three figures in panel 3 marking the unlock order.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

SUPINE SPINAL TWIST

BLOCK	WARMUP / TECHNIQUE
TRAINS	mobility
APPEARS IN	MORN 02 SLOW START: 1 x 20s per side, rest 15s
LOAD	No load
PROGRESSION	Never progressed. Add the active opposite-arm reach once the shoulder stays down on its own.
FIXED	Protocol taken as written. Never scaled by level or energy.



SETUP

1. On your back, arms out wide in a T, palms up.
2. Draw both knees toward the chest.
3. Let the knees fall together to one side while keeping BOTH SHOULDER BLADES ON THE FLOOR.

EXECUTION

1. Rest and breathe for twenty seconds, letting the knees sink a little lower on each exhale.
2. The point is NOT slamming the knees to the floor. It is keeping the far shoulder pinned while the pelvis rotates away from it. That is where the rotation actually happens.
3. Switch sides.
4. REACH VARIATION: actively press the opposite arm along the floor away from the knees, which pulls the stretch further up the obliques and lats.

5. Four or five slow breaths per side.

POSITION CHECKS

- Both shoulder blades stay in contact with the floor.
- The knees sink on the exhale rather than being forced.
- Both sides get equal time.

COMMON MISTAKES

- Letting the top shoulder peel off the mat, which turns it into a shrug and removes the stretch.
- Forcing the knees down.
- Skipping the second side because the first got comfortable.

BREATHING

Slow nasal inhales, long exhales. Four or five breaths per side.

WHY

The point is not slamming the knees to the floor. It is keeping the far shoulder pinned while the pelvis rotates away from it. That is where the rotation actually happens.

Hip and Shoulder Mobility. Combined hip abduction and external rotation is what makes high steps, drop knees and bridging available, and it is measured in climbing assessment batteries via foot-raise height and straddle depth. Cheap to train and it costs no recovery.

Protocol: 45-60s per position, loaded end-range where possible. Rest 15s. 2 rounds. Comfortable end range. 3x per week.

Source: Draper et al., flexibility as a determinant of climbing performance; climbing mobility assessment protocols.

VARIATION LADDER

- | | |
|----|----------------------|
| L1 | Supine Spinal Twist |
| L4 | Supine Twist + Reach |

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 2 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: Climber lying on his back on a mat, arms stretched out WIDE in a T with palms up, both knees bent and fallen together to the LEFT side so they rest toward the floor. BOTH SHOULDER BLADES REMAIN FLAT ON THE MAT, the head turned gently to the right. Viewed from directly above.

PANEL 2: INCORRECT. The same position but with the RIGHT shoulder blade PEELED UP off the mat as the knees pull it over, so the twist has been lost.

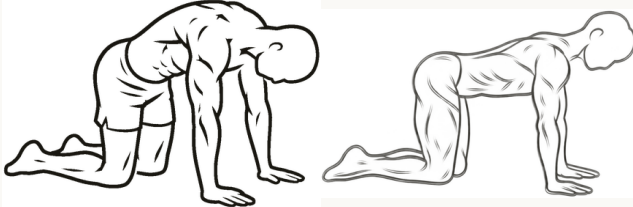
KEY ANGLES AND POSITIONS: Panel 1: both shoulder blades in full contact with the mat, knees stacked and fallen to one side, head turned away from the knees. Panel 2 differs ONLY in the lifted shoulder.

ANNOTATION: An orange downward arrow pressing on the far shoulder in panel 1 with the label 'keep this down'. An orange curved arrow at the pelvis showing it rotating away. An orange X in the corner of panel 2 with a small arrow at the lifted shoulder.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

CAT-COW + THREAD THE NEEDLE

BLOCK	MOBILITY / TECHNIQUE
TRAINS	mobility, shoulder
APPEARS IN	MORN 02 SLOW START: 1 x 12 reps, rest 15s
LOAD	No load
PROGRESSION	Add the overhead reach on the unwind once the thread reaches the mat without forcing.



Stock frames show cat and cow. The thread-through is drawn from the prompt.

SETUP

1. On hands and knees. Hands under the shoulders, knees under the hips.
2. Spine long and neutral to start.
3. Eight cat-cow, then four threads alternating.

EXECUTION

1. COW: let the belly drop, lift the chest and the tailbone. Inhale into it.
2. CAT: press the floor away, round the whole spine, tuck the tailbone. Exhale into it.
3. MOVE ONE VERTEBRA AT A TIME. Move slowly enough to feel each one arrive. Eight full cycles.
4. THREAD THE NEEDLE: from neutral, thread one arm UNDER the body and across until the shoulder and the side of the head rest on the mat.
5. Unwind and reach that same arm up to the ceiling, opening the chest. Alternate sides. Four threads total.

POSITION CHECKS

- The spine moves segment by segment, not as one rigid block.
- The hands stay under the shoulders and do not drift forward.
- The neck follows the spine rather than being cranked.

COMMON MISTAKES

- Moving the spine as one rigid block.
- Cranking the neck at the ends of the range.
- Hands drifting ahead of the shoulders.
- Speeding up. This is the wake-up, not the workout.

BREATHING

Inhale into cow, exhale into cat. Exhale as you thread through.

WHY

This is the segment that gets your spine out of the shape it held all night and out of the shape climbing put it in. Move slowly enough to feel each vertebra arrive.

Hip and Shoulder Mobility. Combined hip abduction and external rotation is what makes high steps, drop knees and bridging available, and it is measured in climbing assessment batteries via foot-raise height and straddle depth. Cheap to train and it costs no recovery.

Protocol: 45-60s per position, loaded end-range where possible. Rest 15s. 2 rounds. Comfortable end range. 3x per week.

Source: Draper et al., flexibility as a determinant of climbing performance; climbing mobility assessment protocols.

NOTE

Eight cat-cow, then four threads alternating

VARIATION LADDER

- L1 Cat-Cow + Thread the Needle
- L4 Cat-Cow + Thread the Needle with Reach

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 4 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3, 4 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: COW. Climber on hands and knees on a mat, hands under shoulders and knees under hips, with the belly dropped, the chest lifted and the tailbone tilted up, so the spine forms a smooth downward curve. Side view.

PANEL 2: CAT. The same quadruped position with the whole spine ROUNDED upward, tailbone tucked under, head released down, forming a smooth upward arc.

PANEL 3: THREAD THE NEEDLE. The same quadruped base, but one arm has threaded UNDER the body and across to the opposite side, so that shoulder and the side of the head rest on the mat, the torso rotated. The other hand stays planted.

PANEL 4: UNWIND. The same base with that arm now reaching straight UP toward the ceiling, chest opened and rotated upward, gaze following the hand.

KEY ANGLES AND POSITIONS: Hands stay under the shoulders and knees under the hips in ALL FOUR panels. In panels 1 and 2 the spinal curve is smooth and continuous, not hinged at one point. The neck follows the line of the spine, never cranked.

ANNOTATION: Orange curved arrows along the spine in panels 1 and 2 showing the direction of the curl. An orange arrow showing the threading path of the arm in panel 3 and the upward reach in panel 4.

NOTE: Four stages of one flow, read left to right. Same figure, same scale.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

90/90 HIP SWITCH

BLOCK	MOBILITY / TECHNIQUE
TRAINS	mobility, legs
APPEARS IN	MORN 02 SLOW START: 1 x 12 reps, rest 15s
LOAD	No load, no hands
PROGRESSION	Lose the hands first, then pause with the knees off the floor mid-rotation.

NO STOCK ILLUSTRATION. No openly-licensed frame shows the 90/90 hip switch. Generate from the prompt.

SETUP

1. Sit with BOTH KNEES BENT AT 90 DEGREES: one leg in front, one out to the side, both shins flat on the floor.
2. Sit TALL. If the lower back rounds, sit up on a folded edge of the mat.
3. Hands light on the floor for balance to start. Lose them as soon as you can.

EXECUTION

1. WITHOUT PUSHING OFF THE HANDS, lift both knees and rotate them through the middle to the mirror-image position on the other side.
2. Slow and controlled. The middle of the rotation is the part that matters and the part everyone rushes.

3. Left plus right is two reps. Twelve reps.
4. LIFT VARIATION: pause with the knees off the floor mid-rotation.
5. Progress: lose the hands first, then add the mid-rotation pause.

POSITION CHECKS

- The lower back stays long and the chest stays tall.
- Both knees stay at 90 degrees throughout.
- The hands are not doing the work.

COMMON MISTAKES

- Pushing off the hands.
- Rounding the lower back to compensate for tight hips. Sit on a folded mat edge instead.
- Rushing the middle rather than controlling it.

BREATHING

Exhale through the rotation, inhale in the settled position.

WHY

Climbers lose internal hip rotation faster than almost any other range, and losing it is what stops you getting your hip into the wall. Three minutes a day beats an hour of stretching once a month.

Hip and Shoulder Mobility. Combined hip abduction and external rotation is what makes high steps, drop knees and bridging available, and it is measured in climbing assessment batteries via foot-raise height and straddle depth. Cheap to train and it costs no recovery.

Protocol: 45-60s per position, loaded end-range where possible. Rest 15s. 2 rounds. Comfortable end range. 3x per week.

Source: Draper et al., flexibility as a determinant of climbing performance; climbing mobility assessment protocols.

NOTE

Left plus right is 2 reps

VARIATION LADDER

L1	90/90 Hip Switch
L4	90/90 Hip Switch + Lift

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: RIGHT SIDE. Climber seated on a mat with BOTH knees bent to 90 degrees: the right leg folded in FRONT with the shin across the body, the left leg folded OUT TO THE SIDE, both shins flat on the floor. Sitting TALL with a long flat lower back, hands resting lightly on the floor. Three-quarter view from above and in front.

PANEL 2: MID ROTATION. The same figure with BOTH KNEES LIFTED off the floor, passing through the middle of the rotation, shins in the air, torso still tall and square, hands NOT pushing off the floor.

PANEL 3: LEFT SIDE. The mirror image of panel 1: the left leg folded in front, the right leg out to the side, both shins flat, sitting tall.

KEY ANGLES AND POSITIONS: BOTH knees at 90 degrees in every panel. The lumbar spine stays LONG and upright in all three, never rounded. In panel 2 the hands are clearly clear of the floor or only lightly touching.

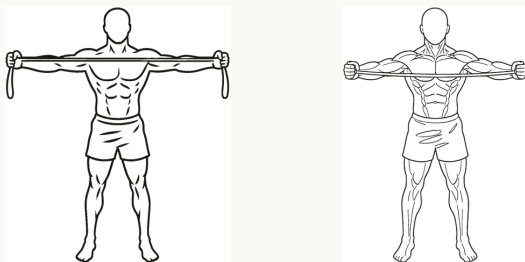
ANNOTATION: A large orange curved arrow sweeping across all three panels showing the knees rotating through the middle from right to left. A small orange arrow pointing at the hands in panel 2 labelled 'no push'.

NOTE: Three stages of one continuous rotation, read left to right.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BAND PULL-APART + OVERHEAD PASS

BLOCK	PREHAB / EASY
TRAINS	shoulder
APPEARS IN	MORN 02 SLOW START: 1 x 15 reps, rest 45s
LOAD	Light band, wide grip - never to failure
PROGRESSION	Narrow the grip before you use a heavier band. Full dislocates only when the overhead pass is painless.



SETUP

1. Standing tall, band in both hands IN FRONT at shoulder height, arms straight.
2. Hands wide enough apart that there is already light tension before the first rep.
3. Ribs down, glutes lightly on. This stops you arching to fake range.

EXECUTION

1. Pull the band APART until the hands are wide and the band touches the chest, SQUEEZING THE SHOULDER BLADES together at the end.
2. Return under control.
3. OVERHEAD PASS: continue the hands UP AND BACK over the head as far as the shoulders allow, then reverse the same path. Widen the grip if anything pinches.
4. Fifteen reps. Never to failure.

5. Progress: NARROW THE GRIP before you use a heavier band. Full dislocates only when the overhead pass is painless.

POSITION CHECKS

- The shoulder blades do the work, not the hands.
- The lower back does not arch on the overhead pass.
- Arms stay straight. Bending the elbows is cheating the pull.

COMMON MISTAKES

- Arching the lower back to fake overhead range.
- Shrugging toward the ears.
- Grip too narrow for your current mobility.
- Bending the elbows to cheat the pull.

BREATHING

Exhale on the pull-apart, inhale on the return.

WHY

Climbing and sitting pull you into the same rounded, internally rotated shoulder. This is the direct antidote and it costs 40 seconds. The shoulder blades do the work, not the hands.

Antagonist and Prehab. Upper-extremity injuries account for the large majority of climbing injuries. Climbing loads the finger and wrist flexors and internal rotators almost exclusively; wrist extensor, supinator/pronator and external rotator work is the standard prevention for medial elbow tendinopathy and shoulder impingement in climbers. This is maintenance, not training - it must never compete with the session that matters.

Protocol: 12-20 reps, controlled tempo. Rest 45-60s. 2-3. Light - never to failure. 2-3x per week.

Source: Climbing injury epidemiology and prevention reviews; medial epicondylalgia rehab protocols for climbers.

VARIATION LADDER

L1 Band Pull-Apart

L4 Band Pull-Apart + Overhead Pass

L6 Band Pull-Apart + Full Dislocate

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain resistance band drawn as two clean parallel lines. Nothing else.

PANEL 1: START. Climber standing tall, holding a resistance band in both hands out in FRONT at shoulder height, arms completely STRAIGHT, hands roughly shoulder width apart with the band under light tension. Ribs down, torso vertical. Front view.

PANEL 2: PULLED APART. The same figure with the arms straight and pulled WIDE apart so the band touches the chest, shoulder blades squeezed together, shoulders DOWN and not shrugged.

PANEL 3: OVERHEAD PASS. The same figure with the arms straight and the band having travelled UP and BACK over the head, hands now behind the head line, torso still VERTICAL with the ribs down and no lower-back arch.

KEY ANGLES AND POSITIONS: Arms COMPLETELY STRAIGHT in all three panels: any elbow bend is the fault this drawing exists to prevent. Torso vertical in all three, especially panel 3, where the temptation is to arch.

ANNOTATION: Orange outward arrows at both hands in panel 2. A long orange arc tracing the hands' path up and over the head in panel 3. A small orange downward arrow at each shoulder marking that they stay depressed. A faint greyed arched-back figure behind panel 3 marked with a small orange X.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

BIRD DOG

BLOCK	PREHAB / EASY
TRAINS	tension
APPEARS IN	MORN 02 SLOW START: 1 x 12 reps, rest 20s
LOAD	Hips level for every rep
PROGRESSION	Add a 3s hold at full extension, then draw elbow to knee underneath before extending again.



SETUP

1. Hands and knees, spine neutral. Hands under shoulders, knees under hips.
2. IMAGINE BALANCING A GLASS OF WATER ON YOUR LOWER BACK that must not spill for the whole set.
3. Head in line with the spine, gaze down at the mat.

EXECUTION

1. Extend ONE ARM forward and the OPPOSITE LEG back until both are level with the torso. No higher.
2. Hold briefly at full extension.
3. Return WITHOUT TOUCHING DOWN, and switch sides.
4. Twelve reps, slow and alternating.

5. Progress: add a 3 second hold at full extension, then draw elbow to knee underneath before extending again.

POSITION CHECKS

- The hips stay LEVEL. They must not rotate open as the leg lifts.
- The leg stops at hip height. Higher means the back has arched.
- The moving limbs are a distraction. The exercise is the trunk refusing to rotate or tip while they move.

COMMON MISTAKES

- Hips rotating open as the leg lifts.
- Kicking the leg above hip height and arching the back.
- Reaching too fast.
- Head lifting out of line with the spine.

BREATHING

Exhale as you extend, inhale as you return.

WHY

The moving limbs are a distraction. The exercise is the trunk refusing to rotate or tip while they move, which is the same demand as the press-out, on the floor, while you are still half asleep.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

NOTE

Alternating - slow

VARIATION LADDER

L1	Bird Dog
L4	Bird Dog, 3s Hold
L6	Bird Dog + Elbow-to-Knee

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Optionally a small glass of water balanced on the lower back in panel 2, upright and unspilled. Nothing else.

PANEL 1: START. Climber on hands and knees on a mat, hands under shoulders, knees under hips, spine long and NEUTRAL, head in line with the spine, gaze down. Side view.

PANEL 2: EXTENDED. The same figure with the RIGHT arm extended straight forward and the LEFT leg extended straight back, BOTH exactly LEVEL with the torso, forming one long straight line from the extended fingertips through the spine to the extended heel. The lower back stays flat and the hips stay level.

PANEL 3: INCORRECT. The same extension but with the leg KICKED UP ABOVE hip height, the lower back visibly ARCHED, and the hips rotated open so one side is higher than the other.

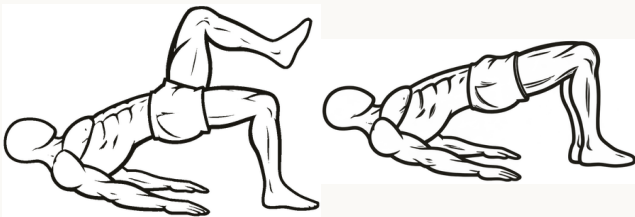
KEY ANGLES AND POSITIONS: In panel 2 the extended arm and leg are HORIZONTAL, level with the torso, no higher. The spine is a straight neutral line. Panel 3 differs by the raised leg, the arched back and the open hip.

ANNOTATION: A dashed orange straight line running from the extended fingertips through the spine to the extended heel in panel 2. A small orange level-bubble or level-line across the hips in panel 2 showing them square. An orange X in the corner of panel 3, with the glass of water shown tipping and spilling.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

GLUTE BRIDGE MARCH

BLOCK	PREHAB / EASY
TRAINS	legs, tension
APPEARS IN	MORN 02 SLOW START: 1 x 16 reps, rest 20s
LOAD	Bodyweight, or a band above the knees
PROGRESSION	Hips dead level for all 16 first, then add the band above the knees.



SETUP

1. On your back, feet hip-width and flat, heels about 30cm from the glutes.
2. Arms at the sides.
3. Band above the knees for the loaded version.

EXECUTION

1. Drive through the HEELS and lift the hips to a straight line from knees to shoulders. Finish with the GLUTES, not by hyperextending the lower back.
2. From there, lift ONE FOOT a few centimetres without letting the hips drop or tilt.
3. Replace it, then lift the other. Left plus right is two reps.
4. HIPS STAY HIGH FOR THE WHOLE SET. Sixteen reps.
5. Progress: hips dead level for all sixteen first, then add the band above the knees.

POSITION CHECKS

- The hips do not drop the moment a foot lifts.
- The pelvis stays level side to side, not tilting toward the lifted leg.
- Drive through the heels, not the toes.

COMMON MISTAKES

- Hips dropping the moment a foot lifts.
- Rushing.
- Pushing through the toes instead of the heels.
- Hyperextending the lower back at the top instead of finishing with the glutes.

BREATHING

Steady at the top, exhale on each foot lift.

WHY

A night of sleep and a day of sitting both switch the glutes off. Marching means the pelvis has to stay level on one leg, which is the same demand as a high step on the wall.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

NOTE

Left plus right is 2 reps

VARIATION LADDER

L1 Glute Bridge Hold

L4 Glute Bridge March

L6 Band Glute Bridge March

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat and a light band above the knees. Nothing else.

PANEL 1: BRIDGE. Climber lying on his back on a mat with both feet flat and hip-width, heels close to the glutes, hips LIFTED so the body forms a straight line from the knees through the hips to the shoulders. Arms flat at the sides. A light band sits above the knees. Side view.

PANEL 2: MARCH. The same bridged position with ONE foot lifted a few centimetres off the mat, knee still bent at 90 degrees. The HIPS REMAIN AT EXACTLY THE SAME HEIGHT as panel 1 and the pelvis is level, not tilted.

PANEL 3: FRONT VIEW. The same marching position seen from the feet end, showing both hip bones LEVEL with each other despite one foot being lifted.

KEY ANGLES AND POSITIONS: The hip height is IDENTICAL in panels 1 and 2: that is the checkpoint. The line from knee to shoulder is straight, with no lower-back hyperextension. In panel 3 the pelvis is level.

ANNOTATION: A dashed orange horizontal line at hip height drawn across BOTH panels 1 and 2 at the same level, proving the hips do not drop. An orange arrow pressing down through the supporting HEEL. In panel 3, a short orange level-line across both hip bones.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

SIDE PLANK HOLD

BLOCK	PREHAB / EASY
TRAINS	tension
APPEARS IN	MORN 02 SLOW START: 1 x 25s per side, rest 20s
LOAD	Bottom shoulder packed, hips high
PROGRESSION	Knees down until 25s is easy, then straight legs, then lift and hold the top leg.



SETUP

1. On your side, ELBOW UNDER THE SHOULDER, forearm pointing forward.
2. EASY: knees bent and stacked. FULL: legs straight, feet stacked or staggered.
3. Lift the hips into one straight line. Top hand on the hip or reaching up.

EXECUTION

1. Hold. Push the floor away with the forearm and LIFT THE BOTTOM RIBS toward the ceiling.
2. Twenty-five seconds. Full time on one side, then the other.
3. ADVANCED: lift and hold the TOP LEG, adding hip abduction on top of the trunk hold.
4. Progress: knees down until 25 seconds is easy, then straight legs, then the top leg lift.

POSITION CHECKS

- The bottom shoulder does not sink. If it does, you are hanging on the joint rather than holding with the trunk.
- You are on a true side position, not rolled forward or back.
- Breathing slow and even. If you cannot breathe evenly, drop to knees down.

COMMON MISTAKES

- Hips sagging.
- Rolling forward or back off the true side position.
- Bottom shoulder collapsing.
- Forty seconds on the strong side and fifteen on the weak one. Match them.

BREATHING

Slow and even. If you cannot breathe evenly, drop to knees down.

WHY

Push the floor away with the forearm and lift the bottom ribs toward the ceiling. If the bottom shoulder sinks you are hanging on the joint rather than holding with the trunk.

Daily Minimum-Dose Trunk. This is the frequency counterpart to the twice-weekly tension work, not a replacement for it. Trunk endurance responds to short, submaximal, frequently repeated bouts, and because no set approaches failure the session carries no recovery cost and needs no long rests. The side bridge and bird dog here are two thirds of the standard core-endurance battery, chosen because they load the trunk without loading the spine in flexion. Hard progressions belong in the twice-weekly session; what this one buys is the habit and the accumulated hours.

Protocol: 10-25s holds, or 10-16 controlled reps. Rest 15-25s. Short on purpose: no set here goes near failure, so there is nothing to recover from between them.. 1-2. Easy to moderate. Nothing in this protocol is a working max.. Daily, climbing days included.

Source: McGill, core endurance training and the side bridge / bird dog standards; Saeterbakken et al. (2018) for the twice-weekly loading this supplements.

VARIATION LADDER

L1	Side Plank Hold - Knees Down
L4	Side Plank Hold
L6	Side Plank Hold + Top Leg Raise

ILLUSTRATION PROMPT

Technical instructional line drawing for a climbing training manual. Dark charcoal ink (#181610) line art on a flat warm parchment background (#E4E2DD). Strictly two colours plus one accent: a single burnt-orange (#E64D19) used ONLY for motion arrows, angle marks and callout numbers, never on the body. No shading, no gradients, no cross-hatching, no textures, no background scenery, no gym interior, no floor shadow. Uniform 2px stroke weight throughout. Anatomically correct adult male climber, lean and light, short dark hair, plain fitted shorts and a plain vest, barefoot unless the panel says otherwise. The whole body inside the frame with a clear margin on every side. Flat orthographic projection, no perspective distortion, no foreshortening of the working limb. Calm, precise and diagrammatic: a coach's manual, not a fitness advert.

COMPOSITION: 3 panels side by side in a single horizontal strip, equal width, thin 1px ink rule between panels, each panel numbered 1, 2, 3 in burnt orange in its top-left corner. The same figure at the same scale in every panel.

EQUIPMENT: A plain exercise mat. Nothing else.

PANEL 1: KNEES DOWN. Climber in a side plank on one forearm with the elbow DIRECTLY under the shoulder, knees bent and stacked, hips lifted so the body is a straight line from knees to head. Top hand on the hip. Front three-quarter view.

PANEL 2: FULL. The same side plank with the legs STRAIGHT and the feet stacked, hips lifted so the body is one straight line from feet to head, the bottom shoulder pushed actively AWAY from the floor so it does not sink.

PANEL 3: INCORRECT. The same full position but with the HIPS SAGGING toward the floor and the bottom shoulder COLLAPSED so the head appears sunk toward it.

KEY ANGLES AND POSITIONS: The supporting elbow is DIRECTLY under the shoulder in all panels. In panels 1 and 2 the body is a straight line and there is a clear gap between the ear and the bottom shoulder. Panel 3 differs by the sag and the collapsed shoulder.

ANNOTATION: A dashed orange straight line through the body in panels 1 and 2. An orange arrow at the bottom shoulder pointing AWAY from the floor in panel 2, and a small orange bracket marking the ear-to-shoulder gap. An orange X in the corner of panel 3.

NOTE: Panels 1 and 2 are a progression, panel 3 is the common failure. Same figure and scale throughout.

NEGATIVE PROMPT: photorealism, 3D render, gym background, mirrors, weight racks, colour photography, dramatic lighting, drop shadows, muscle striation shading, watermark, logo, brand name, text labels other than the numbers specified, extra limbs, warped hands, fused fingers, wrong number of fingers, distorted joints, impossible elbow or knee angles, cropped hands or feet, bodybuilder mass, exaggerated musculature

INDEX

Every name the app uses, alphabetically. Where a session calls a movement something else, the entry points at the page it is written up on.

90/90 Breathing Reset	83	Max Hang - Three-Finger Drag 20mm	77
90/90 Hip Switch	103	Pike Push-Up / Band Overhead Press	25
Band Dead Bug	84	Progressive Finger Loading	44
Band External Rotation	15	Pull Edge No-Hang	96
Band Leg Lowers	88	Pull Edge No-Hang - Open Hand (see Pull Edge No-Hang)	96
Band Pallof Press	12	Pull-Up Strength Endurance	59
Band Pull-Apart + Overhead Pass	105	Pulse Raiser + Shoulder Prep	43
Bird Dog	107	Push-Ups	27
Bulgarian Split Squat	29	Push-Ups / Ring Dips	69
Campus Ladders	50	Quadruped Wrist Rocks	39
Cat-Cow + Thread the Needle	102	Recruitment Pulls	56
Copenhagen Plank	13	Repeaters 7:3	58
Critical Force Block	78	Ring Toe-Hook Hold	53
Cuff Circuit - External Rotation + Y-T-W	33	Russian Twist	73
Dead Hang	22	Side Plank Hold	111
Dead Hang - Grip Rotation (see Dead Hang)	22	Side Plank Reach-Through	90
Density Hangs	80	Single-Leg Romanian Deadlift	31
Explosive Pull-Ups	51	Slow Dead Bug (see Band Dead Bug)	84
Front Lever Progression	6	Standing Band Chop	94
Glute Bridge March	109	Stomach Vacuum	98
Goblet Squat	71	Supine Spinal Twist	100
Half-Kneeling Band Press-Out	92	TEST - 90 Degree Lock-Off	65
Hanging Leg Raise	8	TEST - Foot Raise + Straddle	68
Hip Mobility Flow	37	TEST - Front Lever Hold	66
Hollow Body Hold	86	TEST - Max Hang 20mm Half-Crimp	62
Hollow Hold / Arch Hold	10	TEST - Weighted Pull-Up 5RM	63
Inverted Row	20	TRX Y-T-W	54
KB Overhead Press	60	Warm-Up Flow	5
Limit Bouldering	48	Weighted Pull-Ups	17
Lock-Off Ladder	19	Wrist Conditioning	35
Max Hang - Half-Crimp 20mm	46	Wrist Extensor + Pronator	24

LICENCES AND SOURCES

EXERCISE LINE ART

The figure illustrations in this book are from the Workout Guide library by Bryl Lim, licensed CC BY-SA 4.0, itself derived from Everkinetic artwork also licensed CC BY-SA 4.0. They have been recoloured to the 7Bit ink tone and trimmed for print; no other change has been made. Because the licence is ShareAlike, any redistribution of this book or of those images must carry the same licence and this credit.

Source: github.com/bryllim/workout-guide | github.com/everkinetic/data | creativecommons.org/licenses/by-sa/4.0/

MOVEMENTS USING THAT ART

Band Pull-Apart, Banded Dead Bug, Banded Pallof Press, Banded Woodchop, Bird Dog, Bulgarian Split Squat, Butterfly Stretch, Cat-Cow Stretch, Copenhagen Plank, Dead Hang, Dip, Glute Bridge March, Goblet Squat, Half-Kneeling Pallof Press, Hanging Leg Raise, Hollow Body Hold, Inverted Row, Lying Leg Raise, Pike Push-up, Prone T Raise, Prone Y Raise, Pull-up, Push-up, Russian Twist, Scapular Pull-up, Side Plank, Side Plank Hip Dip, Single-Leg Romanian Deadlift, Standing Dumbbell Press, Torso Twists, Weighted Pull-up, World's Greatest Stretch, Wrist Curl, Wrist Extension.

EVERYTHING ELSE

The written steps, position checks, mistake lists and illustration prompts were written for this book. The prescriptions, protocols, rationales and sources are generated from `src/core` in this repository. Each entry's WHY section names the protocol and the evidence behind it; the full protocol table lives in `src/core/protocols.ts`.

TYPEFACE

Kode Mono by Isa Ozler, SIL Open Font License 1.1. The licence text ships at `tools/form-guide/assets/fonts/OFL.txt`.

FURTHER READING

- Lattice Training, hangboarding and assessment guides - latticeclimbing.com
- Eric Horst, 7/3 repeater protocol - trainingforclimbing.com
- Giles et al., 4-minute all-out finger flexor critical force test
- Effects of Different Loading Programs on Finger Strength in Rock Climbers, Sports Medicine Open (2024) - the density hang evidence
- Saeterbakken et al., core training in highly trained climbers, PLOS ONE (2018)
- Front Physiol (2021), two versus four weekly campus board sessions
- McGill, core endurance standards for the bird dog and side bridge

A CAVEAT WORTH PRINTING

This is a written reference, not coaching. Nothing in it has been checked by a physiotherapist. The finger and campus work in particular carries real injury risk, which is why the app gates the campus board behind a tested benchmark rather than behind experience. Pain that is sharp, that clicks, or that sits in a joint rather than a muscle is a stop signal, not a training cue.